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Last login: Mon Apr 20 10:38:25 on ttys001
apple@MacBook-Pro-10 project-main % source fl-env/bin/activate
(fl-env) apple@MacBook-Pro-10 project-main % python server.py baseline noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
         Instead, use the `flwr run` CLI command to start a local simulation in your Flower app, as shown for example below:

    $ flwr new # Create a new Flower app from a template

    $ flwr run # Run the Flower app in Simulation Mode

Using `start_simulation()` is deprecated.

This is a deprecated feature. It will be removed
entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=5, no round_timeout
2026-04-21 11:56:10,125 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray will
1 no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'node:127.0.0.1': 1.0, 'object_store_memory': 2147483648.0, 'node:__internal_head__': 1.0, 'CPU': 8.0, 'memory': 6171
869184.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13123) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2}
(ClientAppActor pid=13123) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=13123) labels = np.array(trainset.targets)
(ClientAppActor pid=13123) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25}
(ClientAppActor pid=13123) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58}
(ClientAppActor pid=13123) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79}
(ClientAppActor pid=13123) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697}
(ClientAppActor pid=13123) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11}
INFO : aggregate_fit: received 20 results and 0 failures
WARNING : No fit_metrics.aggregation_fn provided
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13121) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
33x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/c
onfigure-logging.html#log-deduplication-for-more-options.)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 11:56:19,477 E 13116 634971] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_11-56-07_918164_13104 is over 95% full, available space: 11
.8035 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13119) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=13119) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13120) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 70x acro
ss cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13120) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 59x
across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 11:56:29,575 E 13116 634971] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_11-56-07_918164_13104 is over 95% full, available space: 11
.8033 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [SUMMARY]
INFO : Run finished 5 round(s) in 17.95s
INFO : History (loss, distributed):
INFO : round 1: 1.9578069638901245
INFO : round 2: 1.325185287303437
INFO : round 3: 0.9078286438704299
INFO : round 4: 0.6865070357490272
INFO : round 5: 0.5639086496144438
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.5654),
INFO : (2, 0.7070999999999999),
INFO : (3, 0.765),
INFO : (4, 0.8182),
INFO : (5, 0.8514)]}
INFO :

--- honest_baseline_noniid ---
Round 1: 0.5654
Round 2: 0.7071
Round 3: 0.7650
Round 4: 0.8182
Round 5: 0.8514
[PLOT] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:
Round 1: 0.5654
Round 2: 0.7071
Round 3: 0.7650
Round 4: 0.8182
Round 5: 0.8514
(pid=gcs_server) [2026-04-21 11:56:38,642 E 13112 634814] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metri
cs will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(raylet) [2026-04-21 11:56:39,605 E 13116 634926] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter a
gent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(raylet) [2026-04-21 11:56:39,647 E 13116 634971] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_11-56-07_918164_13104 is over 95% full, available space: 11
.8026 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13121) [2026-04-21 11:56:40,864 E 13121 635251] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
[2026-04-21 11:56:41,032 E 13104 635047] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agen

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status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=13123) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 7, 4: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 33x
across cluster]
(ClientAppActor pid=13126) [2026-04-21 11:56:41,081 E 13126 635426] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
(f1-env) apple@MacBook-Pro-10 project-main % python server.py attack noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

$ flwr new # Create a new Flower app from a template

$ flwr run # Run the Flower app in Simulation Mode

Using 'start_simulation()' is deprecated.

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entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=5, no_round_timeout
2026-04-21 11:58:24,906 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray wil
l no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'object_store_memory': 2147483648.0, 'CPU': 8.0, 'node:127.0.0.1': 1.0, 'node:__internal_head__': 1.0, 'memory': 7143
817216.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: ('num_cpus': 1)
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
ERROR : Traceback (most recent call last):
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 90, in _submit_job
out_mssg, updated_context = self.actor_pool.get_client_result(
partition_id_str, timeout
)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 400, in get_client_result
return self._fetch_future_result(cid)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 281, in _fetch_future_res
ult
res_cid, out_mssg, updated_context = ray.get(
future
) # type: (str, Message, Context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/auto_init_hook.py", line 22, in auto_init_wrapper
return fn(*args, **kwargs)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/client_mode_hook.py", line 104, in wrapper
return func(*args, **kwargs)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py", line 2961, in get
values, debugger_breakpoint = worker.get_objects(
object_refs, timeout=timeout, _tensor_transport=_tensor_transport
)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py", line 1026, in get_objects
raise value.as_instanceof_cause()
ray.exceptions.RayTaskError(ClientAppException): ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_tran
sport.ray_actor.ClientAppActor object at 0x1122a27b0>)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 144, in __call__
return self._call(message, context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 128, in ffn
out_message = handle_legacy_message_from_msgtype(
client_fn=client_fn, message=message, context=context
)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/client/message_handler/message_handler.py", line 97, in handle_legacy_
message_from_msgtype
client = client_fn(context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 81, in client_fn
AttackClass = Traitor.get(attack_type, Traitor)
AttributeError: type object 'Traitor' has no attribute 'get'

The above exception was the direct cause of the following exception:

ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object at 0x1122a2
7b0>)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 65, in run
raise ClientAppException(str(ex)) from ex
flwr.clientapp.client_app.ClientAppException:
Exception ClientAppException occurred. Message: type object 'Traitor' has no attribute 'get'

ERROR : ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object
at 0x1122a27b0>)
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return self._call(message, context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 128, in ffn
out_message = handle_legacy_message_from_msgtype(
client_fn=client_fn, message=message, context=context
)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/client/message_handler/message_handler.py", line 97, in handle_legacy_
message_from_msgtype
client = client_fn(context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 81, in client_fn
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out_message = handle_legacy_message_from_msgtype(
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File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/client/message_handler/message_handler.py", line 97, in handle_legacy_
message_from_msgtype
client = client_fn(context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 81, in client_fn

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AttackClass = Traitor.get(attack_type, Traitor)
^^^^^^^^^^^^
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```
ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object at 0x1122a27b0>)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 65, in run
    raise ClientAppException(str(ex)) from ex
flwr.clientapp.client_app.ClientAppException:
Exception ClientAppException occurred. Message: type object 'Traitor' has no attribute 'get'
ERROR :      Traceback (most recent call last):
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/legacy_app.py", line 361, in start_simulation
    hist = run_fl(
            server=initialized_server,
            config=initialized_config,
        )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/server/server.py", line 487, in run_fl
    hist, elapsed_time = server.fit(
        num_rounds=config.num_rounds, timeout=config.round_timeout
    )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/server/server.py", line 92, in fit
    self.parameters = self._get_initial_parameters(server_round=0, timeout=timeout)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/server/server.py", line 279, in _get_initial_parameters
    get_parameters_res = random_client.get_parameters(
        ins=ins, timeout=timeout, group_id=server_round
    )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 165, in get_parameters
    message_out = self._submit_job(message, timeout)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 104, in _submit_job
    raise ex
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 90, in _submit_job
    out_mssg, updated_context = self.actor_pool.get_client_result(
        partition_id_str, timeout
    )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 400, in get_client_result
    return self._fetch_future_result(cid)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 281, in _fetch_future_result
    res_cid, out_mssg, updated_context = ray.get(
        future
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    values, debugger_breakpoint = worker.get_objects(
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ray.exceptions.RayTaskError(ClientAppException): ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object at 0x1122a27b0>)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 144, in __call__
    return self._call(message, context)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 128, in _call
    out_message = handle_legacy_message_from_msgtype(
        client_fn=client_fn, message=message, context=context
    )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/client/message_handler/message_handler.py", line 97, in handle_legacy_message_from_msgtype
    client = client_fn(context)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 81, in client_fn
    AttackClass = Traitor.get(attack_type, Traitor)
    ^^^^^^^^^^^^^
AttributeError: type object 'Traitor' has no attribute 'get'
```

The above exception was the direct cause of the following exception:

```
ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object at 0x1122a27b0>)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 65, in run
    raise ClientAppException(str(ex)) from ex
flwr.clientapp.client_app.ClientAppException:
Exception ClientAppException occurred. Message: type object 'Traitor' has no attribute 'get'
ERROR :      Your simulation crashed :(. This could be because of several reasons. The most common are:
  > Sometimes, issues in the simulation code itself can cause crashes. It's always a good idea to double-check your code for any potential bugs or inconsistencies that might be contributing to the problem. For example:
    - You might be using a class attribute in your clients that hasn't been defined.
    - There could be an incorrect method call to a 3rd party library (e.g., PyTorch).
    - The return types of methods in your clients/strategies might be incorrect.
  > Your system couldn't fit a single VirtualClient: try lowering 'client_resources'.
  > All the actors in your pool crashed. This could be because:
    - You clients hit an out-of-memory (OOM) error and actors couldn't recover from it. Try launching your simulation with more generous 'client_resources' settings (i.e. it seems {'num_cpus': 1} is not enough for your run). Use fewer concurrent actors.
    - You were running a multi-node simulation and all worker nodes disconnected. The head node might still be alive but cannot accommodate any actor with resources: {'num_cpus': 1}.
Take a look at the Flower simulation examples for guidance <https://flower.ai/docs/framework/how-to-run-simulations.html>.
Traceback (most recent call last):
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/legacy_app.py", line 361, in start_simulation
    hist = run_fl(
            server=initialized_server,
            config=initialized_config,
        )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/server/server.py", line 487, in run_fl
    hist, elapsed_time = server.fit(
        num_rounds=config.num_rounds, timeout=config.round_timeout
    )
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/server/server.py", line 92, in fit
    self.parameters = self._get_initial_parameters(server_round=0, timeout=timeout)
  File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/server/server.py", line 279, in _get_initial_parameters
    get_parameters_res = random_client.get_parameters(
        ins=ins, timeout=timeout, group_id=server_round
    )
```

```

)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 165, in get_parameters
    message_out = self._submit_job(message, timeout)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 104, in _submit_job
    raise ex
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_client_proxy.py", line 90, in _submit_job
    out_mssg, updated_context = self.actor_pool.get_client_result(
                                ~~~~~^
                                partition_id_str, timeout
                                ^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 400, in get_client_result
    return self._fetch_future_result(cid)
           ~~~~~^
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 281, in _fetch_future_result
    res_cid, out_mssg, updated_context = ray.get(
                                         ~~~~~^
                                         future
                                         ^^^^^
    ) # type: (str, Message, Context)
    ^
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/auto_init_hook.py", line 22, in auto_init_wrapper
    return fn(*args, **kwargs)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/client_mode_hook.py", line 104, in wrapper
    return func(*args, **kwargs)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py", line 2961, in get
    values, debugger_breakpoint = worker.get_objects(
                                ~~~~~^
                                object_refs, timeout=timeout, _tensor_transport=_tensor_transport
                                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py", line 1026, in get_objects
    raise ValueError(f"instanceof_f_cause()
ray.exceptions.RayTaskError(ClientAppException): ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object at 0x1122a27b0>)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 144, in __call__
    return self._call(message, context)
           ~~~~~^
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/clientapp/client_app.py", line 128, in _ffn
    out_message = handle_legacy_message_from_msgtype(
                  client_fn=client_fn, message=message, context=context
    )
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/client/message_handler/message_handler.py", line 97, in handle_legacy_message_from_msgtype
    client = client_fn(context)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 81, in client_fn
    AttackClass = Traitor.get(attack_type, Traitor)
                  ^^^^^^^^^^^
AttributeError: type object 'Traitor' has no attribute 'get'

The above exception was the direct cause of the following exception:

ray::ClientAppActor.run() (pid=13253, ip=127.0.0.1, actor_id=7b27ab91aaca21f3ef615a4101000000, repr=<flwr.simulation.ray_transport.ray_actor.ClientAppActor object at 0x1122a27b0>)
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/ray_transport/ray_actor.py", line 65, in run
    raise ClientAppException(str(ex)) from ex
flwr.clientapp.client_app.ClientAppException:
Exception ClientAppException occurred. Message: type object 'Traitor' has no attribute 'get'

The above exception was the direct cause of the following exception:

Traceback (most recent call last):
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 227, in <module>
    history, results = run_FL(
                        ~~~~~^
                        use_attack=True, use_trust=False,
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
                        num_rounds=NUM_ROUNDS,
                        ^^^^^^^^^^^^^^^^^^^^^^^^^
                        attack_type="label_flip",
                        ^^^^^^^^^^^^^^^^^^^^^^^^^
                        non_iid=non_iid)
                        ^^^^^^^^^^^^^^^^^^^^^
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/server.py", line 155, in run_FL
    history = fl.simulation.start_simulation(
    ...<6 lines>...
        ray_init_args={"ignore_reinit_error": True},
    )
File "/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/flwr/simulation/legacy_app.py", line 397, in start_simulation
    raise RuntimeError("Simulation crashed.") from ex
RuntimeError: Simulation crashed.
(ClientAppActor pid=13253) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13}
(ClientAppActor pid=13253) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migration_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=13253) labels = np.array(trainset.targets)
(raylet) [2026-04-21 11:58:34,889 E 13243 638777] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_11-58-24_125645_13234 is over 95% full, available space: 12.9761 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(fl-env) apple@MacBook-Pro-10 project-main % python server.py attack noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

$ flwr new # Create a new Flower app from a template

$ flwr run # Run the Flower app in Simulation Mode

Using 'start_simulation()' is deprecated.

This is a deprecated feature. It will be removed entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=5, no round_timeout
2026-04-21 12:19:31,201 INFO worker.py:2012 — Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray will no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'memory': 6763331584.0, 'node:127.0.0.1': 1.0, 'node:__internal_head__': 1.0, 'CPU': 8.0, 'object_store_memory': 2147483648.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
(ClientAppActor pid=13758) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migration_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=13758) labels = np.array(trainset.targets)
(ClientAppActor pid=13758) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58}
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters

```

```

INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13758) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25}
(ClientAppActor pid=13758) [SETUP] Client 14 is MALICIOUS (label_flip)
(ClientAppActor pid=13758) [ATTACK] Client 14 submitted poisoned update
(ClientAppActor pid=13758) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13}
(ClientAppActor pid=13758) [SETUP] Client 19 is MALICIOUS (label_flip)
(ClientAppActor pid=13758) [ATTACK] Client 19 submitted poisoned update
(ClientAppActor pid=13758) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697}
(ClientAppActor pid=13758) [SETUP] Client 12 is MALICIOUS (label_flip)
(ClientAppActor pid=13758) [ATTACK] Client 12 submitted poisoned update
INFO : aggregate_fit: received 20 results and 0 failures
WARNING : No fit_metrics_aggregation_fn provided
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13755) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 3
3x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/co
nfigure-logging.html#log-deduplication for more options.)
(ClientAppActor pid=13753) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13755) [ATTACK] Client 18 submitted poisoned update [repeated 5x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:19:41,189 E 13749 653881] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-19-30_356078_13738 is over 95% full, available space: 13
.0055 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13756) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=13756) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13752) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 7
0x across cluster]
(ClientAppActor pid=13754) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=13752) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13755) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeate
d 55x across cluster]
(ClientAppActor pid=13755) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 21x across cluster]
(ClientAppActor pid=13755) [ATTACK] Client 12 submitted poisoned update [repeated 13x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:19:51,225 E 13749 653881] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-19-30_356078_13738 is over 95% full, available space: 11
.906 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [SUMMARY]
INFO : Run finished 5 round(s) in 18.25s
INFO : History (loss, distributed):
INFO : round 1: 1.2871182365539355
INFO : round 2: 1.1095391307014246
INFO : round 3: 0.938342353711113
INFO : round 4: 0.8305672661564982
INFO : round 5: 0.7525412230826795
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.356), (2, 0.4118), (3, 0.4742), (4, 0.5011), (5, 0.5651)]}
INFO :

--- fedavg_attack_noniid ---
Round 1: 0.3560
Round 2: 0.4118
Round 3: 0.4742
Round 4: 0.5011
Round 5: 0.5651
[PLOT] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:
Round 1: 0.3560
Round 2: 0.4118
Round 3: 0.4742
Round 4: 0.5011
Round 5: 0.5651
(pid=gcs_server) [2026-04-21 12:20:00,826 E 13745 653753] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metri
cs will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(raylet) [2026-04-21 12:20:01,253 E 13749 653881] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-19-30_356078_13738 is over 95% full, available space: 11
.9041 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(raylet) [2026-04-21 12:20:01,426 E 13749 653861] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter a
gent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=13755) [2026-04-21 12:20:02,002 E 13755 654134] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
[2026-04-21 12:20:02,269 E 13738 653916] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agen
t status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=13752) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 3
9x across cluster]
(ClientAppActor pid=13757) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 15x across cluster]
(ClientAppActor pid=13753) [ATTACK] Client 18 submitted poisoned update [repeated 3x across cluster]
(ClientAppActor pid=13758) [2026-04-21 12:20:02,277 E 13758 654395] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
(f1-env) appleMacBook-Pro-10 project-main % python server.py attack_noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

    $ flwr new # Create a new Flower app from a template

    $ flwr run # Run the Flower app in Simulation Mode

Using 'start_simulation()' is deprecated.

This is a deprecated feature. It will be removed
entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=30, no round_timeout
2026-04-21 12:22:44,816 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray wil
l no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'CPU': 8.0, 'memory': 8226308096.0, 'node:127.0.0.1': 1.0, 'object_store_memory': 2147483648.0, 'node:__internal_head
__': 1.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html

```



```
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=13868) labels = np.array(trainset.targets)
(ClientAppActor pid=13868) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13}
(ClientAppActor pid=13868) [SETUP] Client 19 is MALICIOUS (label_flip)
(ClientAppActor pid=13868) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58}
(ClientAppActor pid=13868) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9}
(ClientAppActor pid=13868) [SETUP] Client 18 is MALICIOUS (label_flip)
(ClientAppActor pid=13868) [ATTACK] Client 18 submitted poisoned update
(ClientAppActor pid=13868) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79}
(ClientAppActor pid=13868) [SETUP] Client 13 is MALICIOUS (label_flip)
(ClientAppActor pid=13868) [ATTACK] Client 13 submitted poisoned update
(ClientAppActor pid=13868) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010}
(ClientAppActor pid=13868) [SETUP] Client 14 is MALICIOUS (label_flip)
(ClientAppActor pid=13868) [ATTACK] Client 14 submitted poisoned update
INFO : aggregate_fit: received 20 results and 0 failures
WARNING : No fit_metrics_aggregation_fn provided
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 30
x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/con
figure-logging.html#log-deduplication for more options.)
(ClientAppActor pid=13868) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13870) [ATTACK] Client 12 submitted poisoned update [repeated 5x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:22:54,770 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21-12-22-44_038374_13855 is over 95% full, available space: 12
.9042 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13870) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=13870) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
49x across cluster]
(ClientAppActor pid=13869) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 18x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13874) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13875) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 63x
across cluster]
(ClientAppActor pid=13872) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13869) [ATTACK] Client 13 submitted poisoned update [repeated 9x across cluster]
(raylet) [2026-04-21 12:23:04,807 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21-12-22-44_038374_13855 is over 95% full, available space: 12
.8839 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 6]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 7
0x across cluster]
(ClientAppActor pid=13875) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 28x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13873) [ATTACK] Client 18 submitted poisoned update [repeated 15x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 7]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13871) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
1x across cluster]
(ClientAppActor pid=13872) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 8]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13873) [ATTACK] Client 13 submitted poisoned update [repeated 12x across cluster]
(pid=gcs_server) [2026-04-21 12:23:14,594 E 13861 657371] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metri
cs will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(raylet) [2026-04-21 12:23:14,835 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21-12-22-44_038374_13855 is over 95% full, available space: 11
.6821 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(raylet) [2026-04-21 12:23:15,086 E 13865 657479] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter a
gent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13874) [2026-04-21 12:23:15,720 E 13874 657898] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
[2026-04-21 12:23:15,891 E 13855 657537] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agen
t status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 9]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13872) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 4
9x across cluster]
(ClientAppActor pid=13872) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 20x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13875) [ATTACK] Client 13 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 10]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13872) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 67x
```

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across cluster]
(ClientAppActor pid=13875) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 11]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13869) [ATTACK] Client 18 submitted poisoned update [repeated 12x across cluster]
(raylet) [2026-04-21 12:23:24,857 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11.6724 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13868) [2026-04-21 12:23:16,042 E 13868 658001] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 12]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13875) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35} [repeated 63x across cluster]
(ClientAppActor pid=13875) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [ATTACK] Client 12 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 13]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13871) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated 59x across cluster]
(ClientAppActor pid=13871) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 14]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13874) [ATTACK] Client 15 submitted poisoned update [repeated 13x across cluster]
(raylet) [2026-04-21 12:23:34,872 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11.654 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 15]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13869) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 55x across cluster]
(ClientAppActor pid=13868) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 21x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13871) [ATTACK] Client 16 submitted poisoned update [repeated 11x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 16]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13871) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 46x across cluster]
(ClientAppActor pid=13869) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 20x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:23:44,897 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11.575 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13875) [ATTACK] Client 19 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 17]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13873) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 53x across cluster]
(ClientAppActor pid=13871) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 21x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 18]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13875) [ATTACK] Client 17 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 19]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13870) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 52x across cluster]
(ClientAppActor pid=13874) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 20x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13871) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 12:23:54,985 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11.5955 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 20]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13873) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 71x across cluster]
(ClientAppActor pid=13873) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 31x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 21]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [ATTACK] Client 14 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 22]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13873) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 64x across cluster]
(ClientAppActor pid=13868) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 23x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:24:05,046 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11.5932 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13869) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 23]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
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INFO : [ROUND 24]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 65x
across cluster]
(ClientAppActor pid=13868) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=13874) [ATTACK] Client 15 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 25]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13868) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
5x across cluster]
(ClientAppActor pid=13868) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 26x across cluster]
(raylet) [2026-04-21 12:24:15,081 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11
.5913 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13869) [ATTACK] Client 12 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 26]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 27]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13872) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 5
6x across cluster]
(ClientAppActor pid=13869) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 23x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13870) [ATTACK] Client 17 submitted poisoned update [repeated 15x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 28]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13870) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 6
8x across cluster]
(ClientAppActor pid=13869) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 25x across cluster]
(raylet) [2026-04-21 12:24:25,094 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11
.5919 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 29]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13872) [ATTACK] Client 15 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 30]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13875) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
6x across cluster]
(ClientAppActor pid=13869) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 30x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=13874) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [SUMMARY]
INFO : Run finished 30 round(s) in 104.45s
INFO : History (loss, distributed):
INFO : round 1: 1.281678900855799
INFO : round 2: 1.0905228695549523
INFO : round 3: 0.9397350491807104
INFO : round 4: 0.8360259596913006
INFO : round 5: 0.7730379175835144
INFO : round 6: 0.727067707750363
INFO : round 7: 0.6910991291268564
INFO : round 8: 0.6637172751533338
INFO : round 9: 0.6439244402864108
INFO : round 10: 0.6147490493024881
INFO : round 11: 0.5983901303797103
INFO : round 12: 0.5819036554795104
INFO : round 13: 0.5621151223159827
INFO : round 14: 0.5537763508364034
INFO : round 15: 0.5427501632001834
INFO : round 16: 0.5222143964836011
INFO : round 17: 0.5205484269335627
INFO : round 18: 0.5053255472701197
INFO : round 19: 0.5065737544362918
INFO : round 20: 0.4912431971524089
INFO : round 21: 0.48481836147582563
INFO : round 22: 0.47876591164464005
INFO : round 23: 0.4745929421708226
INFO : round 24: 0.4641109710303359
INFO : round 25: 0.4614008014575361
INFO : round 26: 0.4534709038635412
INFO : round 27: 0.449434946539303
INFO : round 28: 0.4431577972329843
INFO : round 29: 0.4426477978404719
INFO : round 30: 0.42995638487438037
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.2146),
INFO : (2, 0.3529),
INFO : (3, 0.4229),
INFO : (4, 0.5061),
INFO : (5, 0.5629),
INFO : (6, 0.5819),
INFO : (7, 0.6053),
INFO : (8, 0.629),
INFO : (9, 0.6264),
INFO : (10, 0.6471),
INFO : (11, 0.6589),
INFO : (12, 0.6654),
INFO : (13, 0.6787),
INFO : (14, 0.6825),
INFO : (15, 0.6892),
INFO : (16, 0.719),
INFO : (17, 0.7124),
INFO : (18, 0.72),
INFO : (19, 0.7201),
INFO : (20, 0.7367),
INFO : (21, 0.7406),
INFO : (22, 0.7482),
INFO : (23, 0.7572),
INFO : (24, 0.763),
INFO : (25, 0.7611),
INFO : (26, 0.7752),
INFO : (27, 0.7712),
```



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INFO : (28, 0.7856999999999998),
INFO : (29, 0.7742),
INFO : (30, 0.7925)}}
INFO :

--- fedavg_attack_noniid ---
Round 1: 0.2146
Round 2: 0.3529
Round 3: 0.4229
Round 4: 0.5061
Round 5: 0.5629
Round 6: 0.5819
Round 7: 0.6053
Round 8: 0.6290
Round 9: 0.6264
Round 10: 0.6471
Round 11: 0.6589
Round 12: 0.6654
Round 13: 0.6787
Round 14: 0.6825
Round 15: 0.6892
Round 16: 0.7190
Round 17: 0.7124
Round 18: 0.7200
Round 19: 0.7201
Round 20: 0.7367
Round 21: 0.7406
Round 22: 0.7482
Round 23: 0.7572
Round 24: 0.7630
Round 25: 0.7611
Round 26: 0.7752
Round 27: 0.7712
Round 28: 0.7857
Round 29: 0.7742
Round 30: 0.7925
[PL0T] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:
Round 1: 0.2146
Round 2: 0.3529
Round 3: 0.4229
Round 4: 0.5061
Round 5: 0.5629
Round 6: 0.5819
Round 7: 0.6053
Round 8: 0.6290
Round 9: 0.6264
Round 10: 0.6471
Round 11: 0.6589
Round 12: 0.6654
Round 13: 0.6787
Round 14: 0.6825
Round 15: 0.6892
Round 16: 0.7190
Round 17: 0.7124
Round 18: 0.7200
Round 19: 0.7201
Round 20: 0.7367
Round 21: 0.7406
Round 22: 0.7482
Round 23: 0.7572
Round 24: 0.7630
Round 25: 0.7611
Round 26: 0.7752
Round 27: 0.7712
Round 28: 0.7857
Round 29: 0.7742
Round 30: 0.7925

(raylet) [2026-04-21 12:24:35,101 E 13865 657495] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-22-44_038374_13855 is over 95% full, available space: 11
.5985 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=13871) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 24
x across cluster]
(ClientAppActor pid=13871) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 8x across cluster]
(fl-env) apple@MacBook-Pro-10 project-main % python server.py trust noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

$ flwr new # Create a new Flower app from a template

$ flwr run # Run the Flower app in Simulation Mode

Using 'start_simulation()' is deprecated.

This is a deprecated feature. It will be removed
entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=30, no round_timeout
2026-04-21 12:35:13,694 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray will
l no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'object_store_memory': 2147483648.0, 'CPU': 8.0, 'memory': 7088832512.0, 'node:127.0.0.1': 1.0, 'node:__internal_head
__': 1.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14192) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=14192) labels = np.array(trainset.targets)
(ClientAppActor pid=14192) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010}
(ClientAppActor pid=14192) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79}
(ClientAppActor pid=14192) [SETUP] Client 13 is MALICIOUS (label_flip)
(ClientAppActor pid=14192) [ATTACK] Client 13 submitted poisoned update
(ClientAppActor pid=14192) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622}
(ClientAppActor pid=14192) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953}
(ClientAppActor pid=14192) [SETUP] Client 15 is MALICIOUS (label_flip)
(ClientAppActor pid=14192) [ATTACK] Client 15 submitted poisoned update
(ClientAppActor pid=14192) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4}
(ClientAppActor pid=14192) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697}
(ClientAppActor pid=14192) [SETUP] Client 12 is MALICIOUS (label_flip)
(ClientAppActor pid=14192) [ATTACK] Client 12 submitted poisoned update
INFO : aggregate_fit: received 20 results and 0 failures

[ROUND 1] CFA Trust Weights:
Client 0: weight=1.000 SKIP (no accuracy yet)
Client 6: weight=1.000 SKIP (no accuracy yet)

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Client 1: weight=1.000 SKIP (no accuracy yet)
Client 16: weight=1.000 SKIP (no accuracy yet)
Client 10: weight=1.000 SKIP (no accuracy yet)
Client 12: weight=1.000 SKIP (no accuracy yet)
Client 13: weight=1.000 SKIP (no accuracy yet)
Client 5: weight=1.000 SKIP (no accuracy yet)
Client 3: weight=1.000 SKIP (no accuracy yet)
Client 7: weight=1.000 SKIP (no accuracy yet)
Client 4: weight=1.000 SKIP (no accuracy yet)
Client 17: weight=1.000 SKIP (no accuracy yet)
Client 14: weight=1.000 SKIP (no accuracy yet)
Client 8: weight=1.000 SKIP (no accuracy yet)
Client 11: weight=1.000 SKIP (no accuracy yet)
Client 2: weight=1.000 SKIP (no accuracy yet)
Client 19: weight=1.000 SKIP (no accuracy yet)
Client 18: weight=1.000 SKIP (no accuracy yet)
Client 9: weight=1.000 SKIP (no accuracy yet)
Client 15: weight=1.000 SKIP (no accuracy yet)
→ Total weight: 20.000 | Avg weight: 1.000
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14189) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 3
1x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/co
nfigure-logging.html#log-deduplication-for-more-options.)
(ClientAppActor pid=14188) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 12x across cluster]
(ClientAppActor pid=14187) [ATTACK] Client 16 submitted poisoned update [repeated 5x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 1 accuracy recorded: 0.3189 (baseline)
INFO :
INFO :      [ROUND 2]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures

[ROUND 2] CFA Trust Weights:
Client 5: weight=1.000 SKIP (round 1)
Client 3: weight=1.000 SKIP (round 1)
Client 0: weight=1.000 SKIP (round 1)
Client 10: weight=1.000 SKIP (round 1)
Client 16: weight=1.000 SKIP (round 1)
Client 15: weight=1.000 SKIP (round 1)
Client 18: weight=1.000 SKIP (round 1)
Client 8: weight=1.000 SKIP (round 1)
Client 7: weight=1.000 SKIP (round 1)
Client 11: weight=1.000 SKIP (round 1)
Client 14: weight=1.000 SKIP (round 1)
Client 17: weight=1.000 SKIP (round 1)
Client 19: weight=1.000 SKIP (round 1)
Client 13: weight=1.000 SKIP (round 1)
Client 12: weight=1.000 SKIP (round 1)
Client 2: weight=1.000 SKIP (round 1)
Client 6: weight=1.000 SKIP (round 1)
Client 9: weight=1.000 SKIP (round 1)
Client 1: weight=1.000 SKIP (round 1)
Client 4: weight=1.000 SKIP (round 1)
→ Total weight: 20.000 | Avg weight: 1.000
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:35:23,748 E 14182 66940] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 12
.8337 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14185) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=14185) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 2 accuracy recorded: 0.4198 | Δacc = 0.1009
INFO :
INFO :      [ROUND 3]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14191) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
64x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.1009 → REWARDING aligned clients

[ROUND 3] CFA Trust Weights:
Client 12: weight=1.004 REWARDED (sim=+0.042)
Client 2: weight=1.005 REWARDED (sim=+0.050)
Client 7: weight=0.980 MILD PENALTY (sim=-0.061)
Client 4: weight=0.980 MILD PENALTY (sim=-0.069)
Client 15: weight=1.000 NEUTRAL (sim=+0.007)
Client 10: weight=1.000 NEUTRAL (sim=-0.008)
Client 0: weight=1.004 REWARDED (sim=+0.038)
Client 3: weight=1.002 REWARDED (sim=+0.021)
Client 6: weight=0.980 MILD PENALTY (sim=-0.056)
Client 1: weight=0.980 MILD PENALTY (sim=-0.010)
Client 18: weight=1.003 REWARDED (sim=+0.027)
Client 14: weight=0.980 MILD PENALTY (sim=-0.026)
Client 17: weight=1.004 REWARDED (sim=+0.041)
Client 16: weight=0.980 MILD PENALTY (sim=-0.021)
Client 8: weight=1.005 REWARDED (sim=+0.055)
Client 11: weight=1.006 REWARDED (sim=+0.060)
Client 19: weight=0.980 MILD PENALTY (sim=-0.028)
Client 13: weight=0.980 MILD PENALTY (sim=-0.157)
Client 5: weight=1.002 REWARDED (sim=+0.017)
Client 9: weight=0.980 MILD PENALTY (sim=-0.070)
→ Total weight: 19.855 | Avg weight: 0.993
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14191) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=14189) [ATTACK] Client 16 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 3 accuracy recorded: 0.5056 | Δacc = 0.0858
INFO :
INFO :      [ROUND 4]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0858 → REWARDING aligned clients

[ROUND 4] CFA Trust Weights:
Client 8: weight=1.012 REWARDED (sim=+0.061)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 12: weight=1.008 REWARDED (sim=+0.041)
Client 14: weight=0.960 MILD PENALTY (sim=-0.030)
Client 13: weight=0.960 MILD PENALTY (sim=-0.151)
Client 7: weight=0.960 MILD PENALTY (sim=-0.071)
Client 0: weight=1.010 REWARDED (sim=+0.060)
Client 11: weight=1.013 REWARDED (sim=+0.071)
Client 6: weight=0.960 MILD PENALTY (sim=-0.047)
Client 4: weight=0.960 MILD PENALTY (sim=-0.083)
Client 16: weight=0.960 MILD PENALTY (sim=-0.027)
Client 10: weight=0.980 MILD PENALTY (sim=-0.012)
Client 3: weight=1.006 REWARDED (sim=+0.034)
Client 18: weight=1.005 REWARDED (sim=+0.021)
Client 9: weight=0.960 MILD PENALTY (sim=-0.064)
Client 15: weight=1.000 NEUTRAL (sim=-0.007)
Client 19: weight=0.960 MILD PENALTY (sim=-0.047)
Client 2: weight=1.011 REWARDED (sim=+0.060)
Client 5: weight=1.004 REWARDED (sim=+0.019)
Client 17: weight=1.007 REWARDED (sim=+0.030)
→ Total weight: 19.718 | Avg weight: 0.986
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)

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INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 4 accuracy recorded: 0.5451 | Δacc = 0.0395
INFO :
INFO :      [ROUND 5]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14190) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 6
1x across cluster]
(ClientAppActor pid=14190) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 22x across cluster]
(ClientAppActor pid=14188) [ATTACK] Client 13 submitted poisoned update [repeated 8x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0395 → REWARDING aligned clients

[ROUND 5] CFA Trust Weights:
Client 13: weight=0.941 MILD PENALTY (sim=-0.132)
Client 1: weight=0.980 NEUTRAL (sim=+0.010)
Client 5: weight=1.006 REWARDED (sim=+0.020)
Client 6: weight=0.941 MILD PENALTY (sim=-0.043)
Client 14: weight=0.941 MILD PENALTY (sim=-0.034)
Client 3: weight=1.010 REWARDED (sim=+0.046)
Client 0: weight=1.016 REWARDED (sim=+0.060)
Client 18: weight=1.006 REWARDED (sim=+0.016)
Client 12: weight=1.012 REWARDED (sim=+0.040)
Client 7: weight=0.941 MILD PENALTY (sim=-0.079)
Client 11: weight=1.021 REWARDED (sim=+0.075)
Client 9: weight=0.941 MILD PENALTY (sim=-0.060)
Client 19: weight=0.941 MILD PENALTY (sim=-0.059)
Client 10: weight=0.980 NEUTRAL (sim=-0.002)
Client 16: weight=0.941 MILD PENALTY (sim=-0.033)
Client 4: weight=0.941 MILD PENALTY (sim=-0.070)
Client 15: weight=0.980 MILD PENALTY (sim=-0.017)
Client 17: weight=1.009 REWARDED (sim=+0.016)
Client 8: weight=1.018 REWARDED (sim=+0.061)
Client 2: weight=1.018 REWARDED (sim=+0.065)
→ Total weight: 19.585 | Avg weight: 0.979
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:35:33,825 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 12
.817 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 5 accuracy recorded: 0.5802 | Δacc = 0.0351
INFO :
INFO :      [ROUND 6]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0351 → REWARDING aligned clients

[ROUND 6] CFA Trust Weights:
Client 2: weight=1.024 REWARDED (sim=+0.060)
Client 15: weight=0.960 MILD PENALTY (sim=-0.019)
Client 8: weight=1.023 REWARDED (sim=+0.055)
Client 9: weight=0.922 MILD PENALTY (sim=-0.060)
Client 16: weight=0.922 MILD PENALTY (sim=-0.034)
Client 0: weight=1.022 REWARDED (sim=+0.056)
Client 17: weight=1.009 NEUTRAL (sim=+0.003)
Client 1: weight=0.981 REWARDED (sim=+0.010)
Client 3: weight=1.016 REWARDED (sim=+0.053)
Client 11: weight=1.029 REWARDED (sim=+0.076)
Client 12: weight=1.016 REWARDED (sim=+0.041)
Client 18: weight=1.008 REWARDED (sim=+0.015)
Client 14: weight=0.922 MILD PENALTY (sim=-0.039)
Client 7: weight=0.922 MILD PENALTY (sim=-0.065)
Client 13: weight=0.922 MILD PENALTY (sim=-0.113)
Client 4: weight=0.922 MILD PENALTY (sim=-0.060)
Client 5: weight=1.008 REWARDED (sim=+0.021)
Client 6: weight=0.922 MILD PENALTY (sim=-0.041)
Client 19: weight=0.922 MILD PENALTY (sim=-0.062)
Client 10: weight=0.980 NEUTRAL (sim=+0.008)
→ Total weight: 19.454 | Avg weight: 0.973
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14186) [NON-IID] Client 9 | Total: 2061 samples | Distribution: {0: 11, 1: 133, 2: 1264, 3: 170, 4: 261, 5: 169, 6: 46, 7: 1, 8: 3, 9: 3} [repeated 74x
across cluster]
(ClientAppActor pid=14185) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=14188) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 6 accuracy recorded: 0.6028 | Δacc = 0.0226
INFO :
INFO :      [ROUND 7]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0226 → REWARDING aligned clients

[ROUND 7] CFA Trust Weights:
Client 7: weight=0.904 MILD PENALTY (sim=-0.060)
Client 16: weight=0.904 MILD PENALTY (sim=-0.037)
Client 9: weight=0.904 MILD PENALTY (sim=-0.055)
Client 11: weight=1.036 REWARDED (sim=+0.074)
Client 17: weight=1.009 NEUTRAL (sim=-0.002)
Client 10: weight=0.981 REWARDED (sim=+0.013)
Client 5: weight=1.009 REWARDED (sim=+0.016)
Client 2: weight=1.029 REWARDED (sim=+0.057)
Client 8: weight=1.029 REWARDED (sim=+0.060)
Client 13: weight=0.904 MILD PENALTY (sim=-0.103)
Client 12: weight=1.020 REWARDED (sim=+0.039)
Client 14: weight=0.904 MILD PENALTY (sim=-0.034)
Client 15: weight=0.941 MILD PENALTY (sim=-0.022)
Client 19: weight=0.904 MILD PENALTY (sim=-0.060)
Client 6: weight=0.904 MILD PENALTY (sim=-0.037)
Client 4: weight=0.904 MILD PENALTY (sim=-0.045)
Client 3: weight=1.021 REWARDED (sim=+0.057)
Client 0: weight=1.027 REWARDED (sim=+0.049)
Client 1: weight=0.981 NEUTRAL (sim=+0.007)
Client 18: weight=1.008 NEUTRAL (sim=+0.008)
→ Total weight: 19.324 | Avg weight: 0.966
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 7 accuracy recorded: 0.6267 | Δacc = 0.0239
INFO :
INFO :      [ROUND 8]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0239 → REWARDING aligned clients

[ROUND 8] CFA Trust Weights:
Client 1: weight=0.981 NEUTRAL (sim=+0.006)
Client 4: weight=0.886 MILD PENALTY (sim=-0.039)
Client 7: weight=0.886 MILD PENALTY (sim=-0.058)
Client 12: weight=1.024 REWARDED (sim=+0.036)
Client 8: weight=1.035 REWARDED (sim=+0.056)
Client 2: weight=1.035 REWARDED (sim=+0.055)
Client 11: weight=1.043 REWARDED (sim=+0.070)
Client 15: weight=0.922 MILD PENALTY (sim=-0.023)
Client 13: weight=0.886 MILD PENALTY (sim=-0.088)
Client 17: weight=1.009 NEUTRAL (sim=-0.009)
Client 18: weight=1.008 NEUTRAL (sim=+0.004)
Client 6: weight=0.886 MILD PENALTY (sim=-0.033)
Client 14: weight=0.886 MILD PENALTY (sim=-0.031)
Client 10: weight=0.983 REWARDED (sim=+0.016)
Client 0: weight=1.031 REWARDED (sim=+0.046)
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Client 19: weight=0.886 MILD PENALTY (sim=-0.059)
Client 16: weight=0.886 MILD PENALTY (sim=-0.033)
Client 3: weight=1.027 REWARDED (sim=+0.058)
Client 9: weight=0.886 MILD PENALTY (sim=-0.048)
Client 5: weight=1.011 REWARDED (sim=+0.012)
→ Total weight: 19.197 | Avg weight: 0.960
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14185) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 70x across cluster]
(ClientAppActor pid=14185) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=14188) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
(pid=gcs_server) [2026-04-21 12:35:43,162 E 14178 669318] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 8 accuracy recorded: 0.6585 | Δacc = 0.0318
INFO :
INFO :      [ROUND 9]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:35:43,903 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11.6819 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(raylet) [2026-04-21 12:35:44,058 E 14182 669424] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=14185) [2026-04-21 12:35:44,603 E 14185 669706] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
[2026-04-21 12:35:44,662 E 14163 669479] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0318 → REWARDING aligned clients

[ROUND 9] CFA Trust Weights:
Client 15: weight=0.904 MILD PENALTY (sim=-0.023)
Client 14: weight=0.868 MILD PENALTY (sim=-0.031)
Client 0: weight=1.036 REWARDED (sim=+0.042)
Client 2: weight=1.041 REWARDED (sim=+0.054)
Client 13: weight=0.868 MILD PENALTY (sim=-0.083)
Client 11: weight=1.050 REWARDED (sim=+0.067)
Client 19: weight=0.868 MILD PENALTY (sim=-0.055)
Client 18: weight=1.008 NEUTRAL (sim=+0.000)
Client 4: weight=0.868 MILD PENALTY (sim=-0.032)
Client 8: weight=1.041 REWARDED (sim=+0.053)
Client 16: weight=0.868 MILD PENALTY (sim=-0.035)
Client 1: weight=0.981 NEUTRAL (sim=+0.006)
Client 17: weight=0.989 MILD PENALTY (sim=-0.012)
Client 10: weight=0.985 REWARDED (sim=+0.020)
Client 3: weight=1.033 REWARDED (sim=+0.056)
Client 7: weight=0.868 MILD PENALTY (sim=-0.046)
Client 12: weight=1.028 REWARDED (sim=+0.037)
Client 6: weight=0.868 MILD PENALTY (sim=-0.031)
Client 9: weight=0.868 MILD PENALTY (sim=-0.045)
Client 5: weight=1.012 REWARDED (sim=+0.012)
→ Total weight: 19.051 | Avg weight: 0.953
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 9 accuracy recorded: 0.6646 | Δacc = 0.0061
INFO :
INFO :      [ROUND 10]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14192) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 56x across cluster]
(ClientAppActor pid=14192) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 21x across cluster]
(ClientAppActor pid=14192) [ATTACK] Client 19 submitted poisoned update [repeated 8x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0061 → REWARDING aligned clients

[ROUND 10] CFA Trust Weights:
Client 2: weight=1.046 REWARDED (sim=+0.047)
Client 14: weight=0.851 MILD PENALTY (sim=-0.025)
Client 3: weight=1.039 REWARDED (sim=+0.057)
Client 16: weight=0.851 MILD PENALTY (sim=-0.036)
Client 19: weight=0.851 MILD PENALTY (sim=-0.052)
Client 1: weight=0.981 NEUTRAL (sim=+0.010)
Client 5: weight=1.012 NEUTRAL (sim=+0.010)
Client 13: weight=0.851 MILD PENALTY (sim=-0.074)
Client 18: weight=1.008 NEUTRAL (sim=-0.001)
Client 4: weight=0.851 MILD PENALTY (sim=-0.033)
Client 10: weight=0.987 REWARDED (sim=+0.020)
Client 8: weight=1.046 REWARDED (sim=+0.054)
Client 12: weight=1.031 REWARDED (sim=+0.034)
Client 7: weight=0.851 MILD PENALTY (sim=-0.044)
Client 0: weight=1.040 REWARDED (sim=+0.039)
Client 9: weight=0.851 MILD PENALTY (sim=-0.042)
Client 17: weight=0.969 MILD PENALTY (sim=-0.015)
Client 11: weight=1.057 REWARDED (sim=+0.065)
Client 6: weight=0.851 MILD PENALTY (sim=-0.031)
Client 15: weight=0.886 MILD PENALTY (sim=-0.024)
→ Total weight: 18.907 | Avg weight: 0.945
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 10 accuracy recorded: 0.6741 | Δacc = 0.0095
INFO :
INFO :      [ROUND 11]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0095 → REWARDING aligned clients

[ROUND 11] CFA Trust Weights:
Client 1: weight=0.982 REWARDED (sim=+0.010)
Client 14: weight=0.834 MILD PENALTY (sim=-0.026)
Client 4: weight=0.834 MILD PENALTY (sim=-0.032)
Client 13: weight=0.834 MILD PENALTY (sim=-0.068)
Client 11: weight=1.064 REWARDED (sim=+0.065)
Client 19: weight=0.834 MILD PENALTY (sim=-0.050)
Client 7: weight=0.834 MILD PENALTY (sim=-0.039)
Client 15: weight=0.868 MILD PENALTY (sim=-0.022)
Client 17: weight=0.949 MILD PENALTY (sim=-0.016)
Client 18: weight=1.008 NEUTRAL (sim=-0.004)
Client 12: weight=1.035 REWARDED (sim=+0.035)
Client 16: weight=0.834 MILD PENALTY (sim=-0.034)
Client 5: weight=1.012 NEUTRAL (sim=+0.004)
Client 10: weight=0.989 REWARDED (sim=+0.022)
Client 0: weight=1.044 REWARDED (sim=+0.038)
Client 3: weight=1.045 REWARDED (sim=+0.058)
Client 2: weight=1.050 REWARDED (sim=+0.046)
Client 9: weight=0.834 MILD PENALTY (sim=-0.039)
Client 6: weight=0.834 MILD PENALTY (sim=-0.027)
Client 8: weight=1.051 REWARDED (sim=+0.046)
→ Total weight: 18.767 | Avg weight: 0.938
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14190) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated 70x across cluster]
(ClientAppActor pid=14190) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=14190) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 11 accuracy recorded: 0.6941 | Δacc = 0.0200
INFO :
INFO :      [ROUND 12]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
```

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(raylet) [2026-04-21 12:35:53,943 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11
.6197 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14190) [2026-04-21 12:35:44,643 E 14190 669780] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0200 → REWARDING aligned clients

[ROUND 12] CFA Trust Weights:
Client 6: weight=0.817 MILD PENALTY (sim=-0.027)
Client 3: weight=1.051 REWARDED (sim=+0.056)
Client 7: weight=0.817 MILD PENALTY (sim=-0.037)
Client 5: weight=1.012 NEUTRAL (sim=+0.004)
Client 8: weight=1.055 REWARDED (sim=+0.034)
Client 18: weight=1.008 NEUTRAL (sim=-0.001)
Client 9: weight=0.817 MILD PENALTY (sim=-0.038)
Client 13: weight=0.817 MILD PENALTY (sim=-0.063)
Client 10: weight=0.991 REWARDED (sim=+0.021)
Client 14: weight=0.817 MILD PENALTY (sim=-0.023)
Client 15: weight=0.851 MILD PENALTY (sim=-0.021)
Client 16: weight=0.817 MILD PENALTY (sim=-0.036)
Client 0: weight=1.047 REWARDED (sim=+0.034)
Client 1: weight=0.982 NEUTRAL (sim=+0.009)
Client 17: weight=0.930 MILD PENALTY (sim=-0.015)
Client 12: weight=1.038 REWARDED (sim=+0.034)
Client 2: weight=1.055 REWARDED (sim=+0.044)
Client 19: weight=0.817 MILD PENALTY (sim=-0.049)
Client 11: weight=1.071 REWARDED (sim=+0.065)
Client 4: weight=0.817 MILD PENALTY (sim=-0.031)
→ Total weight: 18.628 | Avg weight: 0.931
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 12 accuracy recorded: 0.7107 | Δacc = 0.0166
INFO :
INFO : [ROUND 13]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14185) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 6
1x across cluster]
(ClientAppActor pid=14192) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 24x across cluster]
(ClientAppActor pid=14192) [ATTACK] Client 14 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0166 → REWARDING aligned clients

[ROUND 13] CFA Trust Weights:
Client 8: weight=1.059 REWARDED (sim=+0.044)
Client 17: weight=0.912 MILD PENALTY (sim=-0.016)
Client 12: weight=1.042 REWARDED (sim=+0.030)
Client 0: weight=1.051 REWARDED (sim=+0.033)
Client 18: weight=1.008 NEUTRAL (sim=-0.006)
Client 1: weight=0.982 NEUTRAL (sim=+0.009)
Client 15: weight=0.834 MILD PENALTY (sim=-0.020)
Client 16: weight=0.801 MILD PENALTY (sim=-0.035)
Client 9: weight=0.801 MILD PENALTY (sim=-0.036)
Client 19: weight=0.801 MILD PENALTY (sim=-0.049)
Client 3: weight=1.057 REWARDED (sim=+0.059)
Client 5: weight=1.012 NEUTRAL (sim=+0.004)
Client 10: weight=0.993 REWARDED (sim=+0.022)
Client 7: weight=0.801 MILD PENALTY (sim=-0.030)
Client 4: weight=0.801 MILD PENALTY (sim=-0.028)
Client 2: weight=1.060 REWARDED (sim=+0.046)
Client 6: weight=0.801 MILD PENALTY (sim=-0.027)
Client 14: weight=0.801 MILD PENALTY (sim=-0.023)
Client 13: weight=0.801 MILD PENALTY (sim=-0.062)
Client 11: weight=1.078 REWARDED (sim=+0.061)
→ Total weight: 18.492 | Avg weight: 0.925
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 13 accuracy recorded: 0.7177 | Δacc = 0.0070
INFO :
INFO : [ROUND 14]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0070 → REWARDING aligned clients

[ROUND 14] CFA Trust Weights:
Client 11: weight=1.084 REWARDED (sim=+0.057)
Client 4: weight=0.785 MILD PENALTY (sim=-0.023)
Client 15: weight=0.817 MILD PENALTY (sim=-0.018)
Client 19: weight=0.785 MILD PENALTY (sim=-0.047)
Client 18: weight=1.008 NEUTRAL (sim=-0.005)
Client 2: weight=1.064 REWARDED (sim=+0.040)
Client 14: weight=0.785 MILD PENALTY (sim=-0.020)
Client 7: weight=0.785 MILD PENALTY (sim=-0.026)
Client 9: weight=0.785 MILD PENALTY (sim=-0.036)
Client 1: weight=0.982 NEUTRAL (sim=+0.009)
Client 8: weight=1.064 REWARDED (sim=+0.041)
Client 3: weight=1.063 REWARDED (sim=+0.053)
Client 0: weight=1.054 REWARDED (sim=+0.030)
Client 17: weight=0.894 MILD PENALTY (sim=-0.012)
Client 13: weight=0.785 MILD PENALTY (sim=-0.054)
Client 6: weight=0.785 MILD PENALTY (sim=-0.027)
Client 12: weight=1.044 REWARDED (sim=+0.027)
Client 16: weight=0.785 MILD PENALTY (sim=-0.033)
Client 5: weight=1.012 NEUTRAL (sim=+0.002)
Client 10: weight=0.996 REWARDED (sim=+0.023)
→ Total weight: 18.358 | Avg weight: 0.918
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14192) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 65x a
cross cluster]
(ClientAppActor pid=14192) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 25x across cluster]
(ClientAppActor pid=14191) [ATTACK] Client 15 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 14 accuracy recorded: 0.7426 | Δacc = 0.0249
INFO :
INFO : [ROUND 15]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:36:04,038 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11
.6194 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0249 → REWARDING aligned clients

[ROUND 15] CFA Trust Weights:
Client 7: weight=0.769 MILD PENALTY (sim=-0.032)
Client 16: weight=0.769 MILD PENALTY (sim=-0.031)
Client 6: weight=0.769 MILD PENALTY (sim=-0.024)
Client 14: weight=0.769 MILD PENALTY (sim=-0.020)
Client 17: weight=0.876 MILD PENALTY (sim=-0.015)
Client 8: weight=1.068 REWARDED (sim=+0.040)
Client 18: weight=1.008 NEUTRAL (sim=-0.006)
Client 15: weight=0.801 MILD PENALTY (sim=-0.020)
Client 11: weight=1.090 REWARDED (sim=+0.059)
Client 4: weight=0.769 MILD PENALTY (sim=-0.024)
Client 19: weight=0.769 MILD PENALTY (sim=-0.046)
Client 2: weight=1.069 REWARDED (sim=+0.041)
Client 10: weight=0.998 REWARDED (sim=+0.024)
Client 13: weight=0.769 MILD PENALTY (sim=-0.053)
Client 0: weight=1.057 REWARDED (sim=+0.031)
Client 5: weight=1.012 NEUTRAL (sim=+0.003)
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Client 9: weight=0.769 MILD PENALTY (sim=-0.036)
Client 12: weight=1.047 REWARDED (sim=+0.027)
Client 3: weight=1.068 REWARDED (sim=+0.055)
Client 1: weight=0.982 NEUTRAL (sim=+0.006)
→ Total weight: 18.227 | Avg weight: 0.911
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 15 accuracy recorded: 0.7482 | Δacc = 0.0056
INFO :
INFO : [ROUND 16]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0056 → REWARDING aligned clients

[ROUND 16] CFA Trust Weights:
Client 14: weight=0.754 MILD PENALTY (sim=-0.020)
Client 15: weight=0.785 MILD PENALTY (sim=-0.015)
Client 7: weight=0.754 MILD PENALTY (sim=-0.027)
Client 8: weight=1.072 REWARDED (sim=+0.039)
Client 0: weight=1.060 REWARDED (sim=+0.028)
Client 10: weight=1.000 REWARDED (sim=+0.020)
Client 1: weight=0.982 NEUTRAL (sim=+0.009)
Client 12: weight=1.050 REWARDED (sim=+0.027)
Client 4: weight=0.754 MILD PENALTY (sim=-0.022)
Client 19: weight=0.754 MILD PENALTY (sim=-0.046)
Client 17: weight=0.858 MILD PENALTY (sim=-0.014)
Client 6: weight=0.754 MILD PENALTY (sim=-0.020)
Client 9: weight=0.754 MILD PENALTY (sim=-0.037)
Client 18: weight=1.008 NEUTRAL (sim=-0.010)
Client 2: weight=1.073 REWARDED (sim=+0.040)
Client 3: weight=1.074 REWARDED (sim=+0.055)
Client 13: weight=0.754 MILD PENALTY (sim=-0.050)
Client 11: weight=1.096 REWARDED (sim=+0.056)
Client 16: weight=0.754 MILD PENALTY (sim=-0.032)
Client 5: weight=1.012 NEUTRAL (sim=+0.004)
→ Total weight: 18.099 | Avg weight: 0.905
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14192) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 7
2x across cluster]
(ClientAppActor pid=14188) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=14192) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 16 accuracy recorded: 0.7634 | Δacc = 0.0152
INFO :
INFO : [ROUND 17]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0152 → REWARDING aligned clients

[ROUND 17] CFA Trust Weights:
Client 11: weight=1.102 REWARDED (sim=+0.056)
Client 6: weight=0.739 MILD PENALTY (sim=-0.022)
Client 16: weight=0.739 MILD PENALTY (sim=-0.032)
Client 17: weight=0.841 MILD PENALTY (sim=-0.015)
Client 19: weight=0.739 MILD PENALTY (sim=-0.043)
Client 13: weight=0.739 MILD PENALTY (sim=-0.047)
Client 10: weight=1.002 REWARDED (sim=+0.022)
Client 0: weight=1.063 REWARDED (sim=+0.028)
Client 18: weight=0.988 MILD PENALTY (sim=-0.011)
Client 9: weight=0.739 MILD PENALTY (sim=-0.037)
Client 1: weight=0.982 NEUTRAL (sim=+0.006)
Client 15: weight=0.769 MILD PENALTY (sim=-0.018)
Client 2: weight=1.077 REWARDED (sim=+0.035)
Client 4: weight=0.739 MILD PENALTY (sim=-0.020)
Client 3: weight=1.080 REWARDED (sim=+0.053)
Client 5: weight=1.012 NEUTRAL (sim=+0.007)
Client 7: weight=0.739 MILD PENALTY (sim=-0.025)
Client 12: weight=1.053 REWARDED (sim=+0.028)
Client 14: weight=0.739 MILD PENALTY (sim=-0.017)
Client 8: weight=1.076 REWARDED (sim=+0.037)
→ Total weight: 17.953 | Avg weight: 0.898
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 17 accuracy recorded: 0.7754 | Δacc = 0.0120
INFO :
INFO : [ROUND 18]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14189) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
67x across cluster]
(ClientAppActor pid=14189) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14185) [ATTACK] Client 19 submitted poisoned update [repeated 13x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0120 → REWARDING aligned clients

[ROUND 18] CFA Trust Weights:
Client 1: weight=0.982 NEUTRAL (sim=+0.009)
Client 0: weight=1.066 REWARDED (sim=+0.028)
Client 9: weight=0.724 MILD PENALTY (sim=-0.032)
Client 19: weight=0.724 MILD PENALTY (sim=-0.043)
Client 8: weight=1.080 REWARDED (sim=+0.038)
Client 10: weight=1.004 REWARDED (sim=+0.021)
Client 7: weight=0.724 MILD PENALTY (sim=-0.025)
Client 16: weight=0.724 MILD PENALTY (sim=-0.030)
Client 5: weight=1.012 NEUTRAL (sim=+0.001)
Client 17: weight=0.824 MILD PENALTY (sim=-0.015)
Client 14: weight=0.724 MILD PENALTY (sim=-0.017)
Client 2: weight=1.081 REWARDED (sim=+0.037)
Client 3: weight=1.085 REWARDED (sim=+0.051)
Client 11: weight=1.108 REWARDED (sim=+0.053)
Client 13: weight=0.724 MILD PENALTY (sim=-0.050)
Client 12: weight=1.056 REWARDED (sim=+0.026)
Client 4: weight=0.724 MILD PENALTY (sim=-0.021)
Client 18: weight=0.988 NEUTRAL (sim=-0.009)
Client 15: weight=0.754 MILD PENALTY (sim=-0.019)
Client 6: weight=0.724 MILD PENALTY (sim=-0.022)
→ Total weight: 17.830 | Avg weight: 0.891
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:36:14.074 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11
.5119 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 18 accuracy recorded: 0.7915 | Δacc = 0.0161
INFO :
INFO : [ROUND 19]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0161 → REWARDING aligned clients

[ROUND 19] CFA Trust Weights:
Client 11: weight=1.114 REWARDED (sim=+0.054)
Client 16: weight=0.709 MILD PENALTY (sim=-0.034)
Client 2: weight=1.084 REWARDED (sim=+0.035)
Client 18: weight=0.988 NEUTRAL (sim=-0.009)
Client 12: weight=1.058 REWARDED (sim=+0.024)
Client 5: weight=1.012 NEUTRAL (sim=+0.004)
Client 15: weight=0.739 MILD PENALTY (sim=-0.016)
Client 6: weight=0.709 MILD PENALTY (sim=-0.022)
Client 4: weight=0.709 MILD PENALTY (sim=-0.020)
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Client 0: weight=1.068 REWARDED (sim=+0.024)
Client 13: weight=0.709 MILD PENALTY (sim=-0.045)
Client 17: weight=0.808 MILD PENALTY (sim=-0.011)
Client 8: weight=1.084 REWARDED (sim=+0.035)
Client 10: weight=1.006 REWARDED (sim=+0.022)
Client 14: weight=0.709 MILD PENALTY (sim=-0.018)
Client 19: weight=0.709 MILD PENALTY (sim=-0.042)
Client 9: weight=0.709 MILD PENALTY (sim=-0.031)
Client 7: weight=0.709 MILD PENALTY (sim=-0.021)
Client 3: weight=1.091 REWARDED (sim=+0.051)
Client 1: weight=0.982 NEUTRAL (sim=+0.007)
→ Total weight: 17.709 | Avg weight: 0.885
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 19 accuracy recorded: 0.7785 | Δacc = -0.0130
INFO :
INFO :      [ROUND 20]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14190) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 65
x across cluster]
(ClientAppActor pid=14190) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=14191) [ATTACK] Client 12 submitted poisoned update [repeated 11x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = -0.0130 → PENALISING divergent clients

[ROUND 20] CFR Trust Weights:
Client 3: weight=1.091 CLEARED (dev=0.950)
Client 14: weight=0.602 PENALISED (dev=1.018)
Client 0: weight=1.068 CLEARED (dev=0.973)
Client 8: weight=1.084 CLEARED (dev=0.963)
Client 9: weight=0.601 PENALISED (dev=1.031)
Client 15: weight=0.627 PENALISED (dev=1.013)
Client 13: weight=0.600 PENALISED (dev=1.041)
Client 1: weight=0.982 CLEARED (dev=0.995)
Client 2: weight=1.084 CLEARED (dev=0.965)
Client 16: weight=0.601 PENALISED (dev=1.028)
Client 7: weight=0.601 PENALISED (dev=1.023)
Client 6: weight=0.601 PENALISED (dev=1.023)
Client 10: weight=1.006 CLEARED (dev=0.979)
Client 18: weight=0.839 PENALISED (dev=1.008)
Client 12: weight=1.058 CLEARED (dev=0.976)
Client 4: weight=0.601 PENALISED (dev=1.023)
Client 11: weight=1.114 CLEARED (dev=0.947)
Client 17: weight=0.686 PENALISED (dev=1.013)
Client 19: weight=0.600 PENALISED (dev=1.046)
Client 5: weight=1.012 CLEARED (dev=0.995)
→ Total weight: 16.459 | Avg weight: 0.823
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 20 accuracy recorded: 0.8123 | Δacc = 0.0338
INFO :
INFO :      [ROUND 21]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0338 → REWARDING aligned clients

[ROUND 21] CFR Trust Weights:
Client 10: weight=1.008 REWARDED (sim=+0.020)
Client 15: weight=0.614 MILD PENALTY (sim=-0.019)
Client 9: weight=0.589 MILD PENALTY (sim=-0.032)
Client 3: weight=1.096 REWARDED (sim=+0.049)
Client 17: weight=0.672 MILD PENALTY (sim=-0.014)
Client 1: weight=0.982 NEUTRAL (sim=+0.004)
Client 7: weight=0.589 MILD PENALTY (sim=-0.023)
Client 11: weight=1.120 REWARDED (sim=+0.055)
Client 8: weight=1.087 REWARDED (sim=+0.030)
Client 12: weight=1.061 REWARDED (sim=+0.027)
Client 0: weight=1.071 REWARDED (sim=+0.027)
Client 19: weight=0.588 MILD PENALTY (sim=-0.041)
Client 5: weight=1.012 NEUTRAL (sim=+0.007)
Client 2: weight=1.088 REWARDED (sim=+0.034)
Client 13: weight=0.588 MILD PENALTY (sim=-0.042)
Client 4: weight=0.589 MILD PENALTY (sim=-0.017)
Client 14: weight=0.590 MILD PENALTY (sim=-0.016)
Client 16: weight=0.589 MILD PENALTY (sim=-0.033)
Client 18: weight=0.839 NEUTRAL (sim=-0.008)
Client 6: weight=0.589 MILD PENALTY (sim=-0.021)
→ Total weight: 16.362 | Avg weight: 0.818
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14189) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 69
x across cluster]
(ClientAppActor pid=14189) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14187) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 12:36:24,132 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11
.5108 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 21 accuracy recorded: 0.8132 | Δacc = 0.0009
INFO :
INFO :      [ROUND 22]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0009 → REWARDING aligned clients

[ROUND 22] CFR Trust Weights:
Client 18: weight=0.822 MILD PENALTY (sim=-0.011)
Client 8: weight=1.091 REWARDED (sim=+0.035)
Client 10: weight=1.011 REWARDED (sim=+0.022)
Client 1: weight=0.982 NEUTRAL (sim=+0.007)
Client 17: weight=0.658 MILD PENALTY (sim=-0.013)
Client 2: weight=1.092 REWARDED (sim=+0.032)
Client 15: weight=0.602 MILD PENALTY (sim=-0.017)
Client 7: weight=0.577 MILD PENALTY (sim=-0.023)
Client 5: weight=1.012 NEUTRAL (sim=+0.006)
Client 16: weight=0.577 MILD PENALTY (sim=-0.029)
Client 6: weight=0.578 MILD PENALTY (sim=-0.020)
Client 3: weight=1.102 REWARDED (sim=+0.049)
Client 12: weight=1.063 REWARDED (sim=+0.022)
Client 0: weight=1.074 REWARDED (sim=+0.026)
Client 9: weight=0.577 MILD PENALTY (sim=-0.030)
Client 13: weight=0.576 MILD PENALTY (sim=-0.043)
Client 11: weight=1.126 REWARDED (sim=+0.050)
Client 19: weight=0.576 MILD PENALTY (sim=-0.041)
Client 4: weight=0.578 MILD PENALTY (sim=-0.014)
Client 14: weight=0.578 MILD PENALTY (sim=-0.013)
→ Total weight: 16.251 | Avg weight: 0.813
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 22 accuracy recorded: 0.8244 | Δacc = 0.0112
INFO :
INFO :      [ROUND 23]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14189) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 69x
across cluster]
(ClientAppActor pid=14188) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=14191) [ATTACK] Client 14 submitted poisoned update [repeated 15x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0112 → REWARDING aligned clients
```

```
[ROUND 23] CFA Trust Weights:
Client 16: weight=0.566 MILD PENALTY (sim=-0.032)
Client 13: weight=0.565 MILD PENALTY (sim=-0.039)
Client 5: weight=1.012 NEUTRAL (sim=+0.007)
Client 9: weight=0.565 MILD PENALTY (sim=-0.025)
Client 11: weight=1.132 REWARDED (sim=+0.052)
Client 7: weight=0.566 MILD PENALTY (sim=-0.022)
Client 3: weight=1.107 REWARDED (sim=+0.049)
Client 0: weight=1.077 REWARDED (sim=+0.028)
Client 14: weight=0.566 MILD PENALTY (sim=-0.016)
Client 4: weight=0.566 MILD PENALTY (sim=-0.015)
Client 8: weight=1.094 REWARDED (sim=+0.031)
Client 15: weight=0.590 MILD PENALTY (sim=-0.016)
Client 19: weight=0.564 MILD PENALTY (sim=-0.045)
Client 18: weight=0.806 MILD PENALTY (sim=-0.011)
Client 10: weight=1.013 REWARDED (sim=+0.022)
Client 12: weight=1.066 REWARDED (sim=+0.023)
Client 6: weight=0.566 MILD PENALTY (sim=-0.020)
Client 2: weight=1.095 REWARDED (sim=+0.035)
Client 1: weight=0.982 NEUTRAL (sim=+0.008)
Client 17: weight=0.645 MILD PENALTY (sim=-0.016)
→ Total weight: 16.143 | Avg weight: 0.807
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 23 accuracy recorded: 0.8288 | Δacc = 0.0044
INFO :
INFO : [ROUND 24]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0044 → REWARDING aligned clients

[ROUND 24] CFA Trust Weights:
Client 18: weight=0.789 MILD PENALTY (sim=-0.012)
Client 19: weight=0.553 MILD PENALTY (sim=-0.044)
Client 2: weight=1.099 REWARDED (sim=+0.034)
Client 15: weight=0.578 MILD PENALTY (sim=-0.017)
Client 9: weight=0.554 MILD PENALTY (sim=-0.026)
Client 0: weight=1.080 REWARDED (sim=+0.026)
Client 10: weight=1.015 REWARDED (sim=+0.023)
Client 6: weight=0.555 MILD PENALTY (sim=-0.020)
Client 1: weight=0.982 NEUTRAL (sim=+0.007)
Client 13: weight=0.553 MILD PENALTY (sim=-0.043)
Client 8: weight=1.098 REWARDED (sim=+0.035)
Client 16: weight=0.554 MILD PENALTY (sim=-0.032)
Client 4: weight=0.555 MILD PENALTY (sim=-0.013)
Client 12: weight=1.068 REWARDED (sim=+0.020)
Client 17: weight=0.632 MILD PENALTY (sim=-0.016)
Client 3: weight=1.112 REWARDED (sim=+0.047)
Client 5: weight=1.012 NEUTRAL (sim=+0.008)
Client 11: weight=1.138 REWARDED (sim=+0.055)
Client 14: weight=0.555 MILD PENALTY (sim=-0.013)
Client 7: weight=0.555 MILD PENALTY (sim=-0.018)
→ Total weight: 16.038 | Avg weight: 0.802
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14190) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 6
0x across cluster]
(ClientAppActor pid=14185) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 23x across cluster]
(ClientAppActor pid=14188) [ATTACK] Client 15 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 24 accuracy recorded: 0.8228 | Δacc = -0.0060
INFO :
INFO : [ROUND 25]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:36:34,223 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11
.5144 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0060 → PENALISING divergent clients

[ROUND 25] CFA Trust Weights:
Client 14: weight=0.471 PENALISED (dev=1.012)
Client 6: weight=0.470 PENALISED (dev=1.020)
Client 13: weight=0.468 PENALISED (dev=1.039)
Client 7: weight=0.470 PENALISED (dev=1.018)
Client 12: weight=1.068 CLEARED (dev=0.978)
Client 10: weight=1.015 CLEARED (dev=0.979)
Client 9: weight=0.470 PENALISED (dev=1.026)
Client 17: weight=0.537 PENALISED (dev=1.014)
Client 18: weight=0.670 PENALISED (dev=1.013)
Client 5: weight=1.012 CLEARED (dev=0.988)
Client 0: weight=1.080 CLEARED (dev=0.973)
Client 11: weight=1.138 CLEARED (dev=0.950)
Client 8: weight=1.098 CLEARED (dev=0.971)
Client 1: weight=0.982 CLEARED (dev=0.993)
Client 15: weight=0.490 PENALISED (dev=1.021)
Client 19: weight=0.468 PENALISED (dev=1.045)
Client 2: weight=1.099 CLEARED (dev=0.965)
Client 4: weight=0.471 PENALISED (dev=1.013)
Client 16: weight=0.470 PENALISED (dev=1.029)
Client 3: weight=1.112 CLEARED (dev=0.952)
→ Total weight: 15.059 | Avg weight: 0.753
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 25 accuracy recorded: 0.8444 | Δacc = 0.0216
INFO :
INFO : [ROUND 26]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0216 → REWARDING aligned clients

[ROUND 26] CFA Trust Weights:
Client 2: weight=1.103 REWARDED (sim=+0.032)
Client 12: weight=1.070 REWARDED (sim=+0.021)
Client 4: weight=0.461 MILD PENALTY (sim=-0.013)
Client 9: weight=0.460 MILD PENALTY (sim=-0.025)
Client 19: weight=0.458 MILD PENALTY (sim=-0.041)
Client 3: weight=1.118 REWARDED (sim=+0.048)
Client 10: weight=1.018 REWARDED (sim=+0.024)
Client 18: weight=0.670 NEUTRAL (sim=-0.008)
Client 7: weight=0.461 MILD PENALTY (sim=-0.015)
Client 6: weight=0.461 MILD PENALTY (sim=-0.017)
Client 15: weight=0.480 MILD PENALTY (sim=-0.023)
Client 0: weight=1.083 REWARDED (sim=+0.025)
Client 17: weight=0.526 MILD PENALTY (sim=-0.012)
Client 14: weight=0.462 MILD PENALTY (sim=-0.014)
Client 1: weight=0.982 NEUTRAL (sim=+0.008)
Client 11: weight=1.144 REWARDED (sim=+0.051)
Client 13: weight=0.459 MILD PENALTY (sim=-0.038)
Client 8: weight=1.101 REWARDED (sim=+0.029)
Client 16: weight=0.460 MILD PENALTY (sim=-0.033)
Client 5: weight=1.012 NEUTRAL (sim=+0.006)
→ Total weight: 14.988 | Avg weight: 0.749
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14187) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 64x acro
ss cluster]
(ClientAppActor pid=14187) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14189) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
```

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INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA ] Round 26 accuracy recorded: 0.8629 | Δacc = 0.0185
INFO :
INFO :      [ROUND 27]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA ] Δacc = +0.0185 → REWARDING aligned clients

[ROUND 27] CFA Trust Weights:
Client 2: weight=1.107 REWARDED (sim=+0.036)
Client 5: weight=1.013 REWARDED (sim=+0.014)
Client 7: weight=0.452 MILD PENALTY (sim=-0.016)
Client 3: weight=1.123 REWARDED (sim=+0.050)
Client 11: weight=1.149 REWARDED (sim=+0.051)
Client 4: weight=0.461 NEUTRAL (sim=-0.009)
Client 18: weight=0.657 MILD PENALTY (sim=-0.013)
Client 6: weight=0.452 MILD PENALTY (sim=-0.019)
Client 10: weight=1.020 REWARDED (sim=+0.023)
Client 14: weight=0.452 MILD PENALTY (sim=-0.014)
Client 9: weight=0.451 MILD PENALTY (sim=-0.025)
Client 17: weight=0.515 MILD PENALTY (sim=-0.018)
Client 8: weight=1.105 REWARDED (sim=+0.030)
Client 1: weight=0.982 NEUTRAL (sim=+0.006)
Client 12: weight=1.072 REWARDED (sim=+0.021)
Client 16: weight=0.451 MILD PENALTY (sim=-0.029)
Client 13: weight=0.450 MILD PENALTY (sim=-0.041)
Client 0: weight=1.085 REWARDED (sim=+0.024)
Client 15: weight=0.471 MILD PENALTY (sim=-0.021)
Client 19: weight=0.449 MILD PENALTY (sim=-0.041)
→ Total weight: 14.918 | Avg weight: 0.746
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA ] Round 27 accuracy recorded: 0.8598 | Δacc = -0.0031
INFO :
INFO :      [ROUND 28]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      (ClientAppActor pid=14185) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 5
8x across cluster]
INFO :      (ClientAppActor pid=14189) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 23x across cluster]
INFO :      (ClientAppActor pid=14190) [ATTACK] Client 12 submitted poisoned update [repeated 8x across cluster]
(raylet) [2026-04-21 12:36:44.289 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11
.5144 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA ] Δacc = -0.0031 → PENALISING divergent clients

[ROUND 28] CFA Trust Weights:
Client 2: weight=1.107 CLEARED (dev=0.967)
Client 19: weight=0.380 PENALISED (dev=1.044)
Client 8: weight=1.105 CLEARED (dev=0.968)
Client 14: weight=0.384 PENALISED (dev=1.013)
Client 16: weight=0.382 PENALISED (dev=1.030)
Client 15: weight=0.399 PENALISED (dev=1.016)
Client 10: weight=1.020 CLEARED (dev=0.974)
Client 9: weight=0.382 PENALISED (dev=1.021)
Client 3: weight=1.123 CLEARED (dev=0.954)
Client 17: weight=0.437 PENALISED (dev=1.010)
Client 4: weight=0.392 PENALISED (dev=1.011)
Client 6: weight=0.383 PENALISED (dev=1.014)
Client 11: weight=1.149 CLEARED (dev=0.953)
Client 13: weight=0.380 PENALISED (dev=1.043)
Client 1: weight=0.982 CLEARED (dev=0.994)
Client 12: weight=1.072 CLEARED (dev=0.981)
Client 18: weight=0.557 PENALISED (dev=1.017)
Client 7: weight=0.383 PENALISED (dev=1.017)
Client 0: weight=1.085 CLEARED (dev=0.975)
Client 5: weight=1.013 CLEARED (dev=0.985)
→ Total weight: 14.117 | Avg weight: 0.706
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA ] Round 28 accuracy recorded: 0.8678 | Δacc = 0.0080
INFO :
INFO :      [ROUND 29]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      (ClientAppActor pid=14189) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
58x across cluster]
INFO :      (ClientAppActor pid=14188) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO :      (ClientAppActor pid=14191) [ATTACK] Client 14 submitted poisoned update [repeated 14x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA ] Δacc = +0.0080 → REWARDING aligned clients

[ROUND 29] CFA Trust Weights:
Client 10: weight=1.022 REWARDED (sim=+0.024)
Client 14: weight=0.376 MILD PENALTY (sim=-0.017)
Client 5: weight=1.015 REWARDED (sim=+0.013)
Client 12: weight=1.074 REWARDED (sim=+0.017)
Client 7: weight=0.376 MILD PENALTY (sim=-0.012)
Client 6: weight=0.376 MILD PENALTY (sim=-0.016)
Client 4: weight=0.392 NEUTRAL (sim=-0.007)
Client 19: weight=0.372 MILD PENALTY (sim=-0.042)
Client 18: weight=0.546 MILD PENALTY (sim=-0.015)
Client 13: weight=0.373 MILD PENALTY (sim=-0.037)
Client 16: weight=0.374 MILD PENALTY (sim=-0.033)
Client 9: weight=0.375 MILD PENALTY (sim=-0.016)
Client 2: weight=1.110 REWARDED (sim=+0.031)
Client 15: weight=0.391 MILD PENALTY (sim=-0.017)
Client 1: weight=0.982 NEUTRAL (sim=+0.006)
Client 17: weight=0.429 MILD PENALTY (sim=-0.015)
Client 3: weight=1.129 REWARDED (sim=+0.051)
Client 8: weight=1.107 REWARDED (sim=+0.025)
Client 0: weight=1.088 REWARDED (sim=+0.029)
Client 11: weight=1.155 REWARDED (sim=+0.049)
→ Total weight: 14.062 | Avg weight: 0.703
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA ] Round 29 accuracy recorded: 0.8628 | Δacc = -0.0050
INFO :
INFO :      [ROUND 30]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA ] Δacc = -0.0050 → PENALISING divergent clients

[ROUND 30] CFA Trust Weights:
Client 0: weight=1.088 CLEARED (dev=0.971)
Client 11: weight=1.155 CLEARED (dev=0.951)
Client 3: weight=1.129 CLEARED (dev=0.951)
Client 1: weight=0.982 CLEARED (dev=0.989)
Client 15: weight=0.332 PENALISED (dev=1.023)
Client 4: weight=0.332 PENALISED (dev=1.008)
Client 2: weight=1.110 CLEARED (dev=0.967)
Client 6: weight=0.319 PENALISED (dev=1.017)
Client 17: weight=0.364 PENALISED (dev=1.014)
Client 13: weight=0.315 PENALISED (dev=1.040)
Client 5: weight=1.015 CLEARED (dev=0.985)
Client 14: weight=0.319 PENALISED (dev=1.015)
Client 10: weight=1.022 CLEARED (dev=0.976)
Client 18: weight=0.463 PENALISED (dev=1.012)
Client 9: weight=0.318 PENALISED (dev=1.016)
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Client 7: weight=0.319 PENALISED (dev=1.012)
Client 8: weight=1.107 CLEARED (dev=0.980)
Client 19: weight=0.315 PENALISED (dev=1.043)
Client 12: weight=1.074 CLEARED (dev=0.982)
Client 16: weight=0.317 PENALISED (dev=1.030)
→ Total weight: 13.396 | Avg weight: 0.670
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14187) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeated 57x across cluster]
(ClientAppActor pid=14187) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 23x across cluster]
(ClientAppActor pid=14192) [ATTACK] Client 15 submitted poisoned update [repeated 10x across cluster]
(raylet) [2026-04-21 12:36:54,340 E 14182 669440] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-35-12_906556_14163 is over 95% full, available space: 11.5134 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 30 accuracy recorded: 0.8795 | Δacc = 0.0167
INFO :
INFO : [SUMMARY]
INFO : Run finished 30 round(s) in 98.31s
INFO : History (loss, distributed):
INFO : round 1: 0.3189
INFO : round 2: 0.4198
INFO : round 3: 0.5056
INFO : round 4: 0.5451
INFO : round 5: 0.5802
INFO : round 6: 0.6028
INFO : round 7: 0.6267
INFO : round 8: 0.6585
INFO : round 9: 0.6646
INFO : round 10: 0.6741
INFO : round 11: 0.6941000000000002
INFO : round 12: 0.7107
INFO : round 13: 0.7177
INFO : round 14: 0.7426
INFO : round 15: 0.7482
INFO : round 16: 0.7634
INFO : round 17: 0.7754
INFO : round 18: 0.7915
INFO : round 19: 0.7785
INFO : round 20: 0.8123
INFO : round 21: 0.8132
INFO : round 22: 0.8244
INFO : round 23: 0.8288
INFO : round 24: 0.8228
INFO : round 25: 0.8444
INFO : round 26: 0.8629
INFO : round 27: 0.8598
INFO : round 28: 0.8678
INFO : round 29: 0.8628
INFO : round 30: 0.8795
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.3189),
INFO : (2, 0.4198),
INFO : (3, 0.5056),
INFO : (4, 0.5451),
INFO : (5, 0.5802),
INFO : (6, 0.6028),
INFO : (7, 0.6267),
INFO : (8, 0.6585),
INFO : (9, 0.6646),
INFO : (10, 0.6741),
INFO : (11, 0.6941000000000002),
INFO : (12, 0.7107),
INFO : (13, 0.7177),
INFO : (14, 0.7426),
INFO : (15, 0.7482),
INFO : (16, 0.7634),
INFO : (17, 0.7754),
INFO : (18, 0.7915),
INFO : (19, 0.7785),
INFO : (20, 0.8123),
INFO : (21, 0.8132),
INFO : (22, 0.8244),
INFO : (23, 0.8288),
INFO : (24, 0.8228),
INFO : (25, 0.8444),
INFO : (26, 0.8629),
INFO : (27, 0.8598),
INFO : (28, 0.8678),
INFO : (29, 0.8628),
INFO : (30, 0.8795)]}
INFO :
--- cfa_trust_noniid ---
Round 1: 0.3189
Round 2: 0.4198
Round 3: 0.5056
Round 4: 0.5451
Round 5: 0.5802
Round 6: 0.6028
Round 7: 0.6267
Round 8: 0.6585
Round 9: 0.6646
Round 10: 0.6741
Round 11: 0.6941
Round 12: 0.7107
Round 13: 0.7177
Round 14: 0.7426
Round 15: 0.7482
Round 16: 0.7634
Round 17: 0.7754
Round 18: 0.7915
Round 19: 0.7785
Round 20: 0.8123
Round 21: 0.8132
Round 22: 0.8244
Round 23: 0.8288
Round 24: 0.8228
Round 25: 0.8444
Round 26: 0.8629
Round 27: 0.8598
Round 28: 0.8678
Round 29: 0.8628
Round 30: 0.8795
[PLOT] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:
Round 1: 0.3189
Round 2: 0.4198
Round 3: 0.5056
Round 4: 0.5451
Round 5: 0.5802
Round 6: 0.6028
Round 7: 0.6267
Round 8: 0.6585
Round 9: 0.6646
Round 10: 0.6741
Round 11: 0.6941

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Round 12: 0.7107
Round 13: 0.7177
Round 14: 0.7426
Round 15: 0.7482
Round 16: 0.7634
Round 17: 0.7754
Round 18: 0.7915
Round 19: 0.7785
Round 20: 0.8123
Round 21: 0.8132
Round 22: 0.8244
Round 23: 0.8288
Round 24: 0.8228
Round 25: 0.8444
Round 26: 0.8629
Round 27: 0.8598
Round 28: 0.8678
Round 29: 0.8628
Round 30: 0.8795
(ClientAppActor pid=14185) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 4
x across cluster]
(ClientAppActor pid=14188) [SETUP] Client 13 is MALICIOUS (label_flip)
(fl-env) apple@MacBook-Pro-10 project-main % python server.py baseline noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
         Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

$ flwr new # Create a new Flower app from a template

$ flwr run # Run the Flower app in Simulation Mode

Using 'start_simulation()' is deprecated.

This is a deprecated feature. It will be removed
entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=30, no round_timeout
*X2026-04-21 12:56:01,982 INFO worker.py:2012 -- Started a local Ray instance.

/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray will
no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'node:127.0.0.1': 1.0, 'node:__internal_head__': 1.0, 'memory': 6705020928.0, 'object_store_memory': 2147483648.0, 'C
PU': 8.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
*XINFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14613) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=14613) labels = np.array(trainset.targets)
(ClientAppActor pid=14613) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697}
(ClientAppActor pid=14613) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010}
(ClientAppActor pid=14613) [NON-IID] Client 9 | Total: 2061 samples | Distribution: {0: 11, 1: 133, 2: 1264, 3: 170, 4: 261, 5: 169, 6: 46, 7: 1, 8: 3, 9: 3}
(ClientAppActor pid=14613) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79}
(ClientAppActor pid=14613) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11}
(ClientAppActor pid=14613) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91}
INFO : aggregate_fit: received 20 results and 0 failures
WARNING : No fit_metrics_aggregation_fn provided
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14606) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 29x across c
luster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/configure-log
ging.html#log-deduplication for more options.)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:56:12,025 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 12
.823 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14610) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=14610) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14608) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
59x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14610) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 68x acro
ss cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:56:22,099 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11
.8967 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 6]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14611) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 5
2x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 7]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14608) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35} [repeated 64x acr
oss cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 8]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
```

```
(pid=gcs_server) [2026-04-21 12:56:31,815 E 14599 687328] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:56:32,147 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11.8966 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(raylet) [2026-04-21 12:56:32,168 E 14603 687428] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=14611) [2026-04-21 12:56:32,959 E 14611 687782] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
[2026-04-21 12:56:32,989 E 14594 687483] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 9]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14608) [NON-IID] Client 9 | Total: 2061 samples | Distribution: {0: 11, 1: 133, 2: 1264, 3: 170, 4: 261, 5: 169, 6: 46, 7: 1, 8: 3, 9: 3} [repeated 68x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 10]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 11]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14608) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated 56x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:56:42,249 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11.8964 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14607) [2026-04-21 12:56:32,999 E 14607 687905] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 12]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14606) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated 69x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 13]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 14]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14610) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 70x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:56:52,270 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11.5836 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 15]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 16]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14609) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 65x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 17]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 18]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14606) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 76x across cluster]
(raylet) [2026-04-21 12:57:02,324 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11.5837 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 19]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14606) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated 68x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 20]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 21]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14610) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 52x across cluster]
(raylet) [2026-04-21 12:57:12,346 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11.5801 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 22]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14608) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeated 68x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 23]
```

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INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 24]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 12:57:22,369 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11
.5286 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14606) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
72x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 25]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 26]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14610) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
66x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 27]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14609) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
8x across cluster]
(raylet) [2026-04-21 12:57:32,439 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11
.5317 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 28]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 29]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14612) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
66x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 30]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [SUMMARY]
INFO : Run finished 30 round(s) in 97.92s
INFO : History (loss, distributed):
INFO : round 1: 2.0215971073784385
INFO : round 2: 1.423833744594464
INFO : round 3: 0.9552734179999501
INFO : round 4: 0.7131367088697208
INFO : round 5: 0.581944929144253
INFO : round 6: 0.5027966537652686
INFO : round 7: 0.459582558955057
INFO : round 8: 0.42608318901576175
INFO : round 9: 0.3935935949079526
INFO : round 10: 0.38068132590466797
INFO : round 11: 0.35748055517768706
INFO : round 12: 0.35280274117764193
INFO : round 13: 0.3361930998678977
INFO : round 14: 0.3314132608961278
INFO : round 15: 0.3207616012174481
INFO : round 16: 0.3134312622142009
INFO : round 17: 0.30880657275811363
INFO : round 18: 0.3054238168326334
INFO : round 19: 0.2923030183671381
INFO : round 20: 0.2929751542262947
INFO : round 21: 0.2888613886083848
INFO : round 22: 0.28136546656001393
INFO : round 23: 0.2774300002805389
INFO : round 24: 0.27866603896771663
INFO : round 25: 0.27145795095652436
INFO : round 26: 0.2660579904199789
INFO : round 27: 0.26216595231022793
INFO : round 28: 0.2620670170294115
INFO : round 29: 0.2543302844352901
INFO : round 30: 0.2550673183982079
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.3136),
INFO : (2, 0.6374),
INFO : (3, 0.7275),
INFO : (4, 0.8018999999999998),
INFO : (5, 0.843),
INFO : (6, 0.8694),
INFO : (7, 0.8741),
INFO : (8, 0.8803),
INFO : (9, 0.8907),
INFO : (10, 0.8923),
INFO : (11, 0.9009),
INFO : (12, 0.8988),
INFO : (13, 0.9031),
INFO : (14, 0.9041),
INFO : (15, 0.9072),
INFO : (16, 0.9084),
INFO : (17, 0.9096),
INFO : (18, 0.9112),
INFO : (19, 0.9155),
INFO : (20, 0.9141),
INFO : (21, 0.9147),
INFO : (22, 0.9171),
INFO : (23, 0.918),
INFO : (24, 0.9169),
INFO : (25, 0.921),
INFO : (26, 0.9216),
INFO : (27, 0.9226),
INFO : (28, 0.9213),
INFO : (29, 0.9259),
INFO : (30, 0.9252)]}
INFO :
(raylet) [2026-04-21 12:57:42,447 E 14603 687444] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-56-01_142686_14594 is over 95% full, available space: 11
.5305 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
```

```

--- honest_baseline_noniid ---
Round 1: 0.3136
Round 2: 0.6374
Round 3: 0.7275
Round 4: 0.8019
Round 5: 0.8430
Round 6: 0.8694
Round 7: 0.8741
Round 8: 0.8803
Round 9: 0.8907
Round 10: 0.8923
Round 11: 0.9009
Round 12: 0.8988
Round 13: 0.9031
Round 14: 0.9041
Round 15: 0.9072
Round 16: 0.9084
Round 17: 0.9096
Round 18: 0.9112
Round 19: 0.9155
Round 20: 0.9141
Round 21: 0.9147
Round 22: 0.9171
Round 23: 0.9180
Round 24: 0.9169
Round 25: 0.9210
Round 26: 0.9216
Round 27: 0.9226
Round 28: 0.9213
Round 29: 0.9259
Round 30: 0.9252
[PL0T] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:
Round 1: 0.3136
Round 2: 0.6374
Round 3: 0.7275
Round 4: 0.8019
Round 5: 0.8430
Round 6: 0.8694
Round 7: 0.8741
Round 8: 0.8803
Round 9: 0.8907
Round 10: 0.8923
Round 11: 0.9009
Round 12: 0.8988
Round 13: 0.9031
Round 14: 0.9041
Round 15: 0.9072
Round 16: 0.9084
Round 17: 0.9096
Round 18: 0.9112
Round 19: 0.9155
Round 20: 0.9141
Round 21: 0.9147
Round 22: 0.9171
Round 23: 0.9180
Round 24: 0.9169
Round 25: 0.9210
Round 26: 0.9216
Round 27: 0.9226
Round 28: 0.9213
Round 29: 0.9259
Round 30: 0.9252

(ClientAppActor pid=14611) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 59x across cluster]
(fl-env) apple@MacBook-Pro-10 project-main % python server.py baseline_noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
          Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

          $ flwr new # Create a new Flower app from a template

          $ flwr run # Run the Flower app in Simulation Mode

          Using 'start_simulation()' is deprecated.

          This is a deprecated feature. It will be removed
          entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=100, no round_timeout
2026-04-21 12:59:58,902 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray will
1 no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'memory': 8685158400.0, 'node:127.0.0.1': 1.0, 'CPU': 8.0, 'object_store_memory': 2147483648.0, 'node:__internal_head
__': 1.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14724) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=14724) labels = np.array(trainset.targets)
(ClientAppActor pid=14724) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953}
(ClientAppActor pid=14724) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953}
(ClientAppActor pid=14724) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79}
(ClientAppActor pid=14724) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010}
(ClientAppActor pid=14724) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2}
(ClientAppActor pid=14724) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123}
INFO : aggregate_fit: received 20 results and 0 failures
WARNING : No fit_metrics.aggregation_fn provided
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
36x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/
configure-logging.html#log-deduplication for more options.)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:00:08,930 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 12
.9378 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14723) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=14723) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures

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INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14723) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 7
5x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14727) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 53x across c
luster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (raylet) [2026-04-21 13:00:18,948 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.7339 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 6]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 7]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14725) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 7
2x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 8]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14723) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 7
1x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 9]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : (pid=gcs_server) [2026-04-21 13:00:28,567 E 14713 692705] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metri
cs will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : (raylet) [2026-04-21 13:00:28,992 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.7316 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : (raylet) [2026-04-21 13:00:29,156 E 14717 692809] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter a
gent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14723) [2026-04-21 13:00:29,706 E 14723 693040] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : [2026-04-21 13:00:30,144 E 14708 692864] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agen
t status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 10]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14725) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
7x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 11]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 12]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14723) [NON-IID] Client 8 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 62x acro
ss cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (raylet) [2026-04-21 13:00:39,007 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.7318 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : (ClientAppActor pid=14725) [2026-04-21 13:00:30,093 E 14725 693180] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 13]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14723) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 7
2x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 14]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 15]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14721) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
48x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (raylet) [2026-04-21 13:00:49,076 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5926 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 16]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : (ClientAppActor pid=14720) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 72x
across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 17]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
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INFO : [ROUND 18]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
8x across cluster]
(raylet) [2026-04-21 13:00:59,127 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5929 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 19]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 20]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14727) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
60x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 21]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 7
4x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 22]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:01:09,201 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5919 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 23]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14724) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 65x a
cross cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 24]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 25]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 61x acro
ss cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:01:19,253 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5878 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 26]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 9 | Total: 2061 samples | Distribution: {0: 11, 1: 133, 2: 1264, 3: 170, 4: 261, 5: 169, 6: 46, 7: 1, 8: 3, 9: 3} [repeated 73x
across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 27]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 28]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 69
x across cluster]
(raylet) [2026-04-21 13:01:29,304 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5868 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 29]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 30]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 71x across c
luster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 31]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 6
0x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 32]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:01:39,342 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5869 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 33]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14725) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 59
x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 34]
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INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
0x across cluster]
(raylet) [2026-04-21 13:01:49,385 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5863 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 35]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 36]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14727) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 5
3x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 37]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 68x
across cluster]
(raylet) [2026-04-21 13:01:59,433 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5858 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 38]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 39]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
7x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 40]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 41]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 60x
across cluster]
(raylet) [2026-04-21 13:02:09,445 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5856 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 42]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 65
x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 43]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 44]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 7
2x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:02:19,510 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5855 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 45]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 46]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 63
x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 47]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:02:29,612 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5848 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14726) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 72x
across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 48]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 49]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 68x across c
luster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 50]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
```

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INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:02:39,629 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5886 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14727) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeated
d 59x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 51]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 52]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14726) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 58x across c
luster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 53]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:02:49,717 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5884 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14724) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 6
2x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 54]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 55]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 73x acro
ss cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 56]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 57]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:02:59,730 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5877 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14727) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 68x
across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 58]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 59]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14725) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
0x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 60]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:03:09,841 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5858 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14726) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
72x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 61]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14725) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 4
5x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 62]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 63]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:03:19,890 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5369 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 66x acro
ss cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 64]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 65]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
57x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 66]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
```

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INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:03:29,939 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5352 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14727) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
74x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 67]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14722)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 68]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14725) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 63x
across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 69]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:03:40,032 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5346 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14725) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 6
1x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 70]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 71]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14725) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 5
9x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 72]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:03:50,059 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5319 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 73]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeated
d 67x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 74]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14724) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 71x across
cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 75]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 76]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:04:00,129 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5308 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14722) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 7
0x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 77]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 78]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 55x
across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 79]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:04:10,225 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.5301 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 80]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14720) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
80x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 81]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 59
x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 82]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
```

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INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:04:20,279 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.4743 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 83]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 61
x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 84]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14727) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
67x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 85]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:04:30,324 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.4741 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 86]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14726) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
59x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 87]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14725) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
9x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 88]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 89]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:04:40,398 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.4741 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 76x
across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 90]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 91]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
64x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 92]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:04:50,450 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.474 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 93]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 6
5x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 94]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14721) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 7
7x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 95]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:05:00,491 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.4739 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 96]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14727) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 7
4x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 97]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 98]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14727) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 76x
across cluster]
```

```
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 99]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:05:10,586 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.4741 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 100]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14723) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 5
6x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [SUMMARY]
INFO : Run finished 100 round(s) in 314.45s
INFO : History (loss, distributed):
INFO : round 1: 1.974998679023962
INFO : round 2: 1.3853827756814683
INFO : round 3: 0.9742590758366327
INFO : round 4: 0.7363823182857074
INFO : round 5: 0.59805497265281
INFO : round 6: 0.5218424414293453
INFO : round 7: 0.47183061936221565
INFO : round 8: 0.438202812315557
INFO : round 9: 0.40287949081760244
INFO : round 10: 0.3810412438866049
INFO : round 11: 0.369923461026277
INFO : round 12: 0.3536019021163162
INFO : round 13: 0.3392380641791196
INFO : round 14: 0.33314357900890873
INFO : round 15: 0.3213471897696249
INFO : round 16: 0.3175779467049879
INFO : round 17: 0.3093245600382931
INFO : round 18: 0.30534257796911385
INFO : round 19: 0.2992795457581457
INFO : round 20: 0.2970782679895433
INFO : round 21: 0.28952038677605674
INFO : round 22: 0.28482181081078206
INFO : round 23: 0.2817659606240666
INFO : round 24: 0.27912352152894765
INFO : round 25: 0.275415773854588
INFO : round 26: 0.2705134127390223
INFO : round 27: 0.2660883031071375
INFO : round 28: 0.2646512259571507
INFO : round 29: 0.2596944612793077
INFO : round 30: 0.2551094795139834
INFO : round 31: 0.2547980216376412
INFO : round 32: 0.2516363082102503
INFO : round 33: 0.25068135391742274
INFO : round 34: 0.24609262059136228
INFO : round 35: 0.2451048480784193
INFO : round 36: 0.23970922100664183
INFO : round 37: 0.2384655032955627
INFO : round 38: 0.23637198533017795
INFO : round 39: 0.23400971772649085
INFO : round 40: 0.23054756869237644
INFO : round 41: 0.2297612164071359
INFO : round 42: 0.23232980326752362
INFO : round 43: 0.22527199144365073
INFO : round 44: 0.21923490932669504
INFO : round 45: 0.21825844218758061
INFO : round 46: 0.2156500728241504
INFO : round 47: 0.21358021721095086
INFO : round 48: 0.21237204158113313
INFO : round 49: 0.2100624874734079
INFO : round 50: 0.20782215212927033
INFO : round 51: 0.20562539209340946
INFO : round 52: 0.20221434730721027
INFO : round 53: 0.2021388335662266
INFO : round 54: 0.19961858355996615
INFO : round 55: 0.19655178859978675
INFO : round 56: 0.1956009153833095
INFO : round 57: 0.193831558078944
INFO : round 58: 0.19127781802086166
INFO : round 59: 0.18841952738462212
INFO : round 60: 0.18957268117447962
INFO : round 61: 0.1869946811601115
INFO : round 62: 0.184998665622379
INFO : round 63: 0.18292782252309275
INFO : round 64: 0.18187567282962008
INFO : round 65: 0.18058765533874138
INFO : round 66: 0.17980944699408433
INFO : round 67: 0.17704877292457674
INFO : round 68: 0.17493508797073468
INFO : round 69: 0.17645026499869676
INFO : round 70: 0.17217434158908745
INFO : round 71: 0.17080587336195474
INFO : round 72: 0.17037614197217332
INFO : round 73: 0.16666731487560907
INFO : round 74: 0.16539323211873702
INFO : round 75: 0.1651066851314193
INFO : round 76: 0.16324751124086495
INFO : round 77: 0.16193648102208258
INFO : round 78: 0.16143500423222518
INFO : round 79: 0.1601220160170783
INFO : round 80: 0.15864626276262173
INFO : round 81: 0.1580800040358731
INFO : round 82: 0.15616599422025557
INFO : round 83: 0.15553127493584357
INFO : round 84: 0.15440639283377142
INFO : round 85: 0.15373697421162394
INFO : round 86: 0.15158530634741624
INFO : round 87: 0.15129807122801534
INFO : round 88: 0.1520349658458842
INFO : round 89: 0.1581614925816191
INFO : round 90: 0.14997692585970385
INFO : round 91: 0.1463474737816511
INFO : round 92: 0.14481203951545585
INFO : round 93: 0.14486548658862067
INFO : round 94: 0.14504594548409
INFO : round 95: 0.14292182529809327
INFO : round 96: 0.14168505370542525
INFO : round 97: 0.14124547672449064
INFO : round 98: 0.14053035685242102
INFO : round 99: 0.13988993494970778
INFO : round 100: 0.14054965743863476
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.4702),
INFO : (2, 0.6273),
INFO : (3, 0.716),
INFO : (4, 0.7877999999999999),
INFO : (5, 0.8444),
```

```

INFO : (6, 0.8608),
INFO : (7, 0.87),
INFO : (8, 0.8771),
INFO : (9, 0.8886),
INFO : (10, 0.8923),
INFO : (11, 0.8928),
INFO : (12, 0.8979),
INFO : (13, 0.9031),
INFO : (14, 0.9033),
INFO : (15, 0.9072),
INFO : (16, 0.9088),
INFO : (17, 0.9104),
INFO : (18, 0.9097),
INFO : (19, 0.9132),
INFO : (20, 0.9127),
INFO : (21, 0.9157),
INFO : (22, 0.9161),
INFO : (23, 0.9169),
INFO : (24, 0.9166),
INFO : (25, 0.919),
INFO : (26, 0.9201),
INFO : (27, 0.9228),
INFO : (28, 0.9213),
INFO : (29, 0.9229),
INFO : (30, 0.9257),
INFO : (31, 0.9258),
INFO : (32, 0.9272),
INFO : (33, 0.9261),
INFO : (34, 0.9278),
INFO : (35, 0.9276),
INFO : (36, 0.932),
INFO : (37, 0.9304),
INFO : (38, 0.93),
INFO : (39, 0.9311),
INFO : (40, 0.9315),
INFO : (41, 0.9322),
INFO : (42, 0.9296),
INFO : (43, 0.9342),
INFO : (44, 0.9358),
INFO : (45, 0.9356),
INFO : (46, 0.9361),
INFO : (47, 0.936),
INFO : (48, 0.9372),
INFO : (49, 0.938),
INFO : (50, 0.9385),
INFO : (51, 0.9401),
INFO : (52, 0.9416),
INFO : (53, 0.9413),
INFO : (54, 0.9419),
INFO : (55, 0.9427),
INFO : (56, 0.9432),
INFO : (57, 0.9442),
INFO : (58, 0.9447),
INFO : (59, 0.9457),
INFO : (60, 0.945),
INFO : (61, 0.9464),
INFO : (62, 0.9461),
INFO : (63, 0.9469),
INFO : (64, 0.9472),
INFO : (65, 0.9478),
INFO : (66, 0.9479),
INFO : (67, 0.9484),
INFO : (68, 0.9491),
INFO : (69, 0.9489),
INFO : (70, 0.9495),
INFO : (71, 0.9502),
INFO : (72, 0.9503),
INFO : (73, 0.9515),
INFO : (74, 0.9512),
INFO : (75, 0.9514),
INFO : (76, 0.9524),
INFO : (77, 0.953),
INFO : (78, 0.9534),
INFO : (79, 0.9526),
INFO : (80, 0.9533),
INFO : (81, 0.9534),
INFO : (82, 0.9542),
INFO : (83, 0.9541),
INFO : (84, 0.9546),
INFO : (85, 0.9555),
INFO : (86, 0.9551),
INFO : (87, 0.9557),
INFO : (88, 0.9557),
INFO : (89, 0.9542),
INFO : (90, 0.9565),
INFO : (91, 0.9577),
INFO : (92, 0.958),
INFO : (93, 0.9575),
INFO : (94, 0.9573),
INFO : (95, 0.9577),
INFO : (96, 0.9582),
INFO : (97, 0.9584),
INFO : (98, 0.9589),
INFO : (99, 0.9586),
INFO : (100, 0.9587)]}
INFO :

```

--- honest_baseline_noniid ---

```

Round 1: 0.4702
Round 2: 0.6273
Round 3: 0.7160
Round 4: 0.7878
Round 5: 0.8444
Round 6: 0.8608
Round 7: 0.8700
Round 8: 0.8771
Round 9: 0.8886
Round 10: 0.8923
Round 11: 0.8928
Round 12: 0.8979
Round 13: 0.9031
Round 14: 0.9033
Round 15: 0.9072
Round 16: 0.9088
Round 17: 0.9104
Round 18: 0.9097
Round 19: 0.9132
Round 20: 0.9127
Round 21: 0.9157
Round 22: 0.9161
Round 23: 0.9169
Round 24: 0.9166
Round 25: 0.9190
Round 26: 0.9201
Round 27: 0.9228
Round 28: 0.9213
Round 29: 0.9229

```


Round 30: 0.9257
Round 31: 0.9258
Round 32: 0.9272
Round 33: 0.9261
Round 34: 0.9278
Round 35: 0.9276
Round 36: 0.9320
Round 37: 0.9304
Round 38: 0.9300
Round 39: 0.9311
Round 40: 0.9315
Round 41: 0.9322
Round 42: 0.9296
Round 43: 0.9342
Round 44: 0.9358
Round 45: 0.9356
Round 46: 0.9361
Round 47: 0.9360
Round 48: 0.9372
Round 49: 0.9380
Round 50: 0.9385
Round 51: 0.9401
Round 52: 0.9416
Round 53: 0.9413
Round 54: 0.9419
Round 55: 0.9427
Round 56: 0.9432
Round 57: 0.9442
Round 58: 0.9447
Round 59: 0.9457
Round 60: 0.9450
Round 61: 0.9464
Round 62: 0.9461
Round 63: 0.9469
Round 64: 0.9472
Round 65: 0.9478
Round 66: 0.9479
Round 67: 0.9484
Round 68: 0.9491
Round 69: 0.9489
Round 70: 0.9495
Round 71: 0.9502
Round 72: 0.9503
Round 73: 0.9515
Round 74: 0.9512
Round 75: 0.9514
Round 76: 0.9524
Round 77: 0.9530
Round 78: 0.9534
Round 79: 0.9526
Round 80: 0.9533
Round 81: 0.9534
Round 82: 0.9542
Round 83: 0.9541
Round 84: 0.9546
Round 85: 0.9555
Round 86: 0.9551
Round 87: 0.9557
Round 88: 0.9557
Round 89: 0.9542
Round 90: 0.9565
Round 91: 0.9577
Round 92: 0.9580
Round 93: 0.9575
Round 94: 0.9573
Round 95: 0.9577
Round 96: 0.9582
Round 97: 0.9584
Round 98: 0.9589
Round 99: 0.9586
Round 100: 0.9587
[PLOT] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:

Round 1: 0.4702
Round 2: 0.6273
Round 3: 0.7160
Round 4: 0.7878
Round 5: 0.8444
Round 6: 0.8608
Round 7: 0.8700
Round 8: 0.8771
Round 9: 0.8886
Round 10: 0.8923
Round 11: 0.8928
Round 12: 0.8979
Round 13: 0.9031
Round 14: 0.9033
Round 15: 0.9072
Round 16: 0.9088
Round 17: 0.9104
Round 18: 0.9097
Round 19: 0.9132
Round 20: 0.9127
Round 21: 0.9157
Round 22: 0.9161
Round 23: 0.9169
Round 24: 0.9166
Round 25: 0.9190
Round 26: 0.9201
Round 27: 0.9228
Round 28: 0.9213
Round 29: 0.9229
Round 30: 0.9257
Round 31: 0.9258
Round 32: 0.9272
Round 33: 0.9261
Round 34: 0.9278
Round 35: 0.9276
Round 36: 0.9320
Round 37: 0.9304
Round 38: 0.9300
Round 39: 0.9311
Round 40: 0.9315
Round 41: 0.9322
Round 42: 0.9296
Round 43: 0.9342
Round 44: 0.9358
Round 45: 0.9356
Round 46: 0.9361
Round 47: 0.9360
Round 48: 0.9372
Round 49: 0.9380
Round 50: 0.9385
Round 51: 0.9401
Round 52: 0.9416
Round 53: 0.9413

```

Round 54: 0.9419
Round 55: 0.9427
Round 56: 0.9432
Round 57: 0.9442
Round 58: 0.9447
Round 59: 0.9457
Round 60: 0.9450
Round 61: 0.9464
Round 62: 0.9461
Round 63: 0.9469
Round 64: 0.9472
Round 65: 0.9478
Round 66: 0.9479
Round 67: 0.9484
Round 68: 0.9491
Round 69: 0.9489
Round 70: 0.9495
Round 71: 0.9502
Round 72: 0.9503
Round 73: 0.9515
Round 74: 0.9512
Round 75: 0.9514
Round 76: 0.9524
Round 77: 0.9530
Round 78: 0.9534
Round 79: 0.9526
Round 80: 0.9533
Round 81: 0.9534
Round 82: 0.9542
Round 83: 0.9541
Round 84: 0.9546
Round 85: 0.9555
Round 86: 0.9551
Round 87: 0.9557
Round 88: 0.9557
Round 89: 0.9542
Round 90: 0.9565
Round 91: 0.9577
Round 92: 0.9580
Round 93: 0.9575
Round 94: 0.9573
Round 95: 0.9577
Round 96: 0.9582
Round 97: 0.9584
Round 98: 0.9589
Round 99: 0.9586
Round 100: 0.9587

(raylet) [2026-04-21 13:05:20,676 E 14717 692825] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_12-59-58_133282_14708 is over 95% full, available space: 11
.4739 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14724) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 3
6x across cluster]
(fl-env) appleMacBook-Pro-10 project-main % python server.py attack noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
Instead, use the `flwr run` CLI command to start a local simulation in your Flower app, as shown for example below:

$ flwr new # Create a new Flower app from a template

$ flwr run # Run the Flower app in Simulation Mode

Using `start_simulation()` is deprecated.

This is a deprecated feature. It will be removed
entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=100, no round_timeout
2026-04-21 13:08:02,559 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray wil
l no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'CPU': 8.0, 'memory': 8363196416.0, 'node:127.0.0.1': 1.0, 'object_store_memory': 2147483648.0, 'node:__internal_head
__': 1.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58}
(ClientAppActor pid=14922) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=14922) labels = np.array(trainset.targets)
(ClientAppActor pid=14922) [NON-IID] Client 9 | Total: 2061 samples | Distribution: {0: 11, 1: 133, 2: 1264, 3: 170, 4: 261, 5: 169, 6: 46, 7: 1, 8: 3, 9: 3}
(ClientAppActor pid=14922) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11}
(ClientAppActor pid=14922) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2}
(ClientAppActor pid=14922) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35}
(ClientAppActor pid=14922) [SETUP] Client 16 is MALICIOUS (label_flip)
(ClientAppActor pid=14922) [ATTACK] Client 16 submitted poisoned update
(ClientAppActor pid=14922) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91}
INFO : aggregate_fit: received 20 results and 0 failures
WARNING : No fit_metrics_aggregation_fn provided
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 33
x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/con
figure-logging.html#log-deduplication for more options.)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 21x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
(ClientAppActor pid=14917) [ATTACK] Client 18 submitted poisoned update [repeated 15x across cluster]
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:08:12,542 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.2919 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14919) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=14919) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 5
8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 24x across cluster]
(ClientAppActor pid=14920) [ATTACK] Client 14 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :

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```
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 61x across c
luster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14920) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 13:08:22,592 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11
.9396 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 6]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
8x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 7]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 24x across cluster]
(ClientAppActor pid=14916) [ATTACK] Client 13 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 8]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 71x acro
ss cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 31x across cluster]
(pidges_server) [2026-04-21 13:08:32,186 E 14909 704523] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metri
cs will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(raylet) [2026-04-21 13:08:32,613 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11
.9348 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 9]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:08:32,698 E 14913 704641] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter a
gent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=14919) [ATTACK] Client 15 submitted poisoned update [repeated 15x across cluster]
[2026-04-21 13:08:33,617 E 14903 704697] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agen
t status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=14917) [2026-04-21 13:08:33,546 E 14917 704955] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 10]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 28x across cluster]
(ClientAppActor pid=14921) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 11]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 74x acro
ss cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 12]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 30x across cluster]
(raylet) [2026-04-21 13:08:42,667 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11
.8919 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14922) [2026-04-21 13:08:33,652 E 14922 705181] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not
be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
(ClientAppActor pid=14916) [ATTACK] Client 13 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 13]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 6
7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 14]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 24x across cluster]
(ClientAppActor pid=14916) [ATTACK] Client 15 submitted poisoned update [repeated 14x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 15]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeate
d 71x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 29x across cluster]
(raylet) [2026-04-21 13:08:52,749 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11
.4755 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 16]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 12 submitted poisoned update [repeated 10x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
62x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 17]
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INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14919) [ATTACK] Client 15 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 18]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 73x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 19]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 30x across cluster]
(raylet) [2026-04-21 13:09:02,833 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11.4744 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [ATTACK] Client 13 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 20]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated 63x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 21]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 17 submitted poisoned update [repeated 8x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 22]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 66x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 28x across cluster]
(raylet) [2026-04-21 13:09:12,923 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11.4741 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14921) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 23]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated 67x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 24]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 23x across cluster]
(ClientAppActor pid=14918) [ATTACK] Client 13 submitted poisoned update [repeated 14x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 25]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 59x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 26x across cluster]
(raylet) [2026-04-21 13:09:22,938 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11.4735 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 26]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 12 submitted poisoned update [repeated 10x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeated 69x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 27]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 28]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 63x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:09:32,942 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11.4733 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14917) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 29]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [ATTACK] Client 16 submitted poisoned update [repeated 10x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 30]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 62x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14922) [ATTACK] Client 15 submitted poisoned update [repeated 14x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 31]
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INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 981, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
70x across cluster]
(raylet) [2026-04-21 13:09:43,020 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11
.4726 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14923) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 32]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [ATTACK] Client 16 submitted poisoned update [repeated 11x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 33]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 981, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
52x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14920) [ATTACK] Client 14 submitted poisoned update [repeated 13x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 34]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
70x across cluster]
(raylet) [2026-04-21 13:09:53,097 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 11
.473 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 35]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 22x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 36]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 66
x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 29x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 37]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [ATTACK] Client 13 submitted poisoned update [repeated 8x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 58
x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 38]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:10:03,150 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4711 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14920) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 23x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 39]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 64
x across cluster]
(ClientAppActor pid=14920) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 40]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [ATTACK] Client 12 submitted poisoned update [repeated 14x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 41]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
61x across cluster]
(raylet) [2026-04-21 13:10:13,162 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.471 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14922) [ATTACK] Client 15 submitted poisoned update [repeated 10x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 42]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
72x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 43]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 27x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO :      [ROUND 44]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35} [repeated 73x acr
oss cluster]
(raylet) [2026-04-21 13:10:23,199 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4708 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14919) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 29x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
```

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INFO :
INFO : [ROUND 45]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [ATTACK] Client 16 submitted poisoned update [repeated 11x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 46]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 64
x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 27x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 47]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [ATTACK] Client 17 submitted poisoned update [repeated 13x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 66x acro
ss cluster]
(raylet) [2026-04-21 13:10:33,228 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4703 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 48]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 49]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 63
x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 50]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 12 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
58x across cluster]
(raylet) [2026-04-21 13:10:43,237 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4701 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 51]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 22x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [ATTACK] Client 12 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 52]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 13 | Total: 1157 samples | Distribution: {0: 169, 2: 237, 3: 189, 4: 82, 5: 176, 6: 139, 7: 41, 8: 45, 9: 79} [repeated 60x acro
ss cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 53]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [ATTACK] Client 19 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 60x a
cross cluster]
(raylet) [2026-04-21 13:10:53,297 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4697 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 54]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [ATTACK] Client 15 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 55]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 5
7x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 22x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 56]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [ATTACK] Client 16 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 61
x across cluster]
(raylet) [2026-04-21 13:11:03,685 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4686 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 57]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [ATTACK] Client 12 submitted poisoned update [repeated 15x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 58]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 6
5x across cluster]
(ClientAppActor pid=14919) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 27x across cluster]
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INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 59]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 60]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
60x across cluster]
(raylet) [2026-04-21 13:11:13,745 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4782 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 61]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [ATTACK] Client 15 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 7
4x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 62]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 16 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 63]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 71
x across cluster]
(raylet) [2026-04-21 13:11:23,783 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.478 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14922) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 31x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 64]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [ATTACK] Client 16 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 65]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 6
4x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 28x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 66]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [ATTACK] Client 15 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:11:33,886 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4768 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14920) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
69x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 67]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 68]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 6
9x across cluster]
(ClientAppActor pid=14922) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 69]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [ATTACK] Client 15 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 13:11:43,889 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4788 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 70]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeate
d 53x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=14918) [ATTACK] Client 16 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 71]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 70x a
cross cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 72]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [ATTACK] Client 14 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
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INFO : [ROUND 73]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:11:53,928 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4788 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
70x across cluster]
(ClientAppActor pid=14919) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 32x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 74]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [ATTACK] Client 16 submitted poisoned update [repeated 15x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 75]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 66x a
cross cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 28x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 76]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [ATTACK] Client 12 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:12:04,024 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4785 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14919) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 71x a
cross cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 77]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 28x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [ATTACK] Client 15 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 78]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeate
d 68x across cluster]
(ClientAppActor pid=14921) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 79]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [ATTACK] Client 14 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:12:14,039 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4784 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14919) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 49
x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 80]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 20x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [ATTACK] Client 12 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 81]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 6
7x across cluster]
(ClientAppActor pid=14916) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 27x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 82]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 18 submitted poisoned update [repeated 15x across cluster]
(raylet) [2026-04-21 13:12:24,077 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4782 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 83]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 64
x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 28x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 84]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [ATTACK] Client 14 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 6
9x across cluster]
(ClientAppActor pid=14921)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 85]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 86]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:12:34,099 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4775 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14921) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeate
d 65x across cluster]
```

```
(ClientAppActor pid=14920) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 87]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [ATTACK] Client 12 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 5
9x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 88]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14920) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 14 submitted poisoned update [repeated 15x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 89]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:12:44,104 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4772 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 6
2x across cluster]
(ClientAppActor pid=14921) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 90]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [ATTACK] Client 14 submitted poisoned update [repeated 10x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 91]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14918) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
60x across cluster]
(ClientAppActor pid=14918) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 19 submitted poisoned update [repeated 14x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 92]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:12:54,117 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4762 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 7
3x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 93]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [ATTACK] Client 15 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 94]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 6
7x across cluster]
(ClientAppActor pid=14920) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 95]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [ATTACK] Client 16 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:13:04,130 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4759 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 96]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
60x across cluster]
(ClientAppActor pid=14918) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 28x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14919) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 97]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14923) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 71x across c
luster]
(ClientAppActor pid=14919) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 22x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 98]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14922) [ATTACK] Client 18 submitted poisoned update [repeated 10x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:13:14,162 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4747 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 99]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 5
8x across cluster]
(ClientAppActor pid=14919) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 27x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14916) [ATTACK] Client 17 submitted poisoned update [repeated 14x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [ROUND 100]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=14917) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 6
```

```
5x across cluster]
(ClientAppActor pid=14922) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 25x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
INFO :
INFO : [SUMMARY]
INFO : Run finished 100 round(s) in 317.20s
INFO : History (loss, distributed):
INFO : round 1: 1.2957986249710425
INFO : round 2: 1.1123474189648614
INFO : round 3: 0.9443071256430384
INFO : round 4: 0.8205925943752448
INFO : round 5: 0.7438390699819254
INFO : round 6: 0.6917941934003617
INFO : round 7: 0.6554228181656176
INFO : round 8: 0.6293506408651796
INFO : round 9: 0.6047749468693718
INFO : round 10: 0.5940100229586275
INFO : round 11: 0.5767736031034122
INFO : round 12: 0.5633517408713746
INFO : round 13: 0.5416831108137441
INFO : round 14: 0.5375235939178223
INFO : round 15: 0.5263010724664877
INFO : round 16: 0.5171710427576741
INFO : round 17: 0.5135123572410486
INFO : round 18: 0.500646687639407
INFO : round 19: 0.499782900136119
INFO : round 20: 0.48980555637195095
INFO : round 21: 0.48220417585235814
INFO : round 22: 0.4783515937031267
INFO : round 23: 0.46976429719132734
INFO : round 24: 0.4623966634463959
INFO : round 25: 0.46270268995540975
INFO : round 26: 0.4591242500768302
INFO : round 27: 0.4561417211168491
INFO : round 28: 0.4452370711980155
INFO : round 29: 0.45764745378646604
INFO : round 30: 0.4460624266165895
INFO : round 31: 0.43750155886141257
INFO : round 32: 0.4458855003975451
INFO : round 33: 0.428581253274942
INFO : round 34: 0.4269879257146781
INFO : round 35: 0.426507446098371
INFO : round 36: 0.4258185310866505
INFO : round 37: 0.4195509806608621
INFO : round 38: 0.419038270094905
INFO : round 39: 0.4218184802288445
INFO : round 40: 0.4171297205332369
INFO : round 41: 0.41160463275619974
INFO : round 42: 0.41096204791587
INFO : round 43: 0.4057425328717826
INFO : round 44: 0.4059946357823027
INFO : round 45: 0.3995952956592694
INFO : round 46: 0.396823885741706
INFO : round 47: 0.3914686872555425
INFO : round 48: 0.39168184260590766
INFO : round 49: 0.39675395841034833
INFO : round 50: 0.39487230017924085
INFO : round 51: 0.38872102834165284
INFO : round 52: 0.3842617594776824
INFO : round 53: 0.38629775053015153
INFO : round 54: 0.38454143835332827
INFO : round 55: 0.38084491575107027
INFO : round 56: 0.37742942404061464
INFO : round 57: 0.373180763942365
INFO : round 58: 0.3691426237170307
INFO : round 59: 0.371540220762594
INFO : round 60: 0.3603705576814401
INFO : round 61: 0.37138729880031307
INFO : round 62: 0.3750092466227925
INFO : round 63: 0.3654195225657747
INFO : round 64: 0.3659522497806305
INFO : round 65: 0.3645930515310635
INFO : round 66: 0.3619365307088858
INFO : round 67: 0.3638016341974179
INFO : round 68: 0.3700188305621711
INFO : round 69: 0.3843098877337032
INFO : round 70: 0.36091785926026654
INFO : round 71: 0.35308266061182597
INFO : round 72: 0.3453996927212602
INFO : round 73: 0.3612620761981026
INFO : round 74: 0.36221108019542386
INFO : round 75: 0.35272900652580746
INFO : round 76: 0.3616710030423185
INFO : round 77: 0.3595914613705473
INFO : round 78: 0.35607952108017554
INFO : round 79: 0.3508649256473152
INFO : round 80: 0.35918002816054
INFO : round 81: 0.3529517455032458
INFO : round 82: 0.3775201517553
INFO : round 83: 0.3577301160596049
INFO : round 84: 0.3551697909641571
INFO : round 85: 0.35006976144763224
INFO : round 86: 0.348522690252755
INFO : round 87: 0.3446013276950239
INFO : round 88: 0.3542797212973951
INFO : round 89: 0.3391420973756443
INFO : round 90: 0.3535939645462524
INFO : round 91: 0.3435858728405766
INFO : round 92: 0.3478253035118785
INFO : round 93: 0.34852589090792135
INFO : round 94: 0.3392697717815923
INFO : round 95: 0.34713435304431495
INFO : round 96: 0.342835541236134
INFO : round 97: 0.3384312597707438
INFO : round 98: 0.3396553146191679
INFO : round 99: 0.34765997237671675
INFO : round 100: 0.3366224900411721
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.3209),
INFO : (2, 0.3599),
INFO : (3, 0.4387),
INFO : (4, 0.5194),
INFO : (5, 0.5622),
INFO : (6, 0.5925),
INFO : (7, 0.6158),
INFO : (8, 0.6305),
INFO : (9, 0.6487),
INFO : (10, 0.6489),
INFO : (11, 0.6695),
INFO : (12, 0.6805),
INFO : (13, 0.7052000000000002),
INFO : (14, 0.703),
INFO : (15, 0.7138),
INFO : (16, 0.7137),
INFO : (17, 0.7203000000000002),
INFO : (18, 0.7367),
INFO : (19, 0.7425000000000002),
```

```

INFO : (20, 0.7352999999999998),
INFO : (21, 0.7493),
INFO : (22, 0.7483),
INFO : (23, 0.7577),
INFO : (24, 0.7594),
INFO : (25, 0.7524),
INFO : (26, 0.7684),
INFO : (27, 0.7567),
INFO : (28, 0.7677000000000002),
INFO : (29, 0.7433999999999998),
INFO : (30, 0.7613),
INFO : (31, 0.7747000000000002),
INFO : (32, 0.7288),
INFO : (33, 0.762),
INFO : (34, 0.7795),
INFO : (35, 0.771),
INFO : (36, 0.7624),
INFO : (37, 0.7738),
INFO : (38, 0.7768000000000002),
INFO : (39, 0.7653),
INFO : (40, 0.7616000000000002),
INFO : (41, 0.7709),
INFO : (42, 0.7753),
INFO : (43, 0.7899000000000002),
INFO : (44, 0.7775),
INFO : (45, 0.7858000000000002),
INFO : (46, 0.7925),
INFO : (47, 0.8017),
INFO : (48, 0.7891),
INFO : (49, 0.7893),
INFO : (50, 0.7796999999999998),
INFO : (51, 0.7947),
INFO : (52, 0.7907999999999998),
INFO : (53, 0.7901),
INFO : (54, 0.791),
INFO : (55, 0.7925),
INFO : (56, 0.8008),
INFO : (57, 0.7976),
INFO : (58, 0.8105),
INFO : (59, 0.8092),
INFO : (60, 0.8181000000000002),
INFO : (61, 0.8038),
INFO : (62, 0.7967999999999998),
INFO : (63, 0.8113),
INFO : (64, 0.8070000000000002),
INFO : (65, 0.8082),
INFO : (66, 0.8041),
INFO : (67, 0.8065),
INFO : (68, 0.8022),
INFO : (69, 0.7861),
INFO : (70, 0.8082),
INFO : (71, 0.8189),
INFO : (72, 0.834),
INFO : (73, 0.8100000000000002),
INFO : (74, 0.8034),
INFO : (75, 0.8141000000000002),
INFO : (76, 0.8072),
INFO : (77, 0.7929000000000002),
INFO : (78, 0.8058999999999998),
INFO : (79, 0.8129999999999998),
INFO : (80, 0.8033),
INFO : (81, 0.8159),
INFO : (82, 0.7639),
INFO : (83, 0.8026),
INFO : (84, 0.7993),
INFO : (85, 0.8101),
INFO : (86, 0.8116),
INFO : (87, 0.8164),
INFO : (88, 0.8016),
INFO : (89, 0.8235),
INFO : (90, 0.7952),
INFO : (91, 0.8182),
INFO : (92, 0.8135),
INFO : (93, 0.8026),
INFO : (94, 0.8201),
INFO : (95, 0.8052),
INFO : (96, 0.8070000000000002),
INFO : (97, 0.8199),
INFO : (98, 0.8078),
INFO : (99, 0.7974),
INFO : (100, 0.8194)]}
INFO :

```

--- fedavg_attack_noniid ---

```

Round 1: 0.3209
Round 2: 0.3599
Round 3: 0.4387
Round 4: 0.5194
Round 5: 0.5622
Round 6: 0.5925
Round 7: 0.6158
Round 8: 0.6305
Round 9: 0.6487
Round 10: 0.6489
Round 11: 0.6695
Round 12: 0.6805
Round 13: 0.7052
Round 14: 0.7030
Round 15: 0.7138
Round 16: 0.7137
Round 17: 0.7203
Round 18: 0.7367
Round 19: 0.7425
Round 20: 0.7353
Round 21: 0.7493
Round 22: 0.7483
Round 23: 0.7577
Round 24: 0.7594
Round 25: 0.7524
Round 26: 0.7684
Round 27: 0.7567
Round 28: 0.7677
Round 29: 0.7434
Round 30: 0.7613
Round 31: 0.7747
Round 32: 0.7288
Round 33: 0.7620
Round 34: 0.7795
Round 35: 0.7710
Round 36: 0.7624
Round 37: 0.7738
Round 38: 0.7768
Round 39: 0.7653
Round 40: 0.7616
Round 41: 0.7709
Round 42: 0.7753
Round 43: 0.7899

```

Round 44: 0.7775
Round 45: 0.7858
Round 46: 0.7925
Round 47: 0.8017
Round 48: 0.7891
Round 49: 0.7893
Round 50: 0.7797
Round 51: 0.7947
Round 52: 0.7908
Round 53: 0.7901
Round 54: 0.7910
Round 55: 0.7925
Round 56: 0.8008
Round 57: 0.7976
Round 58: 0.8105
Round 59: 0.8092
Round 60: 0.8181
Round 61: 0.8038
Round 62: 0.7968
Round 63: 0.8113
Round 64: 0.8070
Round 65: 0.8082
Round 66: 0.8041
Round 67: 0.8065
Round 68: 0.8022
Round 69: 0.7861
Round 70: 0.8082
Round 71: 0.8189
Round 72: 0.8340
Round 73: 0.8100
Round 74: 0.8034
Round 75: 0.8141
Round 76: 0.8072
Round 77: 0.7929
Round 78: 0.8059
Round 79: 0.8130
Round 80: 0.8033
Round 81: 0.8159
Round 82: 0.7639
Round 83: 0.8026
Round 84: 0.7993
Round 85: 0.8101
Round 86: 0.8116
Round 87: 0.8164
Round 88: 0.8016
Round 89: 0.8235
Round 90: 0.7952
Round 91: 0.8182
Round 92: 0.8135
Round 93: 0.8026
Round 94: 0.8201
Round 95: 0.8052
Round 96: 0.8070
Round 97: 0.8199
Round 98: 0.8078
Round 99: 0.7974
Round 100: 0.8194
[PLOT] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:

Round 1: 0.3209
Round 2: 0.3599
Round 3: 0.4387
Round 4: 0.5194
Round 5: 0.5622
Round 6: 0.5925
Round 7: 0.6158
Round 8: 0.6305
Round 9: 0.6487
Round 10: 0.6489
Round 11: 0.6695
Round 12: 0.6805
Round 13: 0.7052
Round 14: 0.7030
Round 15: 0.7138
Round 16: 0.7137
Round 17: 0.7203
Round 18: 0.7367
Round 19: 0.7425
Round 20: 0.7353
Round 21: 0.7493
Round 22: 0.7483
Round 23: 0.7577
Round 24: 0.7594
Round 25: 0.7524
Round 26: 0.7684
Round 27: 0.7567
Round 28: 0.7677
Round 29: 0.7434
Round 30: 0.7613
Round 31: 0.7747
Round 32: 0.7288
Round 33: 0.7620
Round 34: 0.7795
Round 35: 0.7710
Round 36: 0.7624
Round 37: 0.7738
Round 38: 0.7768
Round 39: 0.7653
Round 40: 0.7616
Round 41: 0.7709
Round 42: 0.7753
Round 43: 0.7899
Round 44: 0.7775
Round 45: 0.7858
Round 46: 0.7925
Round 47: 0.8017
Round 48: 0.7891
Round 49: 0.7893
Round 50: 0.7797
Round 51: 0.7947
Round 52: 0.7908
Round 53: 0.7901
Round 54: 0.7910
Round 55: 0.7925
Round 56: 0.8008
Round 57: 0.7976
Round 58: 0.8105
Round 59: 0.8092
Round 60: 0.8181
Round 61: 0.8038
Round 62: 0.7968
Round 63: 0.8113
Round 64: 0.8070
Round 65: 0.8082
Round 66: 0.8041
Round 67: 0.8065


```

Round 68: 0.8022
Round 69: 0.7861
Round 70: 0.8082
Round 71: 0.8189
Round 72: 0.8340
Round 73: 0.8100
Round 74: 0.8034
Round 75: 0.8141
Round 76: 0.8072
Round 77: 0.7929
Round 78: 0.8059
Round 79: 0.8130
Round 80: 0.8033
Round 81: 0.8159
Round 82: 0.7639
Round 83: 0.8026
Round 84: 0.7993
Round 85: 0.8101
Round 86: 0.8116
Round 87: 0.8164
Round 88: 0.8016
Round 89: 0.8235
Round 90: 0.7952
Round 91: 0.8182
Round 92: 0.8135
Round 93: 0.8026
Round 94: 0.8201
Round 95: 0.8052
Round 96: 0.8070
Round 97: 0.8199
Round 98: 0.8078
Round 99: 0.7974
Round 100: 0.8194
(raylet) [2026-04-21 13:13:24,231 E 14913 704657] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-08-01_768665_14903 is over 95% full, available space: 12
.4747 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=14920) [ATTACK] Client 12 submitted poisoned update [repeated 8x across cluster]
(ClientAppActor pid=14921) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 5x ac
ross cluster]
(f1-env) apple@MacBook-Pro-10 project-main % python server.py trust noniid
WARNING : DEPRECATED FEATURE: flwr.simulation.start_simulation() is deprecated.
          Instead, use the 'flwr run' CLI command to start a local simulation in your Flower app, as shown for example below:

          $ flwr new # Create a new Flower app from a template

          $ flwr run # Run the Flower app in Simulation Mode

Using 'start_simulation()' is deprecated.

This is a deprecated feature. It will be removed
entirely in future versions of Flower.

INFO : Starting Flower simulation, config: num_rounds=100, no round_timeout
2026-04-21 13:24:15,305 INFO worker.py:2012 -- Started a local Ray instance.
/Users/apple/Desktop/Msc Project/Artefact/project-main/fl-env/lib/python3.13/site-packages/ray/_private/worker.py:2051: FutureWarning: Tip: In future versions of Ray, Ray will
no longer override accelerator visible devices env var if num_gpus=0 or num_gpus=None (default). To enable this behavior and turn off this error message, set RAY_ACCEL_ENV_
VAR_OVERRIDE_ON_ZERO=0
warnings.warn(
INFO : Flower VCE: Ray initialized with resources: {'node:127.0.0.1': 1.0, 'node:__internal_head__': 1.0, 'CPU': 8.0, 'memory': 7147470848.0, 'object_store_memory': 2147
483648.0}
INFO : Optimize your simulation with Flower VCE: https://flower.ai/docs/framework/how-to-run-simulations.html
INFO : Flower VCE: Resources for each Virtual Client: {'num_cpus': 1}
INFO : Flower VCE: Creating VirtualClientEngineActorPool with 8 actors
INFO : [INIT]
INFO : Requesting initial parameters from one random client
INFO : Received initial parameters from one random client
INFO : Starting evaluation of initial global parameters
INFO : Evaluation returned no results ('None')
INFO :
INFO : [ROUND 1]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, s
o passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migrati
on_guide.html#adapting-to-changes-in-the-copy-keyword
(ClientAppActor pid=15354) labels = np.array(trainset.targets)
(ClientAppActor pid=15354) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13}
(ClientAppActor pid=15354) [SETUP] Client 19 is MALICIOUS (label_flip)
(ClientAppActor pid=15354) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9}
(ClientAppActor pid=15354) [SETUP] Client 18 is MALICIOUS (label_flip)
(ClientAppActor pid=15354) [ATTACK] Client 18 submitted poisoned update
(ClientAppActor pid=15354) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74}
(ClientAppActor pid=15354) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4}
(ClientAppActor pid=15354) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35}
(ClientAppActor pid=15354) [SETUP] Client 16 is MALICIOUS (label_flip)
(ClientAppActor pid=15354) [ATTACK] Client 16 submitted poisoned update
(ClientAppActor pid=15354) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010}
INFO : aggregate_fit: received 20 results and 0 failures

[ROUND 1] CFA Trust Weights:
Client 10: weight=1.000 SKIP (no accuracy yet)
Client 11: weight=1.000 SKIP (no accuracy yet)
Client 1: weight=1.000 SKIP (no accuracy yet)
Client 13: weight=1.000 SKIP (no accuracy yet)
Client 6: weight=1.000 SKIP (no accuracy yet)
Client 17: weight=1.000 SKIP (no accuracy yet)
Client 12: weight=1.000 SKIP (no accuracy yet)
Client 5: weight=1.000 SKIP (no accuracy yet)
Client 7: weight=1.000 SKIP (no accuracy yet)
Client 15: weight=1.000 SKIP (no accuracy yet)
Client 19: weight=1.000 SKIP (no accuracy yet)
Client 2: weight=1.000 SKIP (no accuracy yet)
Client 8: weight=1.000 SKIP (no accuracy yet)
Client 4: weight=1.000 SKIP (no accuracy yet)
Client 16: weight=1.000 SKIP (no accuracy yet)
Client 18: weight=1.000 SKIP (no accuracy yet)
Client 14: weight=1.000 SKIP (no accuracy yet)
Client 0: weight=1.000 SKIP (no accuracy yet)
Client 9: weight=1.000 SKIP (no accuracy yet)
Client 3: weight=1.000 SKIP (no accuracy yet)
→ Total weight: 20.000 | Avg weight: 1.000
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
33x across cluster] (Ray deduplicates logs by default. Set RAY_DEDUP_LOGS=0 to disable log deduplication, or see https://docs.ray.io/en/master/ray-observability/user-guides/c
onfigure-logging.html#log-deduplication-for-more-options.)
(ClientAppActor pid=15348) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 12x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 1 accuracy recorded: 0.1661 (baseline)
INFO :
INFO : [ROUND 2]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [ATTACK] Client 12 submitted poisoned update [repeated 6x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures

[ROUND 2] CFA Trust Weights:
Client 13: weight=1.000 SKIP (round 1)
Client 7: weight=1.000 SKIP (round 1)
Client 16: weight=1.000 SKIP (round 1)
Client 6: weight=1.000 SKIP (round 1)

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Client 11: weight=1.000 SKIP (round 1)
Client 1: weight=1.000 SKIP (round 1)
Client 8: weight=1.000 SKIP (round 1)
Client 5: weight=1.000 SKIP (round 1)
Client 14: weight=1.000 SKIP (round 1)
Client 19: weight=1.000 SKIP (round 1)
Client 10: weight=1.000 SKIP (round 1)
Client 17: weight=1.000 SKIP (round 1)
Client 4: weight=1.000 SKIP (round 1)
Client 3: weight=1.000 SKIP (round 1)
Client 18: weight=1.000 SKIP (round 1)
Client 2: weight=1.000 SKIP (round 1)
Client 9: weight=1.000 SKIP (round 1)
Client 15: weight=1.000 SKIP (round 1)
Client 0: weight=1.000 SKIP (round 1)
Client 12: weight=1.000 SKIP (round 1)
→ Total weight: 20.000 | Avg weight: 1.000
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:24:25,309 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 13.9035 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15351) /Users/apple/Desktop/Msc Project/Artefact/project-main/dataset.py:69: DeprecationWarning: __array__ implementation doesn't accept a copy keyword, so passing copy=False failed. __array__ must implement 'dtype' and 'copy' keyword arguments. To learn more, see the migration guide https://numpy.org/devdocs/numpy_2_0_migration_guide.html#adapting-to-changes-in-the-copy-keyword [repeated 7x across cluster]
(ClientAppActor pid=15351) labels = np.array(trainset.targets) [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 2 accuracy recorded: 0.4089 | Δacc = 0.2428
INFO :
INFO : [ROUND 3]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.2428 → REWARDING aligned clients

[ROUND 3] CFA Trust Weights:
Client 10: weight=1.000 NEUTRAL (sim=-0.002)
Client 11: weight=1.005 REWARDED (sim=+0.052)
Client 9: weight=0.980 MILD PENALTY (sim=-0.061)
Client 6: weight=0.980 MILD PENALTY (sim=-0.058)
Client 15: weight=1.000 NEUTRAL (sim=+0.005)
Client 12: weight=1.004 REWARDED (sim=+0.039)
Client 17: weight=1.003 REWARDED (sim=+0.031)
Client 7: weight=0.980 MILD PENALTY (sim=-0.060)
Client 14: weight=0.980 MILD PENALTY (sim=-0.026)
Client 2: weight=1.005 REWARDED (sim=+0.051)
Client 13: weight=0.980 MILD PENALTY (sim=-0.156)
Client 19: weight=0.980 MILD PENALTY (sim=-0.030)
Client 1: weight=0.980 MILD PENALTY (sim=-0.015)
Client 3: weight=1.002 REWARDED (sim=+0.020)
Client 8: weight=1.006 REWARDED (sim=+0.058)
Client 16: weight=0.980 MILD PENALTY (sim=-0.015)
Client 0: weight=1.003 REWARDED (sim=+0.025)
Client 18: weight=1.002 REWARDED (sim=+0.022)
Client 5: weight=1.002 REWARDED (sim=+0.024)
Client 4: weight=0.980 MILD PENALTY (sim=-0.051)
→ Total weight: 19.852 | Avg weight: 0.993
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 68x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=15350) [ATTACK] Client 14 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 3 accuracy recorded: 0.5095 | Δacc = 0.1006
INFO :
INFO : [ROUND 4]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.1006 → REWARDING aligned clients

[ROUND 4] CFA Trust Weights:
Client 16: weight=0.960 MILD PENALTY (sim=-0.021)
Client 6: weight=0.960 MILD PENALTY (sim=-0.045)
Client 5: weight=1.005 REWARDED (sim=+0.027)
Client 15: weight=0.980 MILD PENALTY (sim=-0.015)
Client 3: weight=1.005 REWARDED (sim=+0.027)
Client 14: weight=0.960 MILD PENALTY (sim=-0.025)
Client 13: weight=0.960 MILD PENALTY (sim=-0.156)
Client 10: weight=0.980 MILD PENALTY (sim=-0.014)
Client 0: weight=1.009 REWARDED (sim=+0.061)
Client 7: weight=0.960 MILD PENALTY (sim=-0.079)
Client 1: weight=0.980 NEUTRAL (sim=-0.006)
Client 19: weight=0.960 MILD PENALTY (sim=-0.040)
Client 9: weight=0.960 MILD PENALTY (sim=-0.065)
Client 12: weight=1.008 REWARDED (sim=+0.036)
Client 4: weight=0.960 MILD PENALTY (sim=-0.065)
Client 11: weight=1.012 REWARDED (sim=+0.072)
Client 17: weight=1.006 REWARDED (sim=+0.031)
Client 8: weight=1.012 REWARDED (sim=+0.064)
Client 2: weight=1.011 REWARDED (sim=+0.062)
Client 18: weight=1.004 REWARDED (sim=+0.020)
→ Total weight: 19.695 | Avg weight: 0.985
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 4 accuracy recorded: 0.5509 | Δacc = 0.0414
INFO :
INFO : [ROUND 5]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15350) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 7x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 32x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0414 → REWARDING aligned clients

[ROUND 5] CFA Trust Weights:
Client 14: weight=0.941 MILD PENALTY (sim=-0.027)
Client 15: weight=0.960 MILD PENALTY (sim=-0.025)
Client 16: weight=0.941 MILD PENALTY (sim=-0.029)
Client 10: weight=0.980 NEUTRAL (sim=-0.008)
Client 2: weight=1.018 REWARDED (sim=+0.065)
Client 8: weight=1.019 REWARDED (sim=+0.065)
Client 9: weight=0.941 MILD PENALTY (sim=-0.066)
Client 4: weight=0.941 MILD PENALTY (sim=-0.066)
Client 18: weight=1.006 REWARDED (sim=+0.017)
Client 1: weight=0.980 NEUTRAL (sim=+0.001)
Client 3: weight=1.008 REWARDED (sim=+0.034)
Client 0: weight=1.015 REWARDED (sim=+0.067)
Client 12: weight=1.011 REWARDED (sim=+0.038)
Client 19: weight=0.941 MILD PENALTY (sim=-0.049)
Client 11: weight=1.020 REWARDED (sim=+0.078)
Client 7: weight=0.941 MILD PENALTY (sim=-0.078)
Client 5: weight=1.008 REWARDED (sim=+0.028)
Client 13: weight=0.941 MILD PENALTY (sim=-0.146)
Client 6: weight=0.941 MILD PENALTY (sim=-0.037)
Client 17: weight=1.008 REWARDED (sim=+0.021)
→ Total weight: 19.564 | Avg weight: 0.978
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 5 accuracy recorded: 0.5650 | Δacc = 0.0141

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INFO :
INFO : [ROUND 6]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:24:35,349 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 13
.9032 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0141 → REWARDING aligned clients

[ROUND 6] CFA Trust Weights:
Client 6: weight=0.922 MILD PENALTY (sim=-0.037)
Client 2: weight=1.024 REWARDED (sim=+0.060)
Client 18: weight=1.007 REWARDED (sim=+0.014)
Client 3: weight=1.012 REWARDED (sim=+0.041)
Client 17: weight=1.009 REWARDED (sim=+0.010)
Client 4: weight=0.922 MILD PENALTY (sim=-0.057)
Client 14: weight=0.922 MILD PENALTY (sim=-0.028)
Client 19: weight=0.922 MILD PENALTY (sim=-0.052)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 15: weight=0.941 MILD PENALTY (sim=-0.027)
Client 9: weight=0.922 MILD PENALTY (sim=-0.058)
Client 11: weight=1.028 REWARDED (sim=+0.076)
Client 7: weight=0.922 MILD PENALTY (sim=-0.071)
Client 12: weight=1.015 REWARDED (sim=+0.040)
Client 0: weight=1.022 REWARDED (sim=+0.063)
Client 13: weight=0.922 MILD PENALTY (sim=-0.127)
Client 5: weight=1.010 REWARDED (sim=+0.025)
Client 16: weight=0.922 MILD PENALTY (sim=-0.034)
Client 10: weight=0.980 NEUTRAL (sim=+0.002)
Client 8: weight=1.025 REWARDED (sim=+0.061)
→ Total weight: 19.434 | Avg weight: 0.972
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 6 accuracy recorded: 0.5911 | Δacc = 0.0261
INFO :
INFO : [ROUND 7]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15352) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 69
x across cluster]
(ClientAppActor pid=15352) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=15348) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0261 → REWARDING aligned clients

[ROUND 7] CFA Trust Weights:
Client 2: weight=1.030 REWARDED (sim=+0.055)
Client 9: weight=0.904 MILD PENALTY (sim=-0.055)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 18: weight=1.008 REWARDED (sim=+0.011)
Client 14: weight=0.904 MILD PENALTY (sim=-0.027)
Client 0: weight=1.027 REWARDED (sim=+0.056)
Client 13: weight=0.904 MILD PENALTY (sim=-0.112)
Client 17: weight=1.009 NEUTRAL (sim=+0.001)
Client 5: weight=1.013 REWARDED (sim=+0.024)
Client 8: weight=1.031 REWARDED (sim=+0.057)
Client 12: weight=1.020 REWARDED (sim=+0.040)
Client 16: weight=0.904 MILD PENALTY (sim=-0.035)
Client 11: weight=1.035 REWARDED (sim=+0.072)
Client 7: weight=0.904 MILD PENALTY (sim=-0.062)
Client 3: weight=1.017 REWARDED (sim=+0.047)
Client 6: weight=0.904 MILD PENALTY (sim=-0.031)
Client 19: weight=0.904 MILD PENALTY (sim=-0.054)
Client 10: weight=0.980 NEUTRAL (sim=+0.010)
Client 4: weight=0.904 MILD PENALTY (sim=-0.047)
Client 15: weight=0.922 MILD PENALTY (sim=-0.028)
→ Total weight: 19.304 | Avg weight: 0.965
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 7 accuracy recorded: 0.6213 | Δacc = 0.0302
INFO :
INFO : [ROUND 8]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0302 → REWARDING aligned clients

[ROUND 8] CFA Trust Weights:
Client 9: weight=0.886 MILD PENALTY (sim=-0.047)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 14: weight=0.886 MILD PENALTY (sim=-0.028)
Client 19: weight=0.886 MILD PENALTY (sim=-0.053)
Client 13: weight=0.886 MILD PENALTY (sim=-0.096)
Client 8: weight=1.037 REWARDED (sim=+0.059)
Client 15: weight=0.904 MILD PENALTY (sim=-0.027)
Client 6: weight=0.886 MILD PENALTY (sim=-0.030)
Client 0: weight=1.032 REWARDED (sim=+0.049)
Client 5: weight=1.015 REWARDED (sim=+0.020)
Client 18: weight=1.008 NEUTRAL (sim=+0.008)
Client 12: weight=1.024 REWARDED (sim=+0.040)
Client 4: weight=0.886 MILD PENALTY (sim=-0.039)
Client 2: weight=1.035 REWARDED (sim=+0.051)
Client 3: weight=1.022 REWARDED (sim=+0.048)
Client 11: weight=1.043 REWARDED (sim=+0.070)
Client 17: weight=1.009 NEUTRAL (sim=-0.004)
Client 16: weight=0.886 MILD PENALTY (sim=-0.034)
Client 7: weight=0.886 MILD PENALTY (sim=-0.058)
Client 10: weight=0.981 REWARDED (sim=+0.012)
→ Total weight: 19.177 | Avg weight: 0.959
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 7
0x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 29x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 8 accuracy recorded: 0.6241 | Δacc = 0.0028
INFO :
INFO : [ROUND 9]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [ATTACK] Client 12 submitted poisoned update [repeated 8x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0028 → REWARDING aligned clients

[ROUND 9] CFA Trust Weights:
Client 7: weight=0.868 MILD PENALTY (sim=-0.053)
Client 13: weight=0.868 MILD PENALTY (sim=-0.085)
Client 1: weight=0.980 NEUTRAL (sim=+0.001)
Client 10: weight=0.983 REWARDED (sim=+0.015)
Client 5: weight=1.017 REWARDED (sim=+0.017)
Client 16: weight=0.868 MILD PENALTY (sim=-0.034)
Client 3: weight=1.027 REWARDED (sim=+0.050)
Client 15: weight=0.886 MILD PENALTY (sim=-0.026)
Client 9: weight=0.868 MILD PENALTY (sim=-0.049)
Client 12: weight=1.028 REWARDED (sim=+0.041)
Client 4: weight=0.868 MILD PENALTY (sim=-0.037)
Client 11: weight=1.050 REWARDED (sim=+0.067)
Client 18: weight=1.008 NEUTRAL (sim=+0.005)
Client 8: weight=1.042 REWARDED (sim=+0.052)
Client 19: weight=0.868 MILD PENALTY (sim=-0.050)
Client 17: weight=0.868 MILD PENALTY (sim=-0.026)
Client 14: weight=1.009 NEUTRAL (sim=-0.009)
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Client 6: weight=0.868 MILD PENALTY (sim=-0.026)
Client 2: weight=1.040 REWARDED (sim=+0.048)
Client 0: weight=1.037 REWARDED (sim=+0.043)
→ Total weight: 19.051 | Avg weight: 0.953
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(pid=gcs_server) [2026-04-21 13:24:44,864 E 15340 723285] (gcs_server) gcs_server.cc:302: Failed to establish connection to the event+metrics exporter agent. Events and metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(raylet) [2026-04-21 13:24:45,392 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12.5954 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(raylet) [2026-04-21 13:24:45,495 E 15344 723397] (raylet) main.cc:975: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
[2026-04-21 13:24:46,151 E 15335 723452] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
(ClientAppActor pid=15348) [2026-04-21 13:24:46,161 E 15348 723693] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 9 accuracy recorded: 0.6796 | Δacc = 0.0555
INFO :
INFO : [ROUND 10]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated 60x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 24x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0555 → REWARDING aligned clients

[ROUND 10] CFA Trust Weights:
Client 13: weight=0.851 MILD PENALTY (sim=-0.077)
Client 18: weight=1.008 NEUTRAL (sim=+0.004)
Client 5: weight=1.018 REWARDED (sim=+0.016)
Client 9: weight=0.851 MILD PENALTY (sim=-0.046)
Client 8: weight=1.047 REWARDED (sim=+0.050)
Client 15: weight=0.868 MILD PENALTY (sim=-0.026)
Client 19: weight=0.851 MILD PENALTY (sim=-0.050)
Client 0: weight=1.041 REWARDED (sim=+0.038)
Client 7: weight=0.851 MILD PENALTY (sim=-0.044)
Client 4: weight=0.851 MILD PENALTY (sim=-0.035)
Client 12: weight=1.032 REWARDED (sim=+0.040)
Client 17: weight=1.009 NEUTRAL (sim=-0.008)
Client 3: weight=1.032 REWARDED (sim=+0.049)
Client 6: weight=0.851 MILD PENALTY (sim=-0.025)
Client 1: weight=0.980 NEUTRAL (sim=-0.000)
Client 14: weight=0.851 MILD PENALTY (sim=-0.026)
Client 11: weight=1.056 REWARDED (sim=+0.063)
Client 10: weight=0.984 REWARDED (sim=+0.018)
Client 16: weight=0.851 MILD PENALTY (sim=-0.032)
Client 2: weight=1.045 REWARDED (sim=+0.046)
→ Total weight: 18.928 | Avg weight: 0.946
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 10 accuracy recorded: 0.6861 | Δacc = 0.0065
INFO :
INFO : [ROUND 11]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0065 → REWARDING aligned clients

[ROUND 11] CFA Trust Weights:
Client 1: weight=0.980 NEUTRAL (sim=+0.002)
Client 15: weight=0.851 MILD PENALTY (sim=-0.025)
Client 12: weight=1.036 REWARDED (sim=+0.039)
Client 6: weight=0.834 MILD PENALTY (sim=-0.025)
Client 3: weight=1.037 REWARDED (sim=+0.049)
Client 18: weight=1.008 NEUTRAL (sim=+0.000)
Client 13: weight=0.834 MILD PENALTY (sim=-0.071)
Client 8: weight=1.053 REWARDED (sim=+0.048)
Client 7: weight=0.834 MILD PENALTY (sim=-0.039)
Client 4: weight=0.834 MILD PENALTY (sim=-0.030)
Client 10: weight=0.986 REWARDED (sim=+0.019)
Client 11: weight=1.063 REWARDED (sim=+0.062)
Client 17: weight=0.989 MILD PENALTY (sim=-0.010)
Client 9: weight=0.834 MILD PENALTY (sim=-0.045)
Client 0: weight=1.045 REWARDED (sim=+0.037)
Client 5: weight=1.018 NEUTRAL (sim=+0.009)
Client 16: weight=0.834 MILD PENALTY (sim=-0.031)
Client 2: weight=1.049 REWARDED (sim=+0.043)
Client 14: weight=0.834 MILD PENALTY (sim=-0.025)
Client 19: weight=0.834 MILD PENALTY (sim=-0.046)
→ Total weight: 18.785 | Avg weight: 0.939
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 11 accuracy recorded: 0.7047 | Δacc = 0.0186
INFO :
INFO : [ROUND 12]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated 67x across cluster]
(ClientAppActor pid=15351) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 27x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0186 → REWARDING aligned clients

[ROUND 12] CFA Trust Weights:
Client 7: weight=0.817 MILD PENALTY (sim=-0.038)
Client 11: weight=1.070 REWARDED (sim=+0.063)
Client 13: weight=0.817 MILD PENALTY (sim=-0.064)
Client 8: weight=1.058 REWARDED (sim=+0.050)
Client 16: weight=0.817 MILD PENALTY (sim=-0.030)
Client 18: weight=1.008 NEUTRAL (sim=-0.002)
Client 4: weight=0.817 MILD PENALTY (sim=-0.027)
Client 5: weight=1.018 NEUTRAL (sim=+0.008)
Client 14: weight=0.817 MILD PENALTY (sim=-0.029)
Client 12: weight=1.040 REWARDED (sim=+0.036)
Client 17: weight=0.969 MILD PENALTY (sim=-0.012)
Client 0: weight=1.048 REWARDED (sim=+0.032)
Client 3: weight=1.043 REWARDED (sim=+0.053)
Client 9: weight=0.817 MILD PENALTY (sim=-0.039)
Client 15: weight=0.834 MILD PENALTY (sim=-0.025)
Client 10: weight=0.988 REWARDED (sim=+0.018)
Client 1: weight=0.980 NEUTRAL (sim=+0.002)
Client 6: weight=0.817 MILD PENALTY (sim=-0.021)
Client 2: weight=1.053 REWARDED (sim=+0.040)
Client 19: weight=0.817 MILD PENALTY (sim=-0.048)
→ Total weight: 18.645 | Avg weight: 0.932
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 13:24:55,430 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12.5972 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15350) [2026-04-21 13:24:46,147 E 15350 723841] core_worker_process.cc:825: Failed to establish connection to the metrics exporter agent. Metrics will not be exported. Exporter agent status: RpcError: Running out of retries to initialize the metrics agent. rpc_code: 14 [repeated 7x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 12 accuracy recorded: 0.7015 | Δacc = -0.0032
INFO :
INFO : [ROUND 13]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0032 → PENALISING divergent clients
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[ROUND 13] CFA Trust Weights:
Client 4: weight=0.693 PENALISED (dev=1.025)
Client 1: weight=0.980 CLEARED (dev=0.996)
Client 3: weight=1.043 CLEARED (dev=0.948)
Client 7: weight=0.692 PENALISED (dev=1.034)
Client 17: weight=0.823 PENALISED (dev=1.014)
Client 19: weight=0.691 PENALISED (dev=1.045)
Client 9: weight=0.692 PENALISED (dev=1.037)
Client 15: weight=0.707 PENALISED (dev=1.022)
Client 6: weight=0.693 PENALISED (dev=1.020)
Client 2: weight=1.053 CLEARED (dev=0.965)
Client 14: weight=0.693 PENALISED (dev=1.022)
Client 0: weight=1.048 CLEARED (dev=0.971)
Client 5: weight=1.018 CLEARED (dev=0.993)
Client 10: weight=0.988 CLEARED (dev=0.980)
Client 11: weight=1.070 CLEARED (dev=0.939)
Client 13: weight=0.690 PENALISED (dev=1.060)
Client 12: weight=1.040 CLEARED (dev=0.964)
Client 16: weight=0.692 PENALISED (dev=1.035)
Client 18: weight=1.008 CLEARED (dev=1.000)
Client 8: weight=1.058 CLEARED (dev=0.957)
→ Total weight: 17.369 | Avg weight: 0.868
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 13 accuracy recorded: 0.7254 | Δacc = 0.0239
INFO :
INFO : [ROUND 14]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 7
5x across cluster]
(ClientAppActor pid=15351) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=15349) [ATTACK] Client 17 submitted poisoned update [repeated 14x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0239 → REWARDING aligned clients

[ROUND 14] CFA Trust Weights:
Client 16: weight=0.678 MILD PENALTY (sim=-0.031)
Client 19: weight=0.677 MILD PENALTY (sim=-0.044)
Client 3: weight=1.048 REWARDED (sim=+0.051)
Client 18: weight=1.008 NEUTRAL (sim=-0.001)
Client 8: weight=1.061 REWARDED (sim=+0.035)
Client 2: weight=1.057 REWARDED (sim=+0.034)
Client 4: weight=0.679 MILD PENALTY (sim=-0.023)
Client 12: weight=1.043 REWARDED (sim=+0.035)
Client 6: weight=0.679 MILD PENALTY (sim=-0.019)
Client 13: weight=0.676 MILD PENALTY (sim=-0.055)
Client 14: weight=0.679 MILD PENALTY (sim=-0.018)
Client 7: weight=0.678 MILD PENALTY (sim=-0.031)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 11: weight=1.076 REWARDED (sim=+0.057)
Client 10: weight=0.990 REWARDED (sim=+0.020)
Client 0: weight=1.051 REWARDED (sim=+0.030)
Client 9: weight=0.678 MILD PENALTY (sim=-0.041)
Client 5: weight=1.018 NEUTRAL (sim=+0.006)
Client 15: weight=0.693 MILD PENALTY (sim=-0.025)
Client 17: weight=0.806 MILD PENALTY (sim=-0.013)
→ Total weight: 17.256 | Avg weight: 0.863
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 14 accuracy recorded: 0.7527 | Δacc = 0.0273
INFO :
INFO : [ROUND 15]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0273 → REWARDING aligned clients

[ROUND 15] CFA Trust Weights:
Client 6: weight=0.666 MILD PENALTY (sim=-0.020)
Client 13: weight=0.662 MILD PENALTY (sim=-0.055)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 7: weight=0.664 MILD PENALTY (sim=-0.029)
Client 17: weight=0.790 MILD PENALTY (sim=-0.012)
Client 16: weight=0.664 MILD PENALTY (sim=-0.029)
Client 9: weight=0.664 MILD PENALTY (sim=-0.041)
Client 14: weight=0.665 MILD PENALTY (sim=-0.020)
Client 8: weight=1.066 REWARDED (sim=+0.040)
Client 2: weight=1.061 REWARDED (sim=+0.036)
Client 10: weight=0.992 REWARDED (sim=+0.024)
Client 4: weight=0.665 MILD PENALTY (sim=-0.023)
Client 5: weight=1.018 NEUTRAL (sim=+0.007)
Client 18: weight=1.008 NEUTRAL (sim=-0.005)
Client 15: weight=0.679 MILD PENALTY (sim=-0.025)
Client 3: weight=1.053 REWARDED (sim=+0.053)
Client 12: weight=1.047 REWARDED (sim=+0.032)
Client 0: weight=1.054 REWARDED (sim=+0.027)
Client 19: weight=0.664 MILD PENALTY (sim=-0.046)
Client 11: weight=1.082 REWARDED (sim=+0.059)
→ Total weight: 17.146 | Avg weight: 0.857
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35} [repeated 76x acr
oss cluster]
(ClientAppActor pid=15351) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 30x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 15 accuracy recorded: 0.7557 | Δacc = 0.0030
INFO :
INFO : [ROUND 16]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) [ATTACK] Client 13 submitted poisoned update [repeated 10x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0030 → REWARDING aligned clients

[ROUND 16] CFA Trust Weights:
Client 2: weight=1.064 REWARDED (sim=+0.032)
Client 11: weight=1.088 REWARDED (sim=+0.057)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 0: weight=1.057 REWARDED (sim=+0.027)
Client 15: weight=0.665 MILD PENALTY (sim=-0.023)
Client 10: weight=0.995 REWARDED (sim=+0.025)
Client 7: weight=0.651 MILD PENALTY (sim=-0.027)
Client 17: weight=0.774 MILD PENALTY (sim=-0.012)
Client 18: weight=1.008 NEUTRAL (sim=-0.007)
Client 6: weight=0.652 MILD PENALTY (sim=-0.020)
Client 5: weight=1.018 NEUTRAL (sim=+0.003)
Client 14: weight=0.652 MILD PENALTY (sim=-0.021)
Client 13: weight=0.649 MILD PENALTY (sim=-0.052)
Client 19: weight=0.650 MILD PENALTY (sim=-0.045)
Client 9: weight=0.651 MILD PENALTY (sim=-0.037)
Client 3: weight=1.059 REWARDED (sim=+0.056)
Client 12: weight=1.050 REWARDED (sim=+0.032)
Client 16: weight=0.651 MILD PENALTY (sim=-0.030)
Client 4: weight=0.652 MILD PENALTY (sim=-0.025)
Client 8: weight=1.070 REWARDED (sim=+0.040)
→ Total weight: 17.038 | Avg weight: 0.852
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:25:05,448 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.5958 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
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INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 16 accuracy recorded: 0.7786 | Δacc = 0.0229
INFO :
INFO :      [ROUND 17]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0229 → REWARDING aligned clients

[ROUND 17] CFR Trust Weights:
Client 19: weight=0.637 MILD PENALTY (sim=-0.044)
Client 6: weight=0.639 MILD PENALTY (sim=-0.017)
Client 12: weight=1.053 REWARDED (sim=+0.031)
Client 5: weight=1.018 NEUTRAL (sim=+0.003)
Client 13: weight=0.636 MILD PENALTY (sim=-0.055)
Client 1: weight=0.980 NEUTRAL (sim=+0.004)
Client 17: weight=0.759 MILD PENALTY (sim=-0.012)
Client 14: weight=0.639 MILD PENALTY (sim=-0.019)
Client 16: weight=0.638 MILD PENALTY (sim=-0.034)
Client 11: weight=1.094 REWARDED (sim=+0.056)
Client 9: weight=0.638 MILD PENALTY (sim=-0.036)
Client 10: weight=0.997 REWARDED (sim=+0.022)
Client 3: weight=1.065 REWARDED (sim=+0.055)
Client 4: weight=0.639 MILD PENALTY (sim=-0.016)
Client 8: weight=1.074 REWARDED (sim=+0.041)
Client 2: weight=1.068 REWARDED (sim=+0.034)
Client 15: weight=0.652 MILD PENALTY (sim=-0.022)
Client 0: weight=1.060 REWARDED (sim=+0.027)
Client 7: weight=0.638 MILD PENALTY (sim=-0.023)
Client 18: weight=1.008 NEUTRAL (sim=-0.007)
→ Total weight: 16.934 | Avg weight: 0.847
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35} [repeated 72x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 17 accuracy recorded: 0.7905 | Δacc = 0.0119
INFO :
INFO :      [ROUND 18]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0119 → REWARDING aligned clients

[ROUND 18] CFR Trust Weights:
Client 14: weight=0.626 MILD PENALTY (sim=-0.022)
Client 4: weight=0.626 MILD PENALTY (sim=-0.014)
Client 5: weight=1.018 NEUTRAL (sim=+0.007)
Client 9: weight=0.625 MILD PENALTY (sim=-0.032)
Client 3: weight=1.071 REWARDED (sim=+0.054)
Client 0: weight=1.063 REWARDED (sim=+0.027)
Client 18: weight=1.008 NEUTRAL (sim=-0.006)
Client 15: weight=0.639 MILD PENALTY (sim=-0.024)
Client 17: weight=0.744 MILD PENALTY (sim=-0.012)
Client 1: weight=0.980 NEUTRAL (sim=+0.005)
Client 12: weight=1.056 REWARDED (sim=+0.027)
Client 11: weight=1.100 REWARDED (sim=+0.050)
Client 7: weight=0.625 MILD PENALTY (sim=-0.021)
Client 10: weight=0.999 REWARDED (sim=+0.024)
Client 8: weight=1.078 REWARDED (sim=+0.035)
Client 2: weight=1.072 REWARDED (sim=+0.035)
Client 13: weight=0.623 MILD PENALTY (sim=-0.047)
Client 6: weight=0.626 MILD PENALTY (sim=-0.018)
Client 19: weight=0.625 MILD PENALTY (sim=-0.045)
Client 16: weight=0.625 MILD PENALTY (sim=-0.030)
→ Total weight: 16.830 | Avg weight: 0.842
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 18 accuracy recorded: 0.7873 | Δacc = -0.0032
INFO :
INFO :      [ROUND 19]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 64x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 23x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = -0.0032 → PENALISING divergent clients

[ROUND 19] CFR Trust Weights:
Client 2: weight=1.072 CLEARED (dev=0.966)
Client 11: weight=1.100 CLEARED (dev=0.944)
Client 19: weight=0.528 PENALISED (dev=1.043)
Client 3: weight=1.071 CLEARED (dev=0.948)
Client 18: weight=0.856 PENALISED (dev=1.007)
Client 1: weight=0.980 CLEARED (dev=0.993)
Client 16: weight=0.529 PENALISED (dev=1.032)
Client 4: weight=0.531 PENALISED (dev=1.014)
Client 0: weight=1.063 CLEARED (dev=0.974)
Client 13: weight=0.527 PENALISED (dev=1.048)
Client 14: weight=0.531 PENALISED (dev=1.024)
Client 17: weight=0.631 PENALISED (dev=1.010)
Client 12: weight=1.056 CLEARED (dev=0.975)
Client 5: weight=1.018 CLEARED (dev=0.995)
Client 7: weight=0.530 PENALISED (dev=1.021)
Client 8: weight=1.078 CLEARED (dev=0.962)
Client 10: weight=0.999 CLEARED (dev=0.977)
Client 9: weight=0.529 PENALISED (dev=1.031)
Client 6: weight=0.531 PENALISED (dev=1.016)
Client 15: weight=0.542 PENALISED (dev=1.022)
→ Total weight: 15.704 | Avg weight: 0.785
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15350) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 13:25:15,500 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12.496 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 19 accuracy recorded: 0.8184 | Δacc = 0.0311
INFO :
INFO :      [ROUND 20]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0311 → REWARDING aligned clients

[ROUND 20] CFR Trust Weights:
Client 5: weight=1.018 NEUTRAL (sim=+0.008)
Client 16: weight=0.519 MILD PENALTY (sim=-0.029)
Client 19: weight=0.518 MILD PENALTY (sim=-0.043)
Client 15: weight=0.531 MILD PENALTY (sim=-0.021)
Client 7: weight=0.520 MILD PENALTY (sim=-0.021)
Client 2: weight=1.075 REWARDED (sim=+0.029)
Client 12: weight=1.059 REWARDED (sim=+0.028)
Client 9: weight=0.519 MILD PENALTY (sim=-0.034)
Client 8: weight=1.082 REWARDED (sim=+0.037)
Client 0: weight=1.065 REWARDED (sim=+0.024)
Client 13: weight=0.516 MILD PENALTY (sim=-0.051)
Client 1: weight=0.980 NEUTRAL (sim=+0.006)
Client 18: weight=0.856 NEUTRAL (sim=-0.007)
Client 3: weight=1.077 REWARDED (sim=+0.053)
Client 14: weight=0.520 MILD PENALTY (sim=-0.018)
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Client 17: weight=0.619 MILD PENALTY (sim=-0.011)
Client 6: weight=0.521 MILD PENALTY (sim=-0.016)
Client 10: weight=1.002 REWARDED (sim=+0.025)
Client 11: weight=1.106 REWARDED (sim=+0.053)
Client 4: weight=0.521 MILD PENALTY (sim=-0.017)
→ Total weight: 15.622 | Avg weight: 0.781
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 20 accuracy recorded: 0.8256 | Δacc = 0.0072
INFO :
INFO : [ROUND 21]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 6
8x across cluster]
(ClientAppActor pid=15351) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=15348) [ATTACK] Client 12 submitted poisoned update [repeated 11x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0072 → REWARDING aligned clients

[ROUND 21] CFR Trust Weights:
Client 4: weight=0.510 MILD PENALTY (sim=-0.017)
Client 13: weight=0.506 MILD PENALTY (sim=-0.045)
Client 5: weight=1.018 NEUTRAL (sim=+0.009)
Client 17: weight=0.619 NEUTRAL (sim=-0.009)
Client 7: weight=0.509 MILD PENALTY (sim=-0.018)
Client 6: weight=0.510 MILD PENALTY (sim=-0.015)
Client 14: weight=0.510 MILD PENALTY (sim=-0.021)
Client 16: weight=0.509 MILD PENALTY (sim=-0.039)
Client 8: weight=1.086 REWARDED (sim=+0.036)
Client 19: weight=0.507 MILD PENALTY (sim=-0.042)
Client 11: weight=1.112 REWARDED (sim=+0.056)
Client 2: weight=1.078 REWARDED (sim=+0.032)
Client 10: weight=1.004 REWARDED (sim=+0.024)
Client 15: weight=0.520 MILD PENALTY (sim=-0.021)
Client 9: weight=0.508 MILD PENALTY (sim=-0.033)
Client 3: weight=1.083 REWARDED (sim=+0.055)
Client 12: weight=1.062 REWARDED (sim=+0.028)
Client 0: weight=1.068 REWARDED (sim=+0.024)
Client 1: weight=0.980 NEUTRAL (sim=+0.009)
Client 18: weight=0.839 MILD PENALTY (sim=-0.011)
→ Total weight: 15.539 | Avg weight: 0.777
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 21 accuracy recorded: 0.8324 | Δacc = 0.0068
INFO :
INFO : [ROUND 22]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0068 → REWARDING aligned clients

[ROUND 22] CFR Trust Weights:
Client 8: weight=1.090 REWARDED (sim=+0.032)
Client 5: weight=1.018 NEUTRAL (sim=+0.007)
Client 19: weight=0.497 MILD PENALTY (sim=-0.044)
Client 0: weight=1.070 REWARDED (sim=+0.022)
Client 11: weight=1.118 REWARDED (sim=+0.054)
Client 14: weight=0.500 MILD PENALTY (sim=-0.021)
Client 7: weight=0.499 MILD PENALTY (sim=-0.015)
Client 12: weight=1.064 REWARDED (sim=+0.024)
Client 9: weight=0.498 MILD PENALTY (sim=-0.029)
Client 16: weight=0.498 MILD PENALTY (sim=-0.032)
Client 15: weight=0.510 MILD PENALTY (sim=-0.020)
Client 4: weight=0.500 MILD PENALTY (sim=-0.013)
Client 1: weight=0.980 NEUTRAL (sim=+0.009)
Client 3: weight=1.088 REWARDED (sim=+0.053)
Client 10: weight=1.007 REWARDED (sim=+0.024)
Client 17: weight=0.606 MILD PENALTY (sim=-0.011)
Client 18: weight=0.823 MILD PENALTY (sim=-0.012)
Client 6: weight=0.500 MILD PENALTY (sim=-0.013)
Client 2: weight=1.082 REWARDED (sim=+0.035)
Client 13: weight=0.496 MILD PENALTY (sim=-0.045)
→ Total weight: 15.444 | Avg weight: 0.772
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 76
x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 31x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 22 accuracy recorded: 0.8322 | Δacc = -0.0002
INFO :
INFO : [ROUND 23]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [ATTACK] Client 18 submitted poisoned update [repeated 13x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = -0.0002 → PENALISING divergent clients

[ROUND 23] CFR Trust Weights:
Client 13: weight=0.419 PENALISED (dev=1.046)
Client 3: weight=1.088 CLEARED (dev=0.946)
Client 2: weight=1.082 CLEARED (dev=0.967)
Client 19: weight=0.420 PENALISED (dev=1.042)
Client 5: weight=1.018 CLEARED (dev=0.990)
Client 8: weight=1.090 CLEARED (dev=0.966)
Client 0: weight=1.070 CLEARED (dev=0.978)
Client 15: weight=0.432 PENALISED (dev=1.024)
Client 16: weight=0.422 PENALISED (dev=1.033)
Client 4: weight=0.424 PENALISED (dev=1.010)
Client 18: weight=0.698 PENALISED (dev=1.014)
Client 6: weight=0.424 PENALISED (dev=1.015)
Client 7: weight=0.423 PENALISED (dev=1.016)
Client 14: weight=0.424 PENALISED (dev=1.021)
Client 1: weight=0.980 CLEARED (dev=0.992)
Client 11: weight=1.118 CLEARED (dev=0.945)
Client 10: weight=1.007 CLEARED (dev=0.973)
Client 9: weight=0.422 PENALISED (dev=1.031)
Client 17: weight=0.515 PENALISED (dev=1.010)
Client 12: weight=1.064 CLEARED (dev=0.974)
→ Total weight: 14.541 | Avg weight: 0.727
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:25:25.576 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4949 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 23 accuracy recorded: 0.8492 | Δacc = 0.0170
INFO :
INFO : [ROUND 24]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0170 → REWARDING aligned clients

[ROUND 24] CFR Trust Weights:
Client 10: weight=1.009 REWARDED (sim=+0.025)
Client 7: weight=0.415 MILD PENALTY (sim=-0.019)
Client 18: weight=0.698 NEUTRAL (sim=-0.008)
Client 1: weight=0.980 NEUTRAL (sim=+0.007)
Client 19: weight=0.412 MILD PENALTY (sim=-0.044)
Client 14: weight=0.415 MILD PENALTY (sim=-0.020)
Client 13: weight=0.411 MILD PENALTY (sim=-0.041)
Client 8: weight=1.093 REWARDED (sim=+0.036)

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Client 5: weight=1.020 REWARDED (sim=+0.013)
Client 6: weight=0.416 MILD PENALTY (sim=-0.016)
Client 15: weight=0.424 MILD PENALTY (sim=-0.022)
Client 2: weight=1.085 REWARDED (sim=+0.032)
Client 12: weight=1.067 REWARDED (sim=+0.024)
Client 11: weight=1.123 REWARDED (sim=+0.052)
Client 17: weight=0.504 MILD PENALTY (sim=-0.011)
Client 9: weight=0.414 MILD PENALTY (sim=-0.033)
Client 16: weight=0.413 MILD PENALTY (sim=-0.034)
Client 0: weight=1.072 REWARDED (sim=+0.020)
Client 4: weight=0.416 MILD PENALTY (sim=-0.014)
Client 3: weight=1.094 REWARDED (sim=+0.055)
→ Total weight: 14.482 | Avg weight: 0.724
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 7
7x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 30x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 24 accuracy recorded: 0.8438 | Δacc = -0.0054
INFO :
INFO : [ROUND 25]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15350) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0054 → PENALISING divergent clients

[ROUND 25] CFA Trust Weights:
Client 2: weight=1.085 CLEARED (dev=0.969)
Client 15: weight=0.359 PENALISED (dev=1.023)
Client 1: weight=0.980 CLEARED (dev=0.990)
Client 7: weight=0.352 PENALISED (dev=1.012)
Client 16: weight=0.350 PENALISED (dev=1.032)
Client 8: weight=1.093 CLEARED (dev=0.965)
Client 0: weight=1.072 CLEARED (dev=0.976)
Client 6: weight=0.353 PENALISED (dev=1.015)
Client 4: weight=0.353 PENALISED (dev=1.010)
Client 3: weight=1.094 CLEARED (dev=0.945)
Client 11: weight=1.123 CLEARED (dev=0.946)
Client 17: weight=0.428 PENALISED (dev=1.013)
Client 9: weight=0.350 PENALISED (dev=1.028)
Client 5: weight=1.020 CLEARED (dev=0.998)
Client 10: weight=1.009 CLEARED (dev=0.976)
Client 18: weight=0.592 PENALISED (dev=1.011)
Client 19: weight=0.348 PENALISED (dev=1.042)
Client 13: weight=0.347 PENALISED (dev=1.044)
Client 12: weight=1.067 CLEARED (dev=0.976)
Client 14: weight=0.352 PENALISED (dev=1.022)
→ Total weight: 13.730 | Avg weight: 0.687
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 25 accuracy recorded: 0.8610 | Δacc = 0.0172
INFO :
INFO : [ROUND 26]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0172 → REWARDING aligned clients

[ROUND 26] CFA Trust Weights:
Client 4: weight=0.346 MILD PENALTY (sim=-0.012)
Client 6: weight=0.346 MILD PENALTY (sim=-0.012)
Client 10: weight=1.012 REWARDED (sim=+0.025)
Client 16: weight=0.343 MILD PENALTY (sim=-0.033)
Client 11: weight=1.129 REWARDED (sim=+0.052)
Client 12: weight=1.069 REWARDED (sim=+0.020)
Client 9: weight=0.343 MILD PENALTY (sim=-0.026)
Client 13: weight=0.341 MILD PENALTY (sim=-0.044)
Client 5: weight=1.021 REWARDED (sim=+0.013)
Client 15: weight=0.352 MILD PENALTY (sim=-0.023)
Client 1: weight=0.980 NEUTRAL (sim=+0.005)
Client 8: weight=1.097 REWARDED (sim=+0.037)
Client 17: weight=0.420 MILD PENALTY (sim=-0.012)
Client 2: weight=1.089 REWARDED (sim=+0.033)
Client 0: weight=1.075 REWARDED (sim=+0.023)
Client 14: weight=0.345 MILD PENALTY (sim=-0.018)
Client 7: weight=0.345 MILD PENALTY (sim=-0.011)
Client 19: weight=0.342 MILD PENALTY (sim=-0.044)
Client 18: weight=0.581 MILD PENALTY (sim=-0.010)
Client 3: weight=1.100 REWARDED (sim=+0.054)
→ Total weight: 13.674 | Avg weight: 0.684
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
74x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 26 accuracy recorded: 0.8669 | Δacc = 0.0059
INFO :
INFO : [ROUND 27]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:25:35,682 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4948 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0059 → REWARDING aligned clients

[ROUND 27] CFA Trust Weights:
Client 6: weight=0.339 MILD PENALTY (sim=-0.013)
Client 8: weight=1.101 REWARDED (sim=+0.032)
Client 10: weight=1.014 REWARDED (sim=+0.028)
Client 12: weight=1.072 REWARDED (sim=+0.023)
Client 7: weight=0.345 NEUTRAL (sim=-0.010)
Client 0: weight=1.077 REWARDED (sim=+0.025)
Client 5: weight=1.022 REWARDED (sim=+0.014)
Client 11: weight=1.136 REWARDED (sim=+0.056)
Client 19: weight=0.335 MILD PENALTY (sim=-0.047)
Client 18: weight=0.569 MILD PENALTY (sim=-0.015)
Client 16: weight=0.336 MILD PENALTY (sim=-0.036)
Client 13: weight=0.334 MILD PENALTY (sim=-0.048)
Client 17: weight=0.411 MILD PENALTY (sim=-0.012)
Client 15: weight=0.345 MILD PENALTY (sim=-0.020)
Client 14: weight=0.338 MILD PENALTY (sim=-0.020)
Client 3: weight=1.106 REWARDED (sim=+0.055)
Client 2: weight=1.093 REWARDED (sim=+0.034)
Client 4: weight=0.339 MILD PENALTY (sim=-0.012)
Client 9: weight=0.336 MILD PENALTY (sim=-0.024)
Client 1: weight=0.981 REWARDED (sim=+0.013)
→ Total weight: 13.629 | Avg weight: 0.681
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 27 accuracy recorded: 0.8805 | Δacc = 0.0136
INFO :
INFO : [ROUND 28]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 8 | Total: 3682 samples | Distribution: {0: 127, 1: 396, 2: 54, 3: 59, 4: 801, 5: 23, 6: 954, 7: 232, 8: 978, 9: 58} [repeated 6
7x across cluster]
(ClientAppActor pid=15350) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 26x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0136 → REWARDING aligned clients
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[ROUND 28] CFA Trust Weights:
Client 17: weight=0.403 MILD PENALTY (sim=-0.015)
Client 2: weight=1.096 REWARDED (sim=+0.032)
Client 10: weight=1.017 REWARDED (sim=+0.028)
Client 14: weight=0.331 MILD PENALTY (sim=-0.026)
Client 11: weight=1.141 REWARDED (sim=+0.048)
Client 1: weight=0.982 REWARDED (sim=+0.010)
Client 9: weight=0.330 MILD PENALTY (sim=-0.021)
Client 0: weight=1.080 REWARDED (sim=+0.025)
Client 3: weight=1.113 REWARDED (sim=+0.057)
Client 6: weight=0.339 NEUTRAL (sim=-0.010)
Client 19: weight=0.328 MILD PENALTY (sim=-0.045)
Client 4: weight=0.339 NEUTRAL (sim=-0.006)
Client 5: weight=1.024 REWARDED (sim=+0.013)
Client 8: weight=1.105 REWARDED (sim=+0.036)
Client 18: weight=0.558 MILD PENALTY (sim=-0.012)
Client 13: weight=0.327 MILD PENALTY (sim=-0.044)
Client 16: weight=0.329 MILD PENALTY (sim=-0.031)
Client 15: weight=0.338 MILD PENALTY (sim=-0.020)
Client 12: weight=1.074 REWARDED (sim=+0.021)
Client 7: weight=0.345 NEUTRAL (sim=-0.007)
→ Total weight: 13.599 | Avg weight: 0.680
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15352) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 28 accuracy recorded: 0.8793 | Δacc = -0.0012
INFO :
INFO : [ROUND 29]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = -0.0012 → PENALISING divergent clients

[ROUND 29] CFA Trust Weights:
Client 0: weight=1.080 CLEARED (dev=0.977)
Client 5: weight=1.024 CLEARED (dev=0.988)
Client 4: weight=0.288 PENALISED (dev=1.010)
Client 17: weight=0.342 PENALISED (dev=1.011)
Client 16: weight=0.279 PENALISED (dev=1.034)
Client 12: weight=1.074 CLEARED (dev=0.979)
Client 1: weight=0.982 CLEARED (dev=0.991)
Client 7: weight=0.293 PENALISED (dev=1.010)
Client 2: weight=1.096 CLEARED (dev=0.965)
Client 13: weight=0.276 PENALISED (dev=1.045)
Client 19: weight=0.277 PENALISED (dev=1.044)
Client 8: weight=1.105 CLEARED (dev=0.963)
Client 6: weight=0.288 PENALISED (dev=1.009)
Client 18: weight=0.473 PENALISED (dev=1.011)
Client 11: weight=1.141 CLEARED (dev=0.947)
Client 15: weight=0.286 PENALISED (dev=1.021)
Client 14: weight=0.281 PENALISED (dev=1.024)
Client 10: weight=1.017 CLEARED (dev=0.978)
Client 3: weight=1.113 CLEARED (dev=0.948)
Client 9: weight=0.279 PENALISED (dev=1.023)
→ Total weight: 12.995 | Avg weight: 0.650
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 29 accuracy recorded: 0.8801 | Δacc = 0.0008
INFO :
INFO : [ROUND 30]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:25:45,758 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4858 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15349) [NON-IID] Client 12 | Total: 4753 samples | Distribution: {0: 146, 1: 462, 2: 40, 3: 42, 4: 647, 5: 939, 6: 208, 7: 12, 8: 560, 9: 1697} [repeated
d 66x across cluster]
(ClientAppActor pid=15349) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 28x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 18 submitted poisoned update [repeated 14x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0008 → REWARDING aligned clients

[ROUND 30] CFA Trust Weights:
Client 16: weight=0.273 MILD PENALTY (sim=-0.035)
Client 9: weight=0.274 MILD PENALTY (sim=-0.022)
Client 5: weight=1.025 REWARDED (sim=+0.016)
Client 2: weight=1.100 REWARDED (sim=+0.032)
Client 6: weight=0.288 NEUTRAL (sim=-0.009)
Client 14: weight=0.275 MILD PENALTY (sim=-0.025)
Client 13: weight=0.271 MILD PENALTY (sim=-0.043)
Client 12: weight=1.076 REWARDED (sim=+0.021)
Client 11: weight=1.147 REWARDED (sim=+0.052)
Client 18: weight=0.464 MILD PENALTY (sim=-0.015)
Client 7: weight=0.293 NEUTRAL (sim=-0.007)
Client 3: weight=1.119 REWARDED (sim=+0.055)
Client 4: weight=0.288 NEUTRAL (sim=-0.005)
Client 10: weight=1.020 REWARDED (sim=+0.027)
Client 8: weight=1.109 REWARDED (sim=+0.035)
Client 19: weight=0.272 MILD PENALTY (sim=-0.042)
Client 17: weight=0.335 MILD PENALTY (sim=-0.012)
Client 1: weight=0.983 REWARDED (sim=+0.011)
Client 0: weight=1.083 REWARDED (sim=+0.024)
Client 15: weight=0.281 MILD PENALTY (sim=-0.025)
→ Total weight: 12.975 | Avg weight: 0.649
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:25:45,758 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4858 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 30 accuracy recorded: 0.8803 | Δacc = 0.0002
INFO :
INFO : [ROUND 31]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0002 → REWARDING aligned clients

[ROUND 31] CFA Trust Weights:
Client 10: weight=1.023 REWARDED (sim=+0.026)
Client 15: weight=0.275 MILD PENALTY (sim=-0.029)
Client 2: weight=1.103 REWARDED (sim=+0.034)
Client 13: weight=0.266 MILD PENALTY (sim=-0.041)
Client 14: weight=0.270 MILD PENALTY (sim=-0.023)
Client 6: weight=0.288 NEUTRAL (sim=-0.009)
Client 18: weight=0.455 MILD PENALTY (sim=-0.018)
Client 0: weight=1.085 REWARDED (sim=+0.026)
Client 1: weight=0.985 REWARDED (sim=+0.012)
Client 9: weight=0.268 MILD PENALTY (sim=-0.018)
Client 11: weight=1.153 REWARDED (sim=+0.054)
Client 17: weight=0.328 MILD PENALTY (sim=-0.012)
Client 7: weight=0.293 NEUTRAL (sim=-0.007)
Client 8: weight=1.113 REWARDED (sim=+0.034)
Client 16: weight=0.268 MILD PENALTY (sim=-0.033)
Client 19: weight=0.266 MILD PENALTY (sim=-0.044)
Client 4: weight=0.288 NEUTRAL (sim=-0.004)
Client 3: weight=1.125 REWARDED (sim=+0.056)
Client 12: weight=1.078 REWARDED (sim=+0.019)
Client 5: weight=1.027 REWARDED (sim=+0.018)
→ Total weight: 12.956 | Avg weight: 0.648
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
69x across cluster]
(ClientAppActor pid=15352) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 27x across cluster]
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(ClientAppActor pid=15350) [ATTACK] Client 18 submitted poisoned update [repeated 10x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 31 accuracy recorded: 0.8864 | Δacc = 0.0061
INFO :
INFO : [ROUND 32]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0061 → REWARDING aligned clients

[ROUND 32] CFR Trust Weights:
Client 0: weight=1.088 REWARDED (sim=+0.026)
Client 14: weight=0.264 MILD PENALTY (sim=-0.022)
Client 11: weight=1.159 REWARDED (sim=+0.053)
Client 1: weight=0.986 REWARDED (sim=+0.011)
Client 13: weight=0.260 MILD PENALTY (sim=-0.050)
Client 4: weight=0.288 NEUTRAL (sim=-0.007)
Client 8: weight=1.117 REWARDED (sim=+0.038)
Client 9: weight=0.263 MILD PENALTY (sim=-0.018)
Client 15: weight=0.270 MILD PENALTY (sim=-0.019)
Client 18: weight=0.445 MILD PENALTY (sim=-0.019)
Client 5: weight=1.029 REWARDED (sim=+0.018)
Client 19: weight=0.261 MILD PENALTY (sim=-0.044)
Client 17: weight=0.322 MILD PENALTY (sim=-0.011)
Client 3: weight=1.131 REWARDED (sim=+0.053)
Client 6: weight=0.288 NEUTRAL (sim=-0.008)
Client 2: weight=1.108 REWARDED (sim=+0.038)
Client 10: weight=1.026 REWARDED (sim=+0.028)
Client 7: weight=0.293 NEUTRAL (sim=-0.006)
Client 12: weight=1.080 REWARDED (sim=+0.019)
Client 16: weight=0.263 MILD PENALTY (sim=-0.040)
→ Total weight: 12.939 | Avg weight: 0.647
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 46
x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 19x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 32 accuracy recorded: 0.8860 | Δacc = -0.0004
INFO :
INFO : [ROUND 33]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15352) [ATTACK] Client 14 submitted poisoned update [repeated 8x across cluster]
(raylet) [2026-04-21 13:25:55.840 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4845 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = -0.0004 → PENALISING divergent clients

[ROUND 33] CFR Trust Weights:
Client 18: weight=0.378 PENALISED (dev=1.018)
Client 4: weight=0.244 PENALISED (dev=1.005)
Client 19: weight=0.221 PENALISED (dev=1.043)
Client 0: weight=1.088 CLEARED (dev=0.974)
Client 7: weight=0.249 PENALISED (dev=1.004)
Client 15: weight=0.228 PENALISED (dev=1.026)
Client 3: weight=1.131 CLEARED (dev=0.947)
Client 16: weight=0.222 PENALISED (dev=1.041)
Client 9: weight=0.223 PENALISED (dev=1.017)
Client 14: weight=0.224 PENALISED (dev=1.020)
Client 2: weight=1.108 CLEARED (dev=0.968)
Client 5: weight=1.029 CLEARED (dev=0.979)
Client 6: weight=0.244 PENALISED (dev=1.005)
Client 11: weight=1.159 CLEARED (dev=0.945)
Client 1: weight=0.986 CLEARED (dev=0.988)
Client 8: weight=1.117 CLEARED (dev=0.963)
Client 13: weight=0.220 PENALISED (dev=1.042)
Client 10: weight=1.026 CLEARED (dev=0.971)
Client 12: weight=1.080 CLEARED (dev=0.981)
Client 17: weight=0.273 PENALISED (dev=1.017)
→ Total weight: 12.449 | Avg weight: 0.622
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 33 accuracy recorded: 0.8889 | Δacc = 0.0029
INFO :
INFO : [ROUND 34]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0029 → REWARDING aligned clients

[ROUND 34] CFR Trust Weights:
Client 15: weight=0.224 MILD PENALTY (sim=-0.025)
Client 9: weight=0.219 MILD PENALTY (sim=-0.013)
Client 12: weight=1.082 REWARDED (sim=+0.018)
Client 4: weight=0.244 NEUTRAL (sim=-0.003)
Client 6: weight=0.244 NEUTRAL (sim=-0.005)
Client 1: weight=0.987 REWARDED (sim=+0.012)
Client 0: weight=1.091 REWARDED (sim=+0.027)
Client 7: weight=0.249 NEUTRAL (sim=-0.007)
Client 16: weight=0.218 MILD PENALTY (sim=-0.034)
Client 18: weight=0.370 MILD PENALTY (sim=-0.021)
Client 3: weight=1.137 REWARDED (sim=+0.054)
Client 19: weight=0.216 MILD PENALTY (sim=-0.049)
Client 5: weight=1.031 REWARDED (sim=+0.020)
Client 14: weight=0.219 MILD PENALTY (sim=-0.025)
Client 2: weight=1.111 REWARDED (sim=+0.035)
Client 11: weight=1.166 REWARDED (sim=+0.056)
Client 17: weight=0.268 MILD PENALTY (sim=-0.012)
Client 8: weight=1.121 REWARDED (sim=+0.035)
Client 13: weight=0.216 MILD PENALTY (sim=-0.046)
Client 10: weight=1.029 REWARDED (sim=+0.031)
→ Total weight: 12.441 | Avg weight: 0.622
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
70x across cluster]
(ClientAppActor pid=15351) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=15350) [ATTACK] Client 16 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 34 accuracy recorded: 0.8971 | Δacc = 0.0082
INFO :
INFO : [ROUND 35]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0082 → REWARDING aligned clients

[ROUND 35] CFR Trust Weights:
Client 10: weight=1.032 REWARDED (sim=+0.030)
Client 6: weight=0.244 NEUTRAL (sim=-0.006)
Client 7: weight=0.249 NEUTRAL (sim=-0.002)
Client 17: weight=0.262 MILD PENALTY (sim=-0.015)
Client 19: weight=0.212 MILD PENALTY (sim=-0.042)
Client 16: weight=0.213 MILD PENALTY (sim=-0.042)
Client 12: weight=1.084 REWARDED (sim=+0.015)
Client 15: weight=0.219 MILD PENALTY (sim=-0.023)
Client 1: weight=0.988 REWARDED (sim=+0.012)
Client 9: weight=0.214 MILD PENALTY (sim=-0.014)
Client 8: weight=1.125 REWARDED (sim=+0.036)
Client 11: weight=1.172 REWARDED (sim=+0.055)
Client 4: weight=0.244 NEUTRAL (sim=-0.002)
Client 14: weight=0.215 MILD PENALTY (sim=-0.027)
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Client 5: weight=1.033 REWARDED (sim=+0.022)
Client 13: weight=0.211 MILD PENALTY (sim=-0.046)
Client 0: weight=1.094 REWARDED (sim=+0.026)
Client 3: weight=1.144 REWARDED (sim=+0.057)
Client 2: weight=1.115 REWARDED (sim=+0.038)
Client 18: weight=0.363 MILD PENALTY (sim=-0.018)
→ Total weight: 12.434 | Avg weight: 0.622
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 35 accuracy recorded: 0.8985 | Δacc = 0.0014
INFO :
INFO : [ROUND 36]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0014 → REWARDING aligned clients

[ROUND 36] CFA Trust Weights:
Client 13: weight=0.207 MILD PENALTY (sim=-0.046)
Client 0: weight=1.097 REWARDED (sim=+0.029)
Client 2: weight=1.120 REWARDED (sim=+0.037)
Client 7: weight=0.249 NEUTRAL (sim=-0.003)
Client 15: weight=0.215 MILD PENALTY (sim=-0.027)
Client 6: weight=0.244 NEUTRAL (sim=-0.006)
Client 5: weight=1.036 REWARDED (sim=+0.025)
Client 11: weight=1.178 REWARDED (sim=+0.053)
Client 14: weight=0.211 MILD PENALTY (sim=-0.027)
Client 3: weight=1.150 REWARDED (sim=+0.056)
Client 8: weight=1.129 REWARDED (sim=+0.036)
Client 17: weight=0.262 NEUTRAL (sim=-0.009)
Client 18: weight=0.356 MILD PENALTY (sim=-0.022)
Client 12: weight=1.085 REWARDED (sim=+0.016)
Client 9: weight=0.210 MILD PENALTY (sim=-0.016)
Client 4: weight=0.244 NEUTRAL (sim=-0.004)
Client 19: weight=0.208 MILD PENALTY (sim=-0.046)
Client 16: weight=0.209 MILD PENALTY (sim=-0.035)
Client 1: weight=0.989 REWARDED (sim=+0.012)
Client 10: weight=1.035 REWARDED (sim=+0.029)
→ Total weight: 12.434 | Avg weight: 0.622
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 7
5x across cluster]
(ClientAppActor pid=15350) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=15349) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 13:26:05.942 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4881 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 36 accuracy recorded: 0.9022 | Δacc = 0.0037
INFO :
INFO : [ROUND 37]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0037 → REWARDING aligned clients

[ROUND 37] CFA Trust Weights:
Client 4: weight=0.244 NEUTRAL (sim=-0.003)
Client 5: weight=1.038 REWARDED (sim=+0.022)
Client 11: weight=1.184 REWARDED (sim=+0.044)
Client 2: weight=1.124 REWARDED (sim=+0.035)
Client 10: weight=1.038 REWARDED (sim=+0.030)
Client 19: weight=0.204 MILD PENALTY (sim=-0.045)
Client 13: weight=0.203 MILD PENALTY (sim=-0.051)
Client 6: weight=0.244 NEUTRAL (sim=-0.003)
Client 9: weight=0.206 MILD PENALTY (sim=-0.011)
Client 7: weight=0.249 NEUTRAL (sim=-0.005)
Client 1: weight=0.990 REWARDED (sim=+0.013)
Client 16: weight=0.205 MILD PENALTY (sim=-0.034)
Client 8: weight=1.132 REWARDED (sim=+0.025)
Client 18: weight=0.348 MILD PENALTY (sim=-0.015)
Client 0: weight=1.100 REWARDED (sim=+0.029)
Client 14: weight=0.207 MILD PENALTY (sim=-0.027)
Client 12: weight=1.088 REWARDED (sim=+0.021)
Client 15: weight=0.211 MILD PENALTY (sim=-0.027)
Client 17: weight=0.257 MILD PENALTY (sim=-0.013)
Client 3: weight=1.157 REWARDED (sim=+0.057)
→ Total weight: 12.427 | Avg weight: 0.621
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 37 accuracy recorded: 0.9015 | Δacc = -0.0007
INFO :
INFO : [ROUND 38]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 7
8x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 31x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0007 → PENALISING divergent clients

[ROUND 38] CFA Trust Weights:
Client 9: weight=0.175 PENALISED (dev=1.014)
Client 11: weight=1.184 CLEARED (dev=0.950)
Client 16: weight=0.173 PENALISED (dev=1.037)
Client 13: weight=0.172 PENALISED (dev=1.043)
Client 6: weight=0.208 PENALISED (dev=1.005)
Client 2: weight=1.124 CLEARED (dev=0.964)
Client 19: weight=0.172 PENALISED (dev=1.042)
Client 15: weight=0.178 PENALISED (dev=1.026)
Client 0: weight=1.100 CLEARED (dev=0.970)
Client 4: weight=0.208 PENALISED (dev=1.000)
Client 8: weight=1.132 CLEARED (dev=0.963)
Client 18: weight=0.295 PENALISED (dev=1.021)
Client 12: weight=1.088 CLEARED (dev=0.986)
Client 14: weight=0.175 PENALISED (dev=1.028)
Client 17: weight=0.218 PENALISED (dev=1.013)
Client 1: weight=0.990 CLEARED (dev=0.985)
Client 5: weight=1.038 CLEARED (dev=0.975)
Client 7: weight=0.211 PENALISED (dev=1.003)
Client 3: weight=1.157 CLEARED (dev=0.946)
Client 10: weight=1.038 CLEARED (dev=0.969)
→ Total weight: 12.034 | Avg weight: 0.602
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 38 accuracy recorded: 0.8977 | Δacc = -0.0038
INFO :
INFO : [ROUND 39]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0038 → PENALISING divergent clients

[ROUND 39] CFA Trust Weights:
Client 16: weight=0.147 PENALISED (dev=1.035)
Client 3: weight=1.157 CLEARED (dev=0.945)
Client 10: weight=1.038 CLEARED (dev=0.968)
Client 11: weight=1.184 CLEARED (dev=0.949)
Client 8: weight=1.132 CLEARED (dev=0.966)
Client 15: weight=0.151 PENALISED (dev=1.024)
Client 1: weight=0.990 CLEARED (dev=0.987)

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Client 14: weight=0.148 PENALISED (dev=1.025)
Client 0: weight=1.100 CLEARED (dev=0.973)
Client 18: weight=0.250 PENALISED (dev=1.019)
Client 13: weight=0.145 PENALISED (dev=1.045)
Client 9: weight=0.148 PENALISED (dev=1.015)
Client 2: weight=1.124 CLEARED (dev=0.964)
Client 17: weight=0.185 PENALISED (dev=1.015)
Client 12: weight=1.088 CLEARED (dev=0.987)
Client 4: weight=0.176 PENALISED (dev=1.003)
Client 6: weight=0.176 PENALISED (dev=1.005)
Client 5: weight=1.038 CLEARED (dev=0.973)
Client 7: weight=0.180 PENALISED (dev=1.001)
Client 19: weight=0.145 PENALISED (dev=1.047)
→ Total weight: 11.702 | Avg weight: 0.585
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 39 accuracy recorded: 0.9061 | Δacc = 0.0084
INFO :
INFO : [ROUND 40]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 6
2x across cluster]
(ClientAppActor pid=15354) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 25x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 19 submitted poisoned update [repeated 9x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0084 → REWARDING aligned clients

[ROUND 40] CFA Trust Weights:
Client 18: weight=0.245 MILD PENALTY (sim=-0.019)
Client 2: weight=1.128 REWARDED (sim=+0.037)
Client 19: weight=0.143 MILD PENALTY (sim=-0.044)
Client 4: weight=0.176 NEUTRAL (sim=-0.001)
Client 8: weight=1.136 REWARDED (sim=+0.036)
Client 11: weight=1.190 REWARDED (sim=+0.055)
Client 9: weight=0.145 MILD PENALTY (sim=-0.015)
Client 17: weight=0.181 MILD PENALTY (sim=-0.013)
Client 7: weight=0.180 NEUTRAL (sim=+0.001)
Client 12: weight=1.089 REWARDED (sim=+0.016)
Client 15: weight=0.148 MILD PENALTY (sim=-0.030)
Client 3: weight=1.163 REWARDED (sim=+0.056)
Client 1: weight=0.992 REWARDED (sim=+0.017)
Client 16: weight=0.144 MILD PENALTY (sim=-0.047)
Client 6: weight=0.176 NEUTRAL (sim=-0.001)
Client 13: weight=0.142 MILD PENALTY (sim=-0.041)
Client 14: weight=0.145 MILD PENALTY (sim=-0.026)
Client 0: weight=1.103 REWARDED (sim=+0.028)
Client 5: weight=1.041 REWARDED (sim=+0.025)
Client 10: weight=1.041 REWARDED (sim=+0.029)
→ Total weight: 11.709 | Avg weight: 0.585
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:26:16,011 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4949 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 40 accuracy recorded: 0.9044 | Δacc = -0.0017
INFO :
INFO : [ROUND 41]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0017 → PENALISING divergent clients

[ROUND 41] CFA Trust Weights:
Client 6: weight=0.150 PENALISED (dev=1.002)
Client 14: weight=0.123 PENALISED (dev=1.025)
Client 12: weight=1.089 CLEARED (dev=0.990)
Client 16: weight=0.122 PENALISED (dev=1.036)
Client 19: weight=0.121 PENALISED (dev=1.039)
Client 9: weight=0.123 PENALISED (dev=1.013)
Client 8: weight=1.136 CLEARED (dev=0.961)
Client 17: weight=0.154 PENALISED (dev=1.020)
Client 7: weight=0.153 PENALISED (dev=1.001)
Client 10: weight=1.041 CLEARED (dev=0.969)
Client 18: weight=0.208 PENALISED (dev=1.020)
Client 3: weight=1.163 CLEARED (dev=0.948)
Client 2: weight=1.128 CLEARED (dev=0.962)
Client 15: weight=0.126 PENALISED (dev=1.027)
Client 11: weight=1.190 CLEARED (dev=0.947)
Client 0: weight=1.103 CLEARED (dev=0.970)
Client 1: weight=0.992 CLEARED (dev=0.988)
Client 5: weight=1.041 CLEARED (dev=0.971)
Client 4: weight=0.150 PENALISED (dev=1.001)
Client 13: weight=0.120 PENALISED (dev=1.043)
→ Total weight: 11.431 | Avg weight: 0.572
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 41 accuracy recorded: 0.9134 | Δacc = 0.0090
INFO :
INFO : [ROUND 42]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 80x a
cross cluster]
(ClientAppActor pid=15347) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15349) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0090 → REWARDING aligned clients

[ROUND 42] CFA Trust Weights:
Client 4: weight=0.150 NEUTRAL (sim=+0.000)
Client 11: weight=1.196 REWARDED (sim=+0.051)
Client 10: weight=1.044 REWARDED (sim=+0.030)
Client 0: weight=1.106 REWARDED (sim=+0.029)
Client 1: weight=0.994 REWARDED (sim=+0.016)
Client 2: weight=1.132 REWARDED (sim=+0.037)
Client 7: weight=0.153 NEUTRAL (sim=+0.001)
Client 6: weight=0.150 NEUTRAL (sim=-0.003)
Client 19: weight=0.118 MILD PENALTY (sim=-0.049)
Client 5: weight=1.044 REWARDED (sim=+0.030)
Client 18: weight=0.204 MILD PENALTY (sim=-0.024)
Client 9: weight=0.121 MILD PENALTY (sim=-0.012)
Client 13: weight=0.118 MILD PENALTY (sim=-0.042)
Client 14: weight=0.121 MILD PENALTY (sim=-0.021)
Client 16: weight=0.119 MILD PENALTY (sim=-0.035)
Client 15: weight=0.123 MILD PENALTY (sim=-0.028)
Client 3: weight=1.169 REWARDED (sim=+0.055)
Client 8: weight=1.140 REWARDED (sim=+0.038)
Client 17: weight=0.151 MILD PENALTY (sim=-0.020)
Client 12: weight=1.091 REWARDED (sim=+0.013)
→ Total weight: 11.442 | Avg weight: 0.572
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 42 accuracy recorded: 0.9120 | Δacc = -0.0014
INFO :
INFO : [ROUND 43]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0014 → PENALISING divergent clients

[ROUND 43] CFA Trust Weights:
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Client 9: weight=0.102 PENALISED (dev=1.009)
Client 11: weight=1.196 CLEARED (dev=0.956)
Client 5: weight=1.044 CLEARED (dev=0.969)
Client 12: weight=1.091 CLEARED (dev=0.986)
Client 8: weight=1.140 CLEARED (dev=0.963)
Client 2: weight=1.132 CLEARED (dev=0.959)
Client 6: weight=0.127 PENALISED (dev=1.001)
Client 7: weight=0.153 CLEARED (dev=0.995)
Client 1: weight=0.994 CLEARED (dev=0.984)
Client 4: weight=0.127 PENALISED (dev=0.999)
Client 17: weight=0.128 PENALISED (dev=1.017)
Client 10: weight=1.044 CLEARED (dev=0.969)
Client 15: weight=0.104 PENALISED (dev=1.030)
Client 13: weight=0.100 PENALISED (dev=1.048)
Client 0: weight=1.106 CLEARED (dev=0.969)
Client 3: weight=1.169 CLEARED (dev=0.944)
Client 14: weight=0.102 PENALISED (dev=1.029)
Client 19: weight=0.100 PENALISED (dev=1.046)
Client 16: weight=0.101 PENALISED (dev=1.034)
Client 18: weight=0.173 PENALISED (dev=1.025)
→ Total weight: 11.233 | Avg weight: 0.562
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 43 accuracy recorded: 0.9058 | Δacc = -0.0062
INFO :
INFO : [ROUND 44]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
80x across cluster]
(ClientAppActor pid=15351) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 33x across cluster]
(ClientAppActor pid=15348) [ATTACK] Client 15 submitted poisoned update [repeated 15x across cluster]
(raylet) [2026-04-21 13:26:26,063 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4946 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0062 → PENALISING divergent clients

[ROUND 44] CFA Trust Weights:
Client 1: weight=0.994 CLEARED (dev=0.979)
Client 6: weight=0.108 PENALISED (dev=1.000)
Client 7: weight=0.130 PENALISED (dev=0.996)
Client 16: weight=0.100 PENALISED (dev=1.041)
Client 18: weight=0.146 PENALISED (dev=1.029)
Client 11: weight=1.196 CLEARED (dev=0.943)
Client 15: weight=0.100 PENALISED (dev=1.030)
Client 9: weight=0.100 PENALISED (dev=1.005)
Client 0: weight=1.106 CLEARED (dev=0.968)
Client 17: weight=0.108 PENALISED (dev=1.019)
Client 13: weight=0.100 PENALISED (dev=1.040)
Client 19: weight=0.100 PENALISED (dev=1.045)
Client 14: weight=0.100 PENALISED (dev=1.028)
Client 2: weight=1.132 CLEARED (dev=0.962)
Client 3: weight=1.169 CLEARED (dev=0.942)
Client 8: weight=1.140 CLEARED (dev=0.961)
Client 4: weight=0.108 PENALISED (dev=0.999)
Client 12: weight=1.091 CLEARED (dev=0.991)
Client 10: weight=1.044 CLEARED (dev=0.967)
Client 5: weight=1.044 CLEARED (dev=0.967)
→ Total weight: 11.117 | Avg weight: 0.556
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 44 accuracy recorded: 0.9157 | Δacc = 0.0099
INFO :
INFO : [ROUND 45]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0099 → REWARDING aligned clients

[ROUND 45] CFA Trust Weights:
Client 4: weight=0.108 NEUTRAL (sim=+0.006)
Client 8: weight=1.144 REWARDED (sim=+0.036)
Client 19: weight=0.100 MILD PENALTY (sim=-0.048)
Client 0: weight=1.111 REWARDED (sim=+0.037)
Client 16: weight=0.100 MILD PENALTY (sim=-0.043)
Client 1: weight=0.996 REWARDED (sim=+0.020)
Client 10: weight=1.048 REWARDED (sim=+0.035)
Client 6: weight=0.108 NEUTRAL (sim=+0.001)
Client 11: weight=1.203 REWARDED (sim=+0.057)
Client 14: weight=0.100 MILD PENALTY (sim=-0.034)
Client 2: weight=1.136 REWARDED (sim=+0.040)
Client 13: weight=0.100 MILD PENALTY (sim=-0.047)
Client 15: weight=0.100 MILD PENALTY (sim=-0.028)
Client 12: weight=1.091 NEUTRAL (sim=+0.008)
Client 17: weight=0.106 MILD PENALTY (sim=-0.016)
Client 18: weight=0.143 MILD PENALTY (sim=-0.026)
Client 5: weight=1.047 REWARDED (sim=+0.034)
Client 9: weight=0.100 NEUTRAL (sim=-0.006)
Client 7: weight=0.130 NEUTRAL (sim=+0.006)
Client 3: weight=1.176 REWARDED (sim=+0.056)
→ Total weight: 11.147 | Avg weight: 0.557
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 45 accuracy recorded: 0.9169 | Δacc = 0.0012
INFO :
INFO : [ROUND 46]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 16 | Total: 2761 samples | Distribution: {0: 126, 1: 346, 2: 422, 3: 1264, 4: 180, 5: 12, 6: 1, 8: 375, 9: 35} [repeated 80x acr
oss cluster]
(ClientAppActor pid=15348) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0012 → REWARDING aligned clients

[ROUND 46] CFA Trust Weights:
Client 13: weight=0.100 MILD PENALTY (sim=-0.051)
Client 11: weight=1.210 REWARDED (sim=+0.057)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 5: weight=1.051 REWARDED (sim=+0.031)
Client 10: weight=1.051 REWARDED (sim=+0.035)
Client 12: weight=1.091 NEUTRAL (sim=+0.008)
Client 9: weight=0.100 NEUTRAL (sim=-0.006)
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 2: weight=1.141 REWARDED (sim=+0.041)
Client 8: weight=1.148 REWARDED (sim=+0.038)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 6: weight=0.108 NEUTRAL (sim=+0.003)
Client 17: weight=0.104 MILD PENALTY (sim=-0.024)
Client 3: weight=1.183 REWARDED (sim=+0.056)
Client 18: weight=0.140 MILD PENALTY (sim=-0.024)
Client 4: weight=0.108 NEUTRAL (sim=+0.005)
Client 16: weight=0.100 MILD PENALTY (sim=-0.041)
Client 1: weight=0.997 REWARDED (sim=+0.016)
Client 15: weight=0.100 MILD PENALTY (sim=-0.027)
Client 0: weight=1.114 REWARDED (sim=+0.035)
→ Total weight: 11.177 | Avg weight: 0.559
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 46 accuracy recorded: 0.9189 | Δacc = 0.0020

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INFO :
INFO : [ROUND 47]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0020 → REWARDING aligned clients

[ROUND 47] CFR Trust Weights:
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 19: weight=0.100 MILD PENALTY (sim=-0.049)
Client 10: weight=1.055 REWARDED (sim=+0.035)
Client 12: weight=1.091 NEUTRAL (sim=+0.006)
Client 4: weight=0.108 NEUTRAL (sim=+0.006)
Client 1: weight=1.000 REWARDED (sim=+0.022)
Client 2: weight=1.146 REWARDED (sim=+0.041)
Client 17: weight=0.102 MILD PENALTY (sim=-0.024)
Client 3: weight=1.189 REWARDED (sim=+0.057)
Client 18: weight=0.138 MILD PENALTY (sim=-0.029)
Client 5: weight=1.054 REWARDED (sim=+0.035)
Client 9: weight=0.100 NEUTRAL (sim=-0.006)
Client 6: weight=0.108 NEUTRAL (sim=+0.003)
Client 15: weight=0.100 MILD PENALTY (sim=-0.031)
Client 13: weight=0.100 MILD PENALTY (sim=-0.046)
Client 0: weight=1.118 REWARDED (sim=+0.036)
Client 11: weight=1.217 REWARDED (sim=+0.058)
Client 16: weight=0.100 MILD PENALTY (sim=-0.042)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 8: weight=1.153 REWARDED (sim=+0.041)
→ Total weight: 11.209 | Avg weight: 0.560
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 47 accuracy recorded: 0.9159 | Δacc = -0.0030
INFO :
INFO : [ROUND 48]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 8
0x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
(raylet) [2026-04-21 13:26:36,115 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4942 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = -0.0030 → PENALISING divergent clients

[ROUND 48] CFR Trust Weights:
Client 19: weight=0.100 PENALISED (dev=1.052)
Client 15: weight=0.100 PENALISED (dev=1.036)
Client 6: weight=0.100 PENALISED (dev=0.997)
Client 3: weight=1.189 CLEARED (dev=0.945)
Client 10: weight=1.055 CLEARED (dev=0.966)
Client 9: weight=0.100 PENALISED (dev=1.010)
Client 17: weight=0.100 PENALISED (dev=1.018)
Client 4: weight=0.100 PENALISED (dev=0.997)
Client 11: weight=1.217 CLEARED (dev=0.943)
Client 13: weight=0.100 PENALISED (dev=1.042)
Client 1: weight=1.000 CLEARED (dev=0.980)
Client 12: weight=1.091 CLEARED (dev=0.990)
Client 14: weight=0.100 PENALISED (dev=1.025)
Client 18: weight=0.117 PENALISED (dev=1.024)
Client 7: weight=0.130 CLEARED (dev=0.992)
Client 2: weight=1.146 CLEARED (dev=0.961)
Client 5: weight=1.054 CLEARED (dev=0.968)
Client 0: weight=1.118 CLEARED (dev=0.965)
Client 8: weight=1.153 CLEARED (dev=0.965)
Client 16: weight=0.100 PENALISED (dev=1.038)
→ Total weight: 11.170 | Avg weight: 0.558
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 48 accuracy recorded: 0.9178 | Δacc = 0.0019
INFO :
INFO : [ROUND 49]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0019 → REWARDING aligned clients

[ROUND 49] CFR Trust Weights:
Client 12: weight=1.091 NEUTRAL (sim=+0.007)
Client 8: weight=1.157 REWARDED (sim=+0.032)
Client 19: weight=0.100 MILD PENALTY (sim=-0.048)
Client 9: weight=0.100 NEUTRAL (sim=-0.006)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 11: weight=1.224 REWARDED (sim=+0.056)
Client 5: weight=1.058 REWARDED (sim=+0.034)
Client 15: weight=0.100 MILD PENALTY (sim=-0.030)
Client 2: weight=1.151 REWARDED (sim=+0.043)
Client 1: weight=1.002 REWARDED (sim=+0.021)
Client 4: weight=0.100 NEUTRAL (sim=+0.004)
Client 7: weight=0.130 NEUTRAL (sim=+0.007)
Client 17: weight=0.100 MILD PENALTY (sim=-0.021)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 13: weight=0.100 MILD PENALTY (sim=-0.045)
Client 18: weight=0.114 MILD PENALTY (sim=-0.031)
Client 3: weight=1.196 REWARDED (sim=+0.053)
Client 16: weight=0.100 MILD PENALTY (sim=-0.038)
Client 0: weight=1.123 REWARDED (sim=+0.037)
Client 10: weight=1.059 REWARDED (sim=+0.032)
→ Total weight: 11.202 | Avg weight: 0.560
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFR] Round 49 accuracy recorded: 0.9219 | Δacc = 0.0041
INFO :
INFO : [ROUND 50]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 7 | Total: 1701 samples | Distribution: {0: 478, 1: 53, 2: 6, 3: 12, 4: 40, 5: 11, 6: 777, 7: 193, 8: 8, 9: 123} [repeated 80x a
cross cluster]
(ClientAppActor pid=15349) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFR] Δacc = +0.0041 → REWARDING aligned clients

[ROUND 50] CFR Trust Weights:
Client 7: weight=0.130 NEUTRAL (sim=+0.006)
Client 3: weight=1.202 REWARDED (sim=+0.054)
Client 17: weight=0.100 MILD PENALTY (sim=-0.017)
Client 19: weight=0.100 MILD PENALTY (sim=-0.044)
Client 18: weight=0.112 MILD PENALTY (sim=-0.028)
Client 9: weight=0.100 NEUTRAL (sim=-0.008)
Client 14: weight=0.100 MILD PENALTY (sim=-0.028)
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 8: weight=1.161 REWARDED (sim=+0.040)
Client 15: weight=0.100 MILD PENALTY (sim=-0.029)
Client 16: weight=0.100 MILD PENALTY (sim=-0.044)
Client 12: weight=1.091 NEUTRAL (sim=+0.010)
Client 4: weight=0.100 NEUTRAL (sim=+0.002)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 11: weight=1.230 REWARDED (sim=+0.055)
Client 1: weight=1.003 REWARDED (sim=+0.017)
Client 0: weight=1.126 REWARDED (sim=+0.035)
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Client 10: weight=1.062 REWARDED (sim=+0.033)
Client 5: weight=1.062 REWARDED (sim=+0.035)
Client 2: weight=1.155 REWARDED (sim=+0.041)
→ Total weight: 11.235 | Avg weight: 0.562
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 50 accuracy recorded: 0.9235 | Δacc = 0.0016
INFO :
INFO :   [ROUND 51]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0016 → REWARDING aligned clients

[ROUND 51] CFA Trust Weights:
Client 10: weight=1.065 REWARDED (sim=+0.032)
Client 0: weight=1.131 REWARDED (sim=+0.036)
Client 13: weight=0.100 MILD PENALTY (sim=-0.047)
Client 17: weight=0.100 MILD PENALTY (sim=-0.023)
Client 8: weight=1.167 REWARDED (sim=+0.044)
Client 6: weight=0.100 NEUTRAL (sim=+0.002)
Client 1: weight=1.005 REWARDED (sim=+0.017)
Client 9: weight=0.100 NEUTRAL (sim=-0.005)
Client 2: weight=1.160 REWARDED (sim=+0.041)
Client 16: weight=0.100 MILD PENALTY (sim=-0.038)
Client 14: weight=0.100 MILD PENALTY (sim=-0.031)
Client 18: weight=0.110 MILD PENALTY (sim=-0.031)
Client 4: weight=0.100 NEUTRAL (sim=+0.005)
Client 15: weight=0.100 MILD PENALTY (sim=-0.030)
Client 7: weight=0.130 NEUTRAL (sim=+0.005)
Client 11: weight=1.238 REWARDED (sim=+0.060)
Client 3: weight=1.208 REWARDED (sim=+0.051)
Client 5: weight=1.065 REWARDED (sim=+0.034)
Client 12: weight=1.092 REWARDED (sim=+0.015)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
→ Total weight: 11.271 | Avg weight: 0.564
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:26:46,142 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4474 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15348) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated
75x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 51 accuracy recorded: 0.9220 | Δacc = -0.0015
INFO :
INFO :   [ROUND 52]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0015 → PENALISING divergent clients

[ROUND 52] CFA Trust Weights:
Client 5: weight=1.065 CLEARED (dev=0.965)
Client 8: weight=1.167 CLEARED (dev=0.958)
Client 18: weight=0.100 PENALISED (dev=1.025)
Client 19: weight=0.100 PENALISED (dev=1.049)
Client 13: weight=0.100 PENALISED (dev=1.040)
Client 11: weight=1.238 CLEARED (dev=0.943)
Client 10: weight=1.065 CLEARED (dev=0.966)
Client 4: weight=0.100 PENALISED (dev=0.997)
Client 7: weight=0.130 CLEARED (dev=0.992)
Client 0: weight=1.131 CLEARED (dev=0.963)
Client 17: weight=0.100 PENALISED (dev=1.023)
Client 6: weight=0.100 PENALISED (dev=0.995)
Client 3: weight=1.208 CLEARED (dev=0.944)
Client 15: weight=0.100 PENALISED (dev=1.037)
Client 14: weight=0.100 PENALISED (dev=1.028)
Client 2: weight=1.160 CLEARED (dev=0.957)
Client 1: weight=1.005 CLEARED (dev=0.977)
Client 16: weight=0.100 PENALISED (dev=1.043)
Client 9: weight=0.100 PENALISED (dev=1.003)
Client 12: weight=1.092 CLEARED (dev=0.993)
→ Total weight: 11.261 | Avg weight: 0.563
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 52 accuracy recorded: 0.9247 | Δacc = 0.0027
INFO :
INFO :   [ROUND 53]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0027 → REWARDING aligned clients

[ROUND 53] CFA Trust Weights:
Client 5: weight=1.069 REWARDED (sim=+0.035)
Client 10: weight=1.069 REWARDED (sim=+0.032)
Client 8: weight=1.171 REWARDED (sim=+0.039)
Client 14: weight=0.100 MILD PENALTY (sim=-0.028)
Client 13: weight=0.100 MILD PENALTY (sim=-0.041)
Client 2: weight=1.165 REWARDED (sim=+0.040)
Client 12: weight=1.092 NEUTRAL (sim=+0.009)
Client 0: weight=1.134 REWARDED (sim=+0.035)
Client 15: weight=0.100 MILD PENALTY (sim=-0.031)
Client 17: weight=0.100 MILD PENALTY (sim=-0.020)
Client 3: weight=1.215 REWARDED (sim=+0.053)
Client 19: weight=0.100 MILD PENALTY (sim=-0.049)
Client 18: weight=0.100 MILD PENALTY (sim=-0.024)
Client 6: weight=0.100 NEUTRAL (sim=+0.002)
Client 1: weight=1.007 REWARDED (sim=+0.019)
Client 11: weight=1.244 REWARDED (sim=+0.053)
Client 9: weight=0.100 NEUTRAL (sim=-0.005)
Client 7: weight=0.130 NEUTRAL (sim=+0.006)
Client 16: weight=0.100 MILD PENALTY (sim=-0.037)
Client 4: weight=0.100 NEUTRAL (sim=+0.003)
→ Total weight: 11.296 | Avg weight: 0.565
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15352) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 79
x across cluster]
(ClientAppActor pid=15352) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 53 accuracy recorded: 0.9269 | Δacc = 0.0022
INFO :
INFO :   [ROUND 54]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0022 → REWARDING aligned clients

[ROUND 54] CFA Trust Weights:
Client 16: weight=0.100 MILD PENALTY (sim=-0.040)
Client 14: weight=0.100 MILD PENALTY (sim=-0.032)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 9: weight=0.100 NEUTRAL (sim=-0.006)
Client 1: weight=1.009 REWARDED (sim=+0.023)
Client 3: weight=1.222 REWARDED (sim=+0.057)
Client 10: weight=1.073 REWARDED (sim=+0.036)
Client 18: weight=0.100 MILD PENALTY (sim=-0.032)
Client 17: weight=0.100 MILD PENALTY (sim=-0.020)
Client 2: weight=1.170 REWARDED (sim=+0.043)

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Client 12: weight=1.092 NEUTRAL (sim=+0.007)
Client 7: weight=0.130 NEUTRAL (sim=+0.009)
Client 15: weight=0.100 MILD PENALTY (sim=-0.034)
Client 8: weight=1.176 REWARDED (sim=+0.045)
Client 6: weight=0.100 NEUTRAL (sim=+0.002)
Client 0: weight=1.139 REWARDED (sim=+0.036)
Client 11: weight=1.251 REWARDED (sim=+0.057)
Client 4: weight=0.100 NEUTRAL (sim=+0.004)
Client 5: weight=1.073 REWARDED (sim=+0.038)
Client 13: weight=0.100 MILD PENALTY (sim=-0.039)
→ Total weight: 11.335 | Avg weight: 0.567
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 54 accuracy recorded: 0.9257 | Δacc = -0.0012
INFO :
INFO : [ROUND 55]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0012 → PENALISING divergent clients

[ROUND 55] CFA Trust Weights:
Client 14: weight=0.100 PENALISED (dev=1.030)
Client 10: weight=1.073 CLEARED (dev=0.965)
Client 7: weight=0.130 CLEARED (dev=0.993)
Client 12: weight=1.092 CLEARED (dev=0.985)
Client 6: weight=0.100 PENALISED (dev=0.997)
Client 16: weight=0.100 PENALISED (dev=1.043)
Client 17: weight=0.100 PENALISED (dev=1.022)
Client 0: weight=1.139 CLEARED (dev=0.963)
Client 18: weight=0.100 PENALISED (dev=1.029)
Client 15: weight=0.100 PENALISED (dev=1.039)
Client 3: weight=1.222 CLEARED (dev=0.944)
Client 11: weight=1.251 CLEARED (dev=0.941)
Client 1: weight=1.009 CLEARED (dev=0.980)
Client 8: weight=1.176 CLEARED (dev=0.957)
Client 19: weight=0.100 PENALISED (dev=1.044)
Client 13: weight=0.100 PENALISED (dev=1.044)
Client 9: weight=0.100 PENALISED (dev=1.001)
Client 5: weight=1.073 CLEARED (dev=0.965)
Client 2: weight=1.170 CLEARED (dev=0.956)
Client 4: weight=0.100 PENALISED (dev=0.998)
→ Total weight: 11.335 | Avg weight: 0.567
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:26:56.147 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4403 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15353) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 7
9x across cluster]
(ClientAppActor pid=15349) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 34x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 55 accuracy recorded: 0.9281 | Δacc = 0.0024
INFO :
INFO : [ROUND 56]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0024 → REWARDING aligned clients

[ROUND 56] CFA Trust Weights:
Client 3: weight=1.228 REWARDED (sim=+0.056)
Client 0: weight=1.143 REWARDED (sim=+0.039)
Client 7: weight=0.130 NEUTRAL (sim=+0.007)
Client 16: weight=0.100 MILD PENALTY (sim=-0.039)
Client 2: weight=1.175 REWARDED (sim=+0.045)
Client 17: weight=0.100 MILD PENALTY (sim=-0.025)
Client 18: weight=0.100 MILD PENALTY (sim=-0.031)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 10: weight=1.076 REWARDED (sim=+0.036)
Client 6: weight=0.100 NEUTRAL (sim=+0.006)
Client 11: weight=1.257 REWARDED (sim=+0.048)
Client 4: weight=0.100 NEUTRAL (sim=+0.004)
Client 8: weight=1.181 REWARDED (sim=+0.040)
Client 9: weight=0.100 NEUTRAL (sim=-0.005)
Client 1: weight=1.012 REWARDED (sim=+0.022)
Client 5: weight=1.077 REWARDED (sim=+0.034)
Client 13: weight=0.100 MILD PENALTY (sim=-0.039)
Client 12: weight=1.092 NEUTRAL (sim=+0.009)
Client 14: weight=0.100 MILD PENALTY (sim=-0.030)
Client 15: weight=0.100 MILD PENALTY (sim=-0.032)
→ Total weight: 11.372 | Avg weight: 0.569
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 56 accuracy recorded: 0.9084 | Δacc = -0.0197
INFO :
INFO : [ROUND 57]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0197 → PENALISING divergent clients

[ROUND 57] CFA Trust Weights:
Client 8: weight=1.181 CLEARED (dev=0.970)
Client 19: weight=0.100 PENALISED (dev=1.044)
Client 14: weight=0.100 PENALISED (dev=1.030)
Client 5: weight=1.077 CLEARED (dev=0.966)
Client 16: weight=0.100 PENALISED (dev=1.045)
Client 3: weight=1.228 CLEARED (dev=0.947)
Client 12: weight=1.092 CLEARED (dev=0.985)
Client 15: weight=0.100 PENALISED (dev=1.030)
Client 0: weight=1.143 CLEARED (dev=0.964)
Client 17: weight=0.100 PENALISED (dev=1.027)
Client 7: weight=0.130 CLEARED (dev=0.991)
Client 1: weight=1.012 CLEARED (dev=0.979)
Client 11: weight=1.257 CLEARED (dev=0.945)
Client 6: weight=0.100 CLEARED (dev=0.995)
Client 10: weight=1.076 CLEARED (dev=0.967)
Client 4: weight=0.100 CLEARED (dev=0.994)
Client 18: weight=0.100 PENALISED (dev=1.028)
Client 13: weight=0.100 PENALISED (dev=1.041)
Client 2: weight=1.175 CLEARED (dev=0.955)
Client 9: weight=0.100 PENALISED (dev=1.003)
→ Total weight: 11.372 | Avg weight: 0.569
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 7
9x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 33x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 57 accuracy recorded: 0.9274 | Δacc = 0.0190
INFO :
INFO : [ROUND 58]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0190 → REWARDING aligned clients

[ROUND 58] CFA Trust Weights:
Client 7: weight=0.130 NEUTRAL (sim=+0.010)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 11: weight=1.264 REWARDED (sim=+0.055)

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Client 17: weight=0.100 MILD PENALTY (sim=-0.027)
Client 3: weight=1.235 REWARDED (sim=+0.053)
Client 19: weight=0.100 MILD PENALTY (sim=-0.047)
Client 0: weight=1.147 REWARDED (sim=+0.039)
Client 12: weight=1.094 REWARDED (sim=+0.012)
Client 10: weight=1.080 REWARDED (sim=+0.035)
Client 2: weight=1.180 REWARDED (sim=+0.045)
Client 1: weight=1.014 REWARDED (sim=+0.022)
Client 4: weight=0.100 NEUTRAL (sim=+0.002)
Client 5: weight=1.081 REWARDED (sim=+0.037)
Client 16: weight=0.100 MILD PENALTY (sim=-0.045)
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 14: weight=0.100 MILD PENALTY (sim=-0.026)
Client 15: weight=0.100 MILD PENALTY (sim=-0.036)
Client 18: weight=0.100 MILD PENALTY (sim=-0.029)
Client 8: weight=1.186 REWARDED (sim=+0.043)
Client 9: weight=0.100 NEUTRAL (sim=-0.001)
→ Total weight: 11.411 | Avg weight: 0.571
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 58 accuracy recorded: 0.9273 | Δacc = -0.0001
INFO :
INFO : [ROUND 59]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0001 → PENALISING divergent clients

[ROUND 59] CFA Trust Weights:
Client 18: weight=0.100 PENALISED (dev=1.029)
Client 12: weight=1.094 CLEARED (dev=0.994)
Client 19: weight=0.100 PENALISED (dev=1.048)
Client 17: weight=0.100 PENALISED (dev=1.025)
Client 4: weight=0.100 CLEARED (dev=0.993)
Client 6: weight=0.100 PENALISED (dev=0.997)
Client 11: weight=1.264 CLEARED (dev=0.942)
Client 10: weight=1.080 CLEARED (dev=0.968)
Client 3: weight=1.235 CLEARED (dev=0.946)
Client 1: weight=1.014 CLEARED (dev=0.978)
Client 8: weight=1.186 CLEARED (dev=0.959)
Client 14: weight=0.100 PENALISED (dev=1.025)
Client 15: weight=0.100 PENALISED (dev=1.030)
Client 9: weight=0.100 PENALISED (dev=1.002)
Client 2: weight=1.180 CLEARED (dev=0.956)
Client 7: weight=0.130 CLEARED (dev=0.993)
Client 0: weight=1.147 CLEARED (dev=0.963)
Client 16: weight=0.100 PENALISED (dev=1.042)
Client 5: weight=1.081 CLEARED (dev=0.965)
Client 13: weight=0.100 PENALISED (dev=1.041)
→ Total weight: 11.411 | Avg weight: 0.571
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:27:06,245 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session-2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4317 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15352) [NON-IID] Client 6 | Total: 1627 samples | Distribution: {0: 205, 1: 221, 2: 1, 3: 131, 4: 666, 5: 20, 6: 178, 7: 136, 8: 68, 9: 1} [repeated 79x
across cluster]
(ClientAppActor pid=15351) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=15350) [ATTACK] Client 17 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 59 accuracy recorded: 0.9056 | Δacc = -0.0217
INFO :
INFO : [ROUND 60]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0217 → PENALISING divergent clients

[ROUND 60] CFA Trust Weights:
Client 14: weight=0.100 PENALISED (dev=1.026)
Client 5: weight=1.081 CLEARED (dev=0.964)
Client 15: weight=0.100 PENALISED (dev=1.033)
Client 6: weight=0.100 PENALISED (dev=0.996)
Client 18: weight=0.100 PENALISED (dev=1.034)
Client 1: weight=1.014 CLEARED (dev=0.978)
Client 8: weight=1.186 CLEARED (dev=0.975)
Client 3: weight=1.235 CLEARED (dev=0.947)
Client 7: weight=0.130 CLEARED (dev=0.992)
Client 0: weight=1.147 CLEARED (dev=0.963)
Client 4: weight=0.100 CLEARED (dev=0.992)
Client 16: weight=0.100 PENALISED (dev=1.043)
Client 13: weight=0.100 PENALISED (dev=1.044)
Client 11: weight=1.264 CLEARED (dev=0.942)
Client 9: weight=0.100 PENALISED (dev=1.004)
Client 12: weight=1.094 CLEARED (dev=0.994)
Client 10: weight=1.080 CLEARED (dev=0.966)
Client 17: weight=0.100 PENALISED (dev=1.026)
Client 2: weight=1.180 CLEARED (dev=0.954)
Client 19: weight=0.100 PENALISED (dev=1.041)
→ Total weight: 11.411 | Avg weight: 0.571
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 60 accuracy recorded: 0.9297 | Δacc = 0.0241
INFO :
INFO : [ROUND 61]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0241 → REWARDING aligned clients

[ROUND 61] CFA Trust Weights:
Client 11: weight=1.272 REWARDED (sim=+0.058)
Client 17: weight=0.100 MILD PENALTY (sim=-0.028)
Client 19: weight=0.100 MILD PENALTY (sim=-0.042)
Client 5: weight=1.085 REWARDED (sim=+0.038)
Client 2: weight=1.186 REWARDED (sim=+0.046)
Client 18: weight=0.100 MILD PENALTY (sim=-0.030)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 13: weight=0.100 MILD PENALTY (sim=-0.038)
Client 9: weight=0.100 NEUTRAL (sim=-0.004)
Client 10: weight=1.084 REWARDED (sim=+0.033)
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 4: weight=0.100 NEUTRAL (sim=-0.002)
Client 12: weight=1.095 REWARDED (sim=+0.011)
Client 0: weight=1.152 REWARDED (sim=+0.039)
Client 16: weight=0.100 MILD PENALTY (sim=-0.040)
Client 1: weight=1.016 REWARDED (sim=+0.019)
Client 14: weight=0.100 MILD PENALTY (sim=-0.030)
Client 15: weight=0.100 MILD PENALTY (sim=-0.034)
Client 8: weight=1.191 REWARDED (sim=+0.038)
Client 3: weight=1.242 REWARDED (sim=+0.054)
→ Total weight: 11.450 | Avg weight: 0.573
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 7
9x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15352) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 61 accuracy recorded: 0.9305 | Δacc = 0.0008
INFO :
INFO : [ROUND 62]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
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INFO :      aggregate_fit: received 20 results and 0 failures
      [CFA] Δacc = +0.0008 → REWARDING aligned clients

[ROUND 62] CFA Trust Weights:
Client 18: weight=0.100 MILD PENALTY (sim=-0.030)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 0: weight=1.156 REWARDED (sim=+0.038)
Client 3: weight=1.248 REWARDED (sim=+0.054)
Client 10: weight=1.088 REWARDED (sim=+0.034)
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 19: weight=0.100 MILD PENALTY (sim=-0.043)
Client 8: weight=1.196 REWARDED (sim=+0.045)
Client 2: weight=1.191 REWARDED (sim=+0.045)
Client 15: weight=0.100 MILD PENALTY (sim=-0.031)
Client 14: weight=0.100 MILD PENALTY (sim=-0.036)
Client 9: weight=0.100 NEUTRAL (sim=-0.004)
Client 12: weight=1.095 NEUTRAL (sim=+0.010)
Client 1: weight=1.018 REWARDED (sim=+0.024)
Client 7: weight=0.130 NEUTRAL (sim=+0.009)
Client 4: weight=0.100 NEUTRAL (sim=+0.003)
Client 17: weight=0.100 MILD PENALTY (sim=-0.024)
Client 16: weight=0.100 MILD PENALTY (sim=-0.043)
Client 5: weight=1.089 REWARDED (sim=+0.037)
Client 11: weight=1.279 REWARDED (sim=+0.058)
→ Total weight: 11.490 | Avg weight: 0.574
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
      [CFA] Round 62 accuracy recorded: 0.9306 | Δacc = 0.0001
INFO :
INFO :      [ROUND 63]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:27:16,329 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4316 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
      [CFA] Δacc = +0.0001 → REWARDING aligned clients

[ROUND 63] CFA Trust Weights:
Client 10: weight=1.091 REWARDED (sim=+0.033)
Client 2: weight=1.196 REWARDED (sim=+0.044)
Client 5: weight=1.093 REWARDED (sim=+0.036)
Client 7: weight=0.130 REWARDED (sim=+0.010)
Client 6: weight=0.100 NEUTRAL (sim=+0.003)
Client 12: weight=1.096 REWARDED (sim=+0.011)
Client 4: weight=0.100 NEUTRAL (sim=+0.005)
Client 17: weight=0.100 MILD PENALTY (sim=-0.027)
Client 16: weight=0.100 MILD PENALTY (sim=-0.041)
Client 13: weight=0.100 MILD PENALTY (sim=-0.038)
Client 0: weight=1.161 REWARDED (sim=+0.038)
Client 11: weight=1.287 REWARDED (sim=+0.059)
Client 9: weight=0.100 NEUTRAL (sim=-0.006)
Client 15: weight=0.100 MILD PENALTY (sim=-0.035)
Client 18: weight=0.100 MILD PENALTY (sim=-0.029)
Client 14: weight=0.100 MILD PENALTY (sim=-0.037)
Client 1: weight=1.020 REWARDED (sim=+0.023)
Client 3: weight=1.255 REWARDED (sim=+0.054)
Client 19: weight=0.100 MILD PENALTY (sim=-0.040)
Client 8: weight=1.201 REWARDED (sim=+0.043)
→ Total weight: 11.530 | Avg weight: 0.576
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 5 | Total: 3472 samples | Distribution: {0: 314, 1: 26, 2: 430, 3: 298, 4: 97, 5: 62, 6: 3, 7: 469, 8: 1771, 9: 2} [repeated 71x
across cluster]
(ClientAppActor pid=15351) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 28x across cluster]
(ClientAppActor pid=15351) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
      [CFA] Round 63 accuracy recorded: 0.9313 | Δacc = 0.0007
INFO :
INFO :      [ROUND 64]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
      [CFA] Δacc = +0.0007 → REWARDING aligned clients

[ROUND 64] CFA Trust Weights:
Client 6: weight=0.100 NEUTRAL (sim=+0.003)
Client 1: weight=1.023 REWARDED (sim=+0.023)
Client 2: weight=1.202 REWARDED (sim=+0.046)
Client 0: weight=1.165 REWARDED (sim=+0.039)
Client 19: weight=0.100 MILD PENALTY (sim=-0.045)
Client 14: weight=0.100 MILD PENALTY (sim=-0.030)
Client 4: weight=0.100 NEUTRAL (sim=+0.005)
Client 16: weight=0.100 MILD PENALTY (sim=-0.043)
Client 7: weight=0.130 NEUTRAL (sim=+0.007)
Client 11: weight=1.294 REWARDED (sim=+0.056)
Client 3: weight=1.262 REWARDED (sim=+0.055)
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 12: weight=1.098 REWARDED (sim=+0.013)
Client 18: weight=0.100 MILD PENALTY (sim=-0.035)
Client 9: weight=0.100 NEUTRAL (sim=-0.004)
Client 17: weight=0.100 MILD PENALTY (sim=-0.029)
Client 10: weight=1.095 REWARDED (sim=+0.033)
Client 5: weight=1.097 REWARDED (sim=+0.039)
Client 8: weight=1.207 REWARDED (sim=+0.045)
Client 15: weight=0.100 MILD PENALTY (sim=-0.030)
→ Total weight: 11.571 | Avg weight: 0.579
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
      [CFA] Round 64 accuracy recorded: 0.9315 | Δacc = 0.0002
INFO :
INFO :      [ROUND 65]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
      [CFA] Δacc = +0.0002 → REWARDING aligned clients

[ROUND 65] CFA Trust Weights:
Client 17: weight=0.100 MILD PENALTY (sim=-0.031)
Client 15: weight=0.100 MILD PENALTY (sim=-0.034)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 13: weight=0.100 MILD PENALTY (sim=-0.040)
Client 11: weight=1.302 REWARDED (sim=+0.060)
Client 16: weight=0.100 MILD PENALTY (sim=-0.044)
Client 12: weight=1.098 NEUTRAL (sim=+0.009)
Client 18: weight=0.100 MILD PENALTY (sim=-0.032)
Client 0: weight=1.170 REWARDED (sim=+0.038)
Client 9: weight=0.100 NEUTRAL (sim=-0.003)
Client 4: weight=0.100 NEUTRAL (sim=+0.006)
Client 19: weight=0.100 MILD PENALTY (sim=-0.043)
Client 5: weight=1.101 REWARDED (sim=+0.040)
Client 1: weight=1.025 REWARDED (sim=+0.023)
Client 10: weight=1.098 REWARDED (sim=+0.034)
Client 8: weight=1.211 REWARDED (sim=+0.038)
Client 3: weight=1.269 REWARDED (sim=+0.055)
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 6: weight=0.100 NEUTRAL (sim=+0.002)
Client 2: weight=1.208 REWARDED (sim=+0.048)
→ Total weight: 11.611 | Avg weight: 0.581
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
78x across cluster]
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(ClientAppActor pid=15347) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 14 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 65 accuracy recorded: 0.9309 | Δacc = -0.0006
INFO :
INFO : [ROUND 66]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0006 → PENALISING divergent clients

[ROUND 66] CFA Trust Weights:
Client 1: weight=1.025 CLEARED (dev=0.974)
Client 18: weight=0.100 PENALISED (dev=1.031)
Client 6: weight=0.100 PENALISED (dev=0.997)
Client 15: weight=0.100 PENALISED (dev=1.042)
Client 16: weight=0.100 PENALISED (dev=1.042)
Client 13: weight=0.100 PENALISED (dev=1.042)
Client 9: weight=0.100 PENALISED (dev=1.002)
Client 19: weight=0.100 PENALISED (dev=1.043)
Client 2: weight=1.208 CLEARED (dev=0.953)
Client 14: weight=0.100 PENALISED (dev=1.031)
Client 12: weight=1.098 CLEARED (dev=0.993)
Client 8: weight=1.211 CLEARED (dev=0.958)
Client 3: weight=1.269 CLEARED (dev=0.946)
Client 7: weight=0.130 CLEARED (dev=0.991)
Client 0: weight=1.170 CLEARED (dev=0.958)
Client 5: weight=1.101 CLEARED (dev=0.960)
Client 10: weight=1.098 CLEARED (dev=0.965)
Client 4: weight=0.100 CLEARED (dev=0.990)
Client 11: weight=1.302 CLEARED (dev=0.942)
Client 17: weight=0.100 PENALISED (dev=1.026)
→ Total weight: 11.611 | Avg weight: 0.581
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:27:26,348 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.4147 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 66 accuracy recorded: 0.9324 | Δacc = 0.0015
INFO :
INFO : [ROUND 67]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
79x across cluster]
(ClientAppActor pid=15350) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15351) [ATTACK] Client 12 submitted poisoned update [repeated 15x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0015 → REWARDING aligned clients

[ROUND 67] CFA Trust Weights:
Client 8: weight=1.217 REWARDED (sim=+0.046)
Client 4: weight=0.100 NEUTRAL (sim=+0.005)
Client 15: weight=0.100 MILD PENALTY (sim=-0.031)
Client 7: weight=0.130 NEUTRAL (sim=+0.009)
Client 12: weight=1.098 NEUTRAL (sim=+0.009)
Client 14: weight=0.100 MILD PENALTY (sim=-0.028)
Client 3: weight=1.276 REWARDED (sim=+0.054)
Client 2: weight=1.213 REWARDED (sim=+0.047)
Client 9: weight=0.100 NEUTRAL (sim=-0.004)
Client 0: weight=1.175 REWARDED (sim=+0.042)
Client 1: weight=1.028 REWARDED (sim=+0.027)
Client 18: weight=0.100 MILD PENALTY (sim=-0.034)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 16: weight=0.100 MILD PENALTY (sim=-0.047)
Client 10: weight=1.102 REWARDED (sim=+0.036)
Client 11: weight=1.309 REWARDED (sim=+0.059)
Client 5: weight=1.106 REWARDED (sim=+0.041)
Client 13: weight=0.100 MILD PENALTY (sim=-0.044)
Client 17: weight=0.100 MILD PENALTY (sim=-0.028)
Client 19: weight=0.100 MILD PENALTY (sim=-0.051)
→ Total weight: 11.653 | Avg weight: 0.583
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 67 accuracy recorded: 0.9327 | Δacc = 0.0003
INFO :
INFO : [ROUND 68]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0003 → REWARDING aligned clients

[ROUND 68] CFA Trust Weights:
Client 11: weight=1.317 REWARDED (sim=+0.056)
Client 18: weight=0.100 MILD PENALTY (sim=-0.029)
Client 0: weight=1.179 REWARDED (sim=+0.040)
Client 3: weight=1.283 REWARDED (sim=+0.055)
Client 2: weight=1.219 REWARDED (sim=+0.048)
Client 15: weight=0.100 MILD PENALTY (sim=-0.039)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 16: weight=0.100 MILD PENALTY (sim=-0.045)
Client 12: weight=1.099 REWARDED (sim=+0.013)
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 8: weight=1.222 REWARDED (sim=+0.044)
Client 10: weight=1.106 REWARDED (sim=+0.036)
Client 4: weight=0.100 NEUTRAL (sim=+0.004)
Client 9: weight=0.100 NEUTRAL (sim=-0.003)
Client 19: weight=0.100 MILD PENALTY (sim=-0.047)
Client 1: weight=1.031 REWARDED (sim=+0.026)
Client 5: weight=1.110 REWARDED (sim=+0.040)
Client 14: weight=0.100 MILD PENALTY (sim=-0.029)
Client 17: weight=0.100 MILD PENALTY (sim=-0.026)
Client 13: weight=0.100 MILD PENALTY (sim=-0.046)
→ Total weight: 11.696 | Avg weight: 0.585
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 68 accuracy recorded: 0.9342 | Δacc = 0.0015
INFO :
INFO : [ROUND 69]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [NON-IID] Client 2 | Total: 4029 samples | Distribution: {0: 8, 1: 806, 2: 1223, 3: 326, 4: 214, 5: 901, 6: 25, 7: 351, 8: 101, 9: 74} [repeated
75x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=15348) [ATTACK] Client 17 submitted poisoned update [repeated 13x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0015 → REWARDING aligned clients

[ROUND 69] CFA Trust Weights:
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 14: weight=0.100 MILD PENALTY (sim=-0.029)
Client 17: weight=0.100 MILD PENALTY (sim=-0.022)
Client 4: weight=0.100 NEUTRAL (sim=+0.003)
Client 19: weight=0.100 MILD PENALTY (sim=-0.042)
Client 18: weight=0.100 MILD PENALTY (sim=-0.029)
Client 2: weight=1.225 REWARDED (sim=+0.046)
Client 1: weight=1.033 REWARDED (sim=+0.025)
Client 11: weight=1.324 REWARDED (sim=+0.059)
Client 16: weight=0.100 MILD PENALTY (sim=-0.044)
Client 8: weight=1.227 REWARDED (sim=+0.042)
Client 5: weight=1.115 REWARDED (sim=+0.043)
Client 9: weight=0.100 NEUTRAL (sim=-0.003)

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Client 13: weight=0.100 MILD PENALTY (sim=-0.046)
Client 10: weight=1.110 REWARDED (sim=+0.036)
Client 12: weight=1.099 NEUTRAL (sim=+0.003)
Client 0: weight=1.184 REWARDED (sim=+0.039)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 15: weight=0.100 MILD PENALTY (sim=-0.034)
Client 3: weight=1.290 REWARDED (sim=+0.055)
→ Total weight: 11.737 | Avg weight: 0.587
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 69 accuracy recorded: 0.9324 | Δacc = -0.0018
INFO :
INFO : [ROUND 70]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0018 → PENALISING divergent clients

[ROUND 70] CFA Trust Weights:
Client 13: weight=0.100 PENALISED (dev=1.039)
Client 2: weight=1.225 CLEARED (dev=0.955)
Client 1: weight=1.033 CLEARED (dev=0.976)
Client 6: weight=0.100 PENALISED (dev=0.997)
Client 10: weight=1.110 CLEARED (dev=0.969)
Client 19: weight=0.100 PENALISED (dev=1.043)
Client 15: weight=0.100 PENALISED (dev=1.036)
Client 4: weight=0.100 PENALISED (dev=0.996)
Client 14: weight=0.100 PENALISED (dev=1.030)
Client 3: weight=1.290 CLEARED (dev=0.946)
Client 9: weight=0.100 PENALISED (dev=1.004)
Client 11: weight=1.324 CLEARED (dev=0.940)
Client 8: weight=1.227 CLEARED (dev=0.954)
Client 17: weight=0.100 PENALISED (dev=1.029)
Client 18: weight=0.100 PENALISED (dev=1.027)
Client 16: weight=0.100 PENALISED (dev=1.046)
Client 12: weight=1.099 CLEARED (dev=0.988)
Client 5: weight=1.115 CLEARED (dev=0.958)
Client 7: weight=0.130 CLEARED (dev=0.994)
Client 0: weight=1.184 CLEARED (dev=0.961)
→ Total weight: 11.737 | Avg weight: 0.587
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [raylet] [2026-04-21 13:27:36.351 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12.3955 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 70 accuracy recorded: 0.9335 | Δacc = 0.0011
INFO :
INFO : [ROUND 71]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 7
8x across cluster]
(ClientAppActor pid=15352) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=15353) [ATTACK] Client 18 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0011 → REWARDING aligned clients

[ROUND 71] CFA Trust Weights:
Client 4: weight=0.100 NEUTRAL (sim=+0.005)
Client 18: weight=0.100 MILD PENALTY (sim=-0.031)
Client 8: weight=1.233 REWARDED (sim=+0.047)
Client 9: weight=0.100 NEUTRAL (sim=-0.003)
Client 16: weight=0.100 MILD PENALTY (sim=-0.043)
Client 19: weight=0.100 MILD PENALTY (sim=-0.045)
Client 5: weight=1.120 REWARDED (sim=+0.043)
Client 14: weight=0.100 MILD PENALTY (sim=-0.031)
Client 17: weight=0.100 MILD PENALTY (sim=-0.030)
Client 12: weight=1.099 NEUTRAL (sim=+0.008)
Client 10: weight=1.114 REWARDED (sim=+0.034)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 15: weight=0.100 MILD PENALTY (sim=-0.032)
Client 0: weight=1.189 REWARDED (sim=+0.041)
Client 2: weight=1.231 REWARDED (sim=+0.048)
Client 7: weight=0.130 NEUTRAL (sim=+0.010)
Client 1: weight=1.036 REWARDED (sim=+0.025)
Client 13: weight=0.100 MILD PENALTY (sim=-0.045)
Client 11: weight=1.332 REWARDED (sim=+0.058)
Client 3: weight=1.297 REWARDED (sim=+0.053)
→ Total weight: 11.779 | Avg weight: 0.589
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 71 accuracy recorded: 0.9347 | Δacc = 0.0012
INFO :
INFO : [ROUND 72]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0012 → REWARDING aligned clients

[ROUND 72] CFA Trust Weights:
Client 15: weight=0.100 MILD PENALTY (sim=-0.031)
Client 6: weight=0.100 NEUTRAL (sim=+0.002)
Client 16: weight=0.100 MILD PENALTY (sim=-0.044)
Client 17: weight=0.100 MILD PENALTY (sim=-0.022)
Client 3: weight=1.304 REWARDED (sim=+0.053)
Client 13: weight=0.100 MILD PENALTY (sim=-0.039)
Client 5: weight=1.124 REWARDED (sim=+0.040)
Client 7: weight=0.130 NEUTRAL (sim=+0.006)
Client 8: weight=1.238 REWARDED (sim=+0.042)
Client 1: weight=1.038 REWARDED (sim=+0.022)
Client 4: weight=0.100 NEUTRAL (sim=+0.001)
Client 18: weight=0.100 MILD PENALTY (sim=-0.029)
Client 2: weight=1.236 REWARDED (sim=+0.042)
Client 9: weight=0.100 NEUTRAL (sim=-0.003)
Client 10: weight=1.118 REWARDED (sim=+0.032)
Client 11: weight=1.340 REWARDED (sim=+0.057)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 12: weight=1.099 NEUTRAL (sim=+0.009)
Client 0: weight=1.193 REWARDED (sim=+0.036)
Client 19: weight=0.100 MILD PENALTY (sim=-0.040)
→ Total weight: 11.819 | Avg weight: 0.591
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 72 accuracy recorded: 0.9354 | Δacc = 0.0007
INFO :
INFO : [ROUND 73]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 7
7x across cluster]
(ClientAppActor pid=15347) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 14 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0007 → REWARDING aligned clients

[ROUND 73] CFA Trust Weights:
Client 9: weight=0.100 NEUTRAL (sim=-0.001)
Client 18: weight=0.100 MILD PENALTY (sim=-0.037)
Client 11: weight=1.347 REWARDED (sim=+0.056)
Client 10: weight=1.121 REWARDED (sim=+0.034)
Client 12: weight=1.100 REWARDED (sim=+0.011)
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Client 15: weight=0.100 MILD PENALTY (sim=-0.032)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 14: weight=0.100 MILD PENALTY (sim=-0.032)
Client 13: weight=0.100 MILD PENALTY (sim=-0.036)
Client 1: weight=1.040 REWARDED (sim=+0.024)
Client 2: weight=1.242 REWARDED (sim=+0.050)
Client 0: weight=1.198 REWARDED (sim=+0.040)
Client 8: weight=1.244 REWARDED (sim=+0.046)
Client 16: weight=0.100 MILD PENALTY (sim=-0.041)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 17: weight=0.100 MILD PENALTY (sim=-0.030)
Client 4: weight=0.100 NEUTRAL (sim=+0.004)
Client 3: weight=1.311 REWARDED (sim=+0.055)
Client 5: weight=1.129 REWARDED (sim=+0.039)
Client 7: weight=0.130 REWARDED (sim=+0.010)
→ Total weight: 11.862 | Avg weight: 0.593
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 73 accuracy recorded: 0.9379 | Δacc = 0.0025
INFO :
INFO :      [ROUND 74]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:27:46,379 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3979 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0025 → REWARDING aligned clients

[ROUND 74] CFA Trust Weights:
Client 3: weight=1.318 REWARDED (sim=+0.055)
Client 8: weight=1.250 REWARDED (sim=+0.046)
Client 2: weight=1.248 REWARDED (sim=+0.047)
Client 14: weight=0.100 MILD PENALTY (sim=-0.035)
Client 1: weight=1.043 REWARDED (sim=+0.025)
Client 9: weight=0.100 NEUTRAL (sim=+0.000)
Client 19: weight=0.100 MILD PENALTY (sim=-0.039)
Client 18: weight=0.100 MILD PENALTY (sim=-0.028)
Client 16: weight=0.100 MILD PENALTY (sim=-0.049)
Client 12: weight=1.102 REWARDED (sim=+0.013)
Client 10: weight=1.125 REWARDED (sim=+0.034)
Client 11: weight=1.355 REWARDED (sim=+0.055)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 7: weight=0.130 NEUTRAL (sim=+0.009)
Client 17: weight=0.100 MILD PENALTY (sim=-0.030)
Client 4: weight=0.100 NEUTRAL (sim=+0.007)
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 0: weight=1.202 REWARDED (sim=+0.039)
Client 15: weight=0.100 MILD PENALTY (sim=-0.034)
Client 5: weight=1.133 REWARDED (sim=+0.042)
→ Total weight: 11.905 | Avg weight: 0.595
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) [NON-IID] Client 0 | Total: 2581 samples | Distribution: {0: 572, 1: 346, 2: 274, 3: 417, 4: 301, 5: 3, 6: 224, 7: 210, 8: 230, 9: 4} [repeated 7
7x across cluster]
(ClientAppActor pid=15352) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15350) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 74 accuracy recorded: 0.9342 | Δacc = -0.0037
INFO :
INFO :      [ROUND 75]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0037 → PENALISING divergent clients

[ROUND 75] CFA Trust Weights:
Client 11: weight=1.355 CLEARED (dev=0.939)
Client 17: weight=0.100 PENALISED (dev=1.032)
Client 15: weight=0.100 PENALISED (dev=1.032)
Client 10: weight=1.125 CLEARED (dev=0.967)
Client 14: weight=0.100 PENALISED (dev=1.035)
Client 8: weight=1.250 CLEARED (dev=0.956)
Client 12: weight=1.102 CLEARED (dev=0.989)
Client 7: weight=0.130 CLEARED (dev=0.992)
Client 16: weight=0.100 PENALISED (dev=1.045)
Client 4: weight=0.100 PENALISED (dev=0.996)
Client 5: weight=1.133 CLEARED (dev=0.957)
Client 3: weight=1.318 CLEARED (dev=0.943)
Client 0: weight=1.202 CLEARED (dev=0.961)
Client 19: weight=0.100 PENALISED (dev=1.044)
Client 13: weight=0.100 PENALISED (dev=1.043)
Client 9: weight=0.100 PENALISED (dev=1.002)
Client 1: weight=1.043 CLEARED (dev=0.976)
Client 18: weight=0.100 PENALISED (dev=1.029)
Client 2: weight=1.248 CLEARED (dev=0.951)
Client 6: weight=0.100 PENALISED (dev=0.994)
→ Total weight: 11.905 | Avg weight: 0.595
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 75 accuracy recorded: 0.9348 | Δacc = 0.0006
INFO :
INFO :      [ROUND 76]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0006 → REWARDING aligned clients

[ROUND 76] CFA Trust Weights:
Client 0: weight=1.207 REWARDED (sim=+0.039)
Client 6: weight=0.100 NEUTRAL (sim=+0.006)
Client 16: weight=0.100 MILD PENALTY (sim=-0.040)
Client 12: weight=1.103 REWARDED (sim=+0.015)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 8: weight=1.255 REWARDED (sim=+0.043)
Client 1: weight=1.046 REWARDED (sim=+0.028)
Client 11: weight=1.363 REWARDED (sim=+0.059)
Client 18: weight=0.100 MILD PENALTY (sim=-0.033)
Client 10: weight=1.129 REWARDED (sim=+0.033)
Client 4: weight=0.100 NEUTRAL (sim=+0.006)
Client 2: weight=1.253 REWARDED (sim=+0.047)
Client 13: weight=0.100 MILD PENALTY (sim=-0.048)
Client 7: weight=0.130 NEUTRAL (sim=+0.009)
Client 3: weight=1.325 REWARDED (sim=+0.055)
Client 5: weight=1.139 REWARDED (sim=+0.045)
Client 17: weight=0.100 MILD PENALTY (sim=-0.028)
Client 15: weight=0.100 MILD PENALTY (sim=-0.035)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 9: weight=0.100 NEUTRAL (sim=-0.002)
→ Total weight: 11.950 | Avg weight: 0.597
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 7
8x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15351) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 76 accuracy recorded: 0.9339 | Δacc = -0.0009
INFO :
INFO :      [ROUND 77]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0009 → PENALISING divergent clients
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[ROUND 77] CFA Trust Weights:
Client 11: weight=1.363 CLEARED (dev=0.942)
Client 1: weight=1.046 CLEARED (dev=0.973)
Client 5: weight=1.139 CLEARED (dev=0.958)
Client 2: weight=1.253 CLEARED (dev=0.953)
Client 19: weight=0.100 PENALISED (dev=1.043)
Client 9: weight=0.100 PENALISED (dev=1.001)
Client 13: weight=0.100 PENALISED (dev=1.040)
Client 6: weight=0.100 PENALISED (dev=0.996)
Client 8: weight=1.255 CLEARED (dev=0.954)
Client 16: weight=0.100 PENALISED (dev=1.047)
Client 7: weight=0.130 CLEARED (dev=0.989)
Client 4: weight=0.100 PENALISED (dev=0.995)
Client 18: weight=0.100 PENALISED (dev=1.029)
Client 3: weight=1.325 CLEARED (dev=0.946)
Client 0: weight=1.207 CLEARED (dev=0.961)
Client 17: weight=0.100 PENALISED (dev=1.026)
Client 10: weight=1.129 CLEARED (dev=0.965)
Client 14: weight=0.100 PENALISED (dev=1.031)
Client 12: weight=0.937 PENALISED (dev=0.997)
Client 15: weight=0.100 PENALISED (dev=1.031)
→ Total weight: 11.784 | Avg weight: 0.589
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 77 accuracy recorded: 0.9357 | Δacc = 0.0018
INFO :
INFO : [ROUND 78]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:27:56,441 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3968 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0018 → REWARDING aligned clients

[ROUND 78] CFA Trust Weights:
Client 10: weight=1.133 REWARDED (sim=+0.035)
Client 7: weight=0.130 NEUTRAL (sim=+0.008)
Client 0: weight=1.212 REWARDED (sim=+0.039)
Client 19: weight=0.100 MILD PENALTY (sim=-0.044)
Client 16: weight=0.100 MILD PENALTY (sim=-0.042)
Client 13: weight=0.100 MILD PENALTY (sim=-0.038)
Client 18: weight=0.100 MILD PENALTY (sim=-0.034)
Client 1: weight=1.049 REWARDED (sim=+0.026)
Client 3: weight=1.332 REWARDED (sim=+0.052)
Client 4: weight=0.100 NEUTRAL (sim=+0.009)
Client 2: weight=1.259 REWARDED (sim=+0.045)
Client 11: weight=1.370 REWARDED (sim=+0.058)
Client 14: weight=0.100 MILD PENALTY (sim=-0.027)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 12: weight=0.938 REWARDED (sim=+0.010)
Client 15: weight=0.100 MILD PENALTY (sim=-0.034)
Client 17: weight=0.100 MILD PENALTY (sim=-0.027)
Client 8: weight=1.261 REWARDED (sim=+0.044)
Client 5: weight=1.143 REWARDED (sim=+0.040)
Client 9: weight=0.100 NEUTRAL (sim=-0.001)
→ Total weight: 11.827 | Avg weight: 0.591
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) [NON-IID] Client 11 | Total: 4036 samples | Distribution: {0: 1, 1: 640, 2: 14, 3: 216, 4: 1552, 5: 29, 6: 259, 7: 163, 8: 540, 9: 622} [repeated
78x across cluster]
(ClientAppActor pid=15349) [SETUP] Client 13 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=15350) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 78 accuracy recorded: 0.9352 | Δacc = -0.0005
INFO :
INFO : [ROUND 79]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0005 → PENALISING divergent clients

[ROUND 79] CFA Trust Weights:
Client 7: weight=0.130 CLEARED (dev=0.989)
Client 5: weight=1.143 CLEARED (dev=0.959)
Client 18: weight=0.100 PENALISED (dev=1.031)
Client 14: weight=0.100 PENALISED (dev=1.029)
Client 9: weight=0.100 PENALISED (dev=1.000)
Client 15: weight=0.100 PENALISED (dev=1.033)
Client 4: weight=0.100 PENALISED (dev=0.995)
Client 19: weight=0.100 PENALISED (dev=1.048)
Client 6: weight=0.100 CLEARED (dev=0.993)
Client 2: weight=1.259 CLEARED (dev=0.954)
Client 1: weight=1.049 CLEARED (dev=0.975)
Client 3: weight=1.332 CLEARED (dev=0.944)
Client 0: weight=1.212 CLEARED (dev=0.957)
Client 16: weight=0.100 PENALISED (dev=1.047)
Client 11: weight=1.370 CLEARED (dev=0.944)
Client 12: weight=0.797 PENALISED (dev=0.995)
Client 8: weight=1.261 CLEARED (dev=0.953)
Client 10: weight=1.133 CLEARED (dev=0.964)
Client 17: weight=0.100 PENALISED (dev=1.028)
Client 13: weight=0.100 PENALISED (dev=1.039)
→ Total weight: 11.686 | Avg weight: 0.584
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 79 accuracy recorded: 0.9381 | Δacc = 0.0029
INFO :
INFO : [ROUND 80]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0029 → REWARDING aligned clients

[ROUND 80] CFA Trust Weights:
Client 13: weight=0.100 MILD PENALTY (sim=-0.038)
Client 0: weight=1.216 REWARDED (sim=+0.039)
Client 6: weight=0.100 NEUTRAL (sim=+0.004)
Client 16: weight=0.100 MILD PENALTY (sim=-0.042)
Client 18: weight=0.100 MILD PENALTY (sim=-0.032)
Client 19: weight=0.100 MILD PENALTY (sim=-0.047)
Client 4: weight=0.100 NEUTRAL (sim=+0.004)
Client 9: weight=0.100 NEUTRAL (sim=-0.001)
Client 7: weight=0.130 NEUTRAL (sim=+0.009)
Client 2: weight=1.265 REWARDED (sim=+0.048)
Client 17: weight=0.100 MILD PENALTY (sim=-0.027)
Client 3: weight=1.339 REWARDED (sim=+0.055)
Client 5: weight=1.148 REWARDED (sim=+0.042)
Client 1: weight=1.051 REWARDED (sim=+0.027)
Client 11: weight=1.378 REWARDED (sim=+0.056)
Client 15: weight=0.100 MILD PENALTY (sim=-0.031)
Client 8: weight=1.266 REWARDED (sim=+0.047)
Client 14: weight=0.100 MILD PENALTY (sim=-0.038)
Client 12: weight=0.797 NEUTRAL (sim=+0.010)
Client 10: weight=1.137 REWARDED (sim=+0.036)
→ Total weight: 11.729 | Avg weight: 0.586
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [NON-IID] Client 3 | Total: 4304 samples | Distribution: {0: 769, 1: 141, 2: 96, 3: 84, 4: 128, 5: 116, 6: 2, 7: 940, 8: 18, 9: 2010} [repeated 7
7x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15352) [ATTACK] Client 15 submitted poisoned update [repeated 16x across cluster]
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INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA] Round 80 accuracy recorded: 0.9399 | Δacc = 0.0018
INFO :
INFO :      [ROUND 81]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA] Δacc = +0.0018 → REWARDING aligned clients

[ROUND 81] CFA Trust Weights:
Client 11: weight=1.387 REWARDED (sim=+0.062)
Client 14: weight=0.100 MILD PENALTY (sim=-0.033)
Client 15: weight=0.100 MILD PENALTY (sim=-0.036)
Client 1: weight=1.054 REWARDED (sim=+0.024)
Client 18: weight=0.100 MILD PENALTY (sim=-0.034)
Client 9: weight=0.100 NEUTRAL (sim=-0.001)
Client 8: weight=1.272 REWARDED (sim=+0.046)
Client 12: weight=0.797 NEUTRAL (sim=+0.008)
Client 19: weight=0.100 MILD PENALTY (sim=-0.043)
Client 5: weight=1.153 REWARDED (sim=+0.043)
Client 16: weight=0.100 MILD PENALTY (sim=-0.042)
Client 3: weight=1.347 REWARDED (sim=+0.055)
Client 7: weight=0.130 REWARDED (sim=+0.010)
Client 4: weight=0.100 NEUTRAL (sim=+0.007)
Client 2: weight=1.271 REWARDED (sim=+0.046)
Client 17: weight=0.100 MILD PENALTY (sim=-0.028)
Client 0: weight=1.221 REWARDED (sim=+0.039)
Client 10: weight=1.141 REWARDED (sim=+0.035)
Client 6: weight=0.100 NEUTRAL (sim=+0.005)
Client 13: weight=0.100 MILD PENALTY (sim=-0.038)
→ Total weight: 11.773 | Avg weight: 0.589
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:28:06,465 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3967 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA] Round 81 accuracy recorded: 0.9409 | Δacc = 0.0010
INFO :
INFO :      [ROUND 82]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA] Δacc = +0.0010 → REWARDING aligned clients

[ROUND 82] CFA Trust Weights:
Client 3: weight=1.354 REWARDED (sim=+0.055)
Client 1: weight=1.057 REWARDED (sim=+0.030)
Client 2: weight=1.277 REWARDED (sim=+0.051)
Client 0: weight=1.226 REWARDED (sim=+0.043)
Client 16: weight=0.100 MILD PENALTY (sim=-0.042)
Client 14: weight=0.100 MILD PENALTY (sim=-0.037)
Client 13: weight=0.100 MILD PENALTY (sim=-0.036)
Client 6: weight=0.100 NEUTRAL (sim=+0.007)
Client 10: weight=1.145 REWARDED (sim=+0.037)
Client 9: weight=0.100 NEUTRAL (sim=+0.002)
Client 8: weight=1.279 REWARDED (sim=+0.050)
Client 4: weight=0.100 NEUTRAL (sim=+0.008)
Client 12: weight=0.797 NEUTRAL (sim=-0.001)
Client 18: weight=0.100 MILD PENALTY (sim=-0.031)
Client 15: weight=0.100 MILD PENALTY (sim=-0.033)
Client 17: weight=0.100 MILD PENALTY (sim=-0.032)
Client 5: weight=1.158 REWARDED (sim=+0.043)
Client 11: weight=1.395 REWARDED (sim=+0.059)
Client 19: weight=0.100 MILD PENALTY (sim=-0.049)
Client 7: weight=0.130 REWARDED (sim=+0.013)
→ Total weight: 11.819 | Avg weight: 0.591
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 15 | Total: 3188 samples | Distribution: {0: 790, 1: 47, 2: 729, 3: 38, 4: 2, 5: 135, 6: 417, 7: 48, 8: 29, 9: 953} [repeated 77
x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 15 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15351) [ATTACK] Client 19 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA] Round 82 accuracy recorded: 0.9411 | Δacc = 0.0002
INFO :
INFO :      [ROUND 83]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA] Δacc = +0.0002 → REWARDING aligned clients

[ROUND 83] CFA Trust Weights:
Client 9: weight=0.100 NEUTRAL (sim=-0.000)
Client 14: weight=0.100 MILD PENALTY (sim=-0.036)
Client 4: weight=0.100 NEUTRAL (sim=+0.007)
Client 16: weight=0.100 MILD PENALTY (sim=-0.045)
Client 10: weight=1.149 REWARDED (sim=+0.037)
Client 6: weight=0.100 NEUTRAL (sim=+0.006)
Client 18: weight=0.100 MILD PENALTY (sim=-0.035)
Client 1: weight=1.060 REWARDED (sim=+0.028)
Client 11: weight=1.403 REWARDED (sim=+0.059)
Client 17: weight=0.100 MILD PENALTY (sim=-0.027)
Client 12: weight=0.797 NEUTRAL (sim=+0.006)
Client 5: weight=1.163 REWARDED (sim=+0.046)
Client 15: weight=0.100 MILD PENALTY (sim=-0.039)
Client 19: weight=0.100 MILD PENALTY (sim=-0.038)
Client 7: weight=0.130 REWARDED (sim=+0.012)
Client 2: weight=1.284 REWARDED (sim=+0.051)
Client 3: weight=1.362 REWARDED (sim=+0.054)
Client 13: weight=0.100 MILD PENALTY (sim=-0.044)
Client 0: weight=1.231 REWARDED (sim=+0.041)
Client 8: weight=1.285 REWARDED (sim=+0.050)
→ Total weight: 11.865 | Avg weight: 0.593
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[ CFA] Round 83 accuracy recorded: 0.9418 | Δacc = 0.0007
INFO :
INFO :      [ROUND 84]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 1 | Total: 3670 samples | Distribution: {0: 440, 1: 12, 2: 234, 3: 86, 4: 212, 5: 24, 6: 181, 7: 2374, 8: 16, 9: 91} [repeated 6
8x across cluster]
(ClientAppActor pid=15354) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 14 submitted poisoned update [repeated 13x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[ CFA] Δacc = +0.0007 → REWARDING aligned clients

[ROUND 84] CFA Trust Weights:
Client 4: weight=0.100 NEUTRAL (sim=+0.009)
Client 0: weight=1.237 REWARDED (sim=+0.042)
Client 9: weight=0.100 NEUTRAL (sim=+0.002)
Client 10: weight=1.154 REWARDED (sim=+0.038)
Client 6: weight=0.100 NEUTRAL (sim=+0.010)
Client 18: weight=0.100 MILD PENALTY (sim=-0.031)
Client 19: weight=0.100 MILD PENALTY (sim=-0.044)
Client 13: weight=0.100 MILD PENALTY (sim=-0.046)
Client 11: weight=1.411 REWARDED (sim=+0.060)
Client 1: weight=1.063 REWARDED (sim=+0.031)
Client 3: weight=1.369 REWARDED (sim=+0.056)
Client 17: weight=0.100 MILD PENALTY (sim=-0.027)
Client 16: weight=0.100 MILD PENALTY (sim=-0.046)
Client 15: weight=0.100 MILD PENALTY (sim=-0.041)
Client 7: weight=0.131 REWARDED (sim=+0.011)
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Client 8: weight=1.291 REWARDED (sim=+0.048)
Client 12: weight=0.797 NEUTRAL (sim=+0.009)
Client 2: weight=1.290 REWARDED (sim=+0.049)
Client 5: weight=1.168 REWARDED (sim=+0.046)
Client 14: weight=0.100 MILD PENALTY (sim=-0.038)
→ Total weight: 11.912 | Avg weight: 0.596
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 84 accuracy recorded: 0.9428 | Δacc = 0.0010
INFO :
INFO :   [ROUND 85]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0010 → REWARDING aligned clients

[ROUND 85] CFA Trust Weights:
Client 15: weight=0.100 MILD PENALTY (sim=-0.039)
Client 1: weight=1.066 REWARDED (sim=+0.029)
Client 16: weight=0.100 MILD PENALTY (sim=-0.049)
Client 9: weight=0.100 NEUTRAL (sim=-0.000)
Client 2: weight=1.297 REWARDED (sim=+0.050)
Client 17: weight=0.100 MILD PENALTY (sim=-0.029)
Client 8: weight=1.298 REWARDED (sim=+0.049)
Client 5: weight=1.174 REWARDED (sim=+0.044)
Client 0: weight=1.242 REWARDED (sim=+0.044)
Client 13: weight=0.100 MILD PENALTY (sim=-0.036)
Client 14: weight=0.100 MILD PENALTY (sim=-0.031)
Client 10: weight=1.158 REWARDED (sim=+0.036)
Client 3: weight=1.377 REWARDED (sim=+0.056)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 11: weight=1.420 REWARDED (sim=+0.060)
Client 4: weight=0.100 REWARDED (sim=+0.012)
Client 12: weight=0.797 NEUTRAL (sim=+0.003)
Client 6: weight=0.100 NEUTRAL (sim=+0.009)
Client 7: weight=0.131 REWARDED (sim=+0.011)
Client 18: weight=0.100 MILD PENALTY (sim=-0.029)
→ Total weight: 11.959 | Avg weight: 0.598
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:28:16,498 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3946 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 85 accuracy recorded: 0.9425 | Δacc = -0.0003
INFO :
INFO :   [ROUND 86]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15353) [NON-IID] Client 4 | Total: 1981 samples | Distribution: {1: 219, 2: 42, 3: 725, 4: 91, 5: 39, 6: 6, 7: 69, 8: 779, 9: 11} [repeated 70x across c
luster]
(ClientAppActor pid=15347) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=15351) [ATTACK] Client 13 submitted poisoned update [repeated 12x across cluster]
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0003 → PENALISING divergent clients

[ROUND 86] CFA Trust Weights:
Client 17: weight=0.100 PENALISED (dev=1.030)
Client 9: weight=0.100 PENALISED (dev=0.999)
Client 0: weight=1.242 CLEARED (dev=0.961)
Client 15: weight=0.100 PENALISED (dev=1.031)
Client 11: weight=1.420 CLEARED (dev=0.938)
Client 7: weight=0.131 CLEARED (dev=0.988)
Client 6: weight=0.100 CLEARED (dev=0.992)
Client 5: weight=1.174 CLEARED (dev=0.955)
Client 18: weight=0.100 PENALISED (dev=1.036)
Client 14: weight=0.100 PENALISED (dev=1.039)
Client 13: weight=0.100 PENALISED (dev=1.037)
Client 1: weight=1.066 CLEARED (dev=0.968)
Client 19: weight=0.100 PENALISED (dev=1.047)
Client 12: weight=0.797 CLEARED (dev=0.990)
Client 8: weight=1.298 CLEARED (dev=0.949)
Client 3: weight=1.377 CLEARED (dev=0.943)
Client 10: weight=1.158 CLEARED (dev=0.962)
Client 4: weight=0.100 CLEARED (dev=0.991)
Client 16: weight=0.100 PENALISED (dev=1.048)
Client 2: weight=1.297 CLEARED (dev=0.950)
→ Total weight: 11.959 | Avg weight: 0.598
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 86 accuracy recorded: 0.9419 | Δacc = -0.0006
INFO :
INFO :   [ROUND 87]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0006 → PENALISING divergent clients

[ROUND 87] CFA Trust Weights:
Client 0: weight=1.242 CLEARED (dev=0.957)
Client 11: weight=1.420 CLEARED (dev=0.938)
Client 6: weight=0.100 CLEARED (dev=0.993)
Client 17: weight=0.100 PENALISED (dev=1.029)
Client 13: weight=0.100 PENALISED (dev=1.036)
Client 3: weight=1.377 CLEARED (dev=0.944)
Client 1: weight=1.066 CLEARED (dev=0.969)
Client 7: weight=0.131 CLEARED (dev=0.987)
Client 19: weight=0.100 PENALISED (dev=1.046)
Client 12: weight=0.678 PENALISED (dev=0.993)
Client 2: weight=1.297 CLEARED (dev=0.951)
Client 5: weight=1.174 CLEARED (dev=0.958)
Client 10: weight=1.158 CLEARED (dev=0.961)
Client 18: weight=0.100 PENALISED (dev=1.038)
Client 16: weight=0.100 PENALISED (dev=1.043)
Client 15: weight=0.100 PENALISED (dev=1.046)
Client 8: weight=1.298 CLEARED (dev=0.951)
Client 4: weight=0.100 CLEARED (dev=0.993)
Client 9: weight=0.100 PENALISED (dev=0.996)
Client 14: weight=0.100 PENALISED (dev=1.031)
→ Total weight: 11.840 | Avg weight: 0.592
INFO :   configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :   aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 87 accuracy recorded: 0.9448 | Δacc = 0.0029
INFO :
INFO :   [ROUND 88]
INFO :   configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15351) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 75
x across cluster]
(ClientAppActor pid=15351) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15353) [ATTACK] Client 12 submitted poisoned update [repeated 15x across cluster]
INFO :   aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0029 → REWARDING aligned clients

[ROUND 88] CFA Trust Weights:
Client 18: weight=0.100 MILD PENALTY (sim=-0.041)
Client 13: weight=0.100 MILD PENALTY (sim=-0.040)
Client 11: weight=1.428 REWARDED (sim=+0.060)
Client 17: weight=0.100 MILD PENALTY (sim=-0.028)
Client 7: weight=0.131 REWARDED (sim=+0.011)
Client 15: weight=0.100 MILD PENALTY (sim=-0.033)
Client 19: weight=0.100 MILD PENALTY (sim=-0.046)
Client 4: weight=0.100 REWARDED (sim=+0.011)

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Client 3: weight=1.385 REWARDED (sim=+0.057)
Client 8: weight=1.303 REWARDED (sim=+0.044)
Client 6: weight=0.100 NEUTRAL (sim=+0.007)
Client 16: weight=0.100 MILD PENALTY (sim=-0.039)
Client 2: weight=1.303 REWARDED (sim=+0.049)
Client 9: weight=0.100 NEUTRAL (sim=+0.000)
Client 12: weight=0.678 NEUTRAL (sim=+0.000)
Client 5: weight=1.178 REWARDED (sim=+0.041)
Client 14: weight=0.100 MILD PENALTY (sim=-0.030)
Client 0: weight=1.247 REWARDED (sim=+0.042)
Client 10: weight=1.162 REWARDED (sim=+0.035)
Client 1: weight=1.070 REWARDED (sim=+0.030)
→ Total weight: 11.886 | Avg weight: 0.594
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 88 accuracy recorded: 0.9447 | Δacc = -0.0001
INFO :
INFO : [ROUND 89]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:28:26,583 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3873 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0001 → PENALISING divergent clients

[ROUND 89] CFA Trust Weights:
Client 3: weight=1.385 CLEARED (dev=0.944)
Client 14: weight=0.100 PENALISED (dev=1.032)
Client 15: weight=0.100 PENALISED (dev=1.036)
Client 6: weight=0.100 PENALISED (dev=0.994)
Client 0: weight=1.247 CLEARED (dev=0.958)
Client 8: weight=1.303 CLEARED (dev=0.950)
Client 13: weight=0.100 PENALISED (dev=1.047)
Client 18: weight=0.100 PENALISED (dev=1.030)
Client 19: weight=0.100 PENALISED (dev=1.046)
Client 11: weight=1.428 CLEARED (dev=0.941)
Client 9: weight=0.100 PENALISED (dev=1.002)
Client 2: weight=1.303 CLEARED (dev=0.949)
Client 12: weight=0.576 PENALISED (dev=0.990)
Client 10: weight=1.162 CLEARED (dev=0.963)
Client 5: weight=1.178 CLEARED (dev=0.954)
Client 4: weight=0.100 CLEARED (dev=0.992)
Client 1: weight=1.070 CLEARED (dev=0.960)
Client 17: weight=0.100 PENALISED (dev=1.029)
Client 7: weight=0.131 CLEARED (dev=0.988)
Client 16: weight=0.100 PENALISED (dev=1.042)
→ Total weight: 11.784 | Avg weight: 0.589
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 89 accuracy recorded: 0.9442 | Δacc = -0.0005
INFO :
INFO : [ROUND 90]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 19 | Total: 2333 samples | Distribution: {0: 213, 1: 911, 2: 183, 3: 28, 4: 12, 5: 18, 6: 702, 7: 239, 8: 14, 9: 13} [repeated 8
0x across cluster]
(ClientAppActor pid=15348) [SETUP] Client 19 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15347) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0005 → PENALISING divergent clients

[ROUND 90] CFA Trust Weights:
Client 17: weight=0.100 PENALISED (dev=1.031)
Client 16: weight=0.100 PENALISED (dev=1.043)
Client 4: weight=0.100 PENALISED (dev=0.994)
Client 12: weight=0.489 PENALISED (dev=0.994)
Client 15: weight=0.100 PENALISED (dev=1.034)
Client 3: weight=1.385 CLEARED (dev=0.942)
Client 14: weight=0.100 PENALISED (dev=1.036)
Client 5: weight=1.178 CLEARED (dev=0.959)
Client 11: weight=1.428 CLEARED (dev=0.943)
Client 2: weight=1.303 CLEARED (dev=0.950)
Client 9: weight=0.100 PENALISED (dev=1.000)
Client 7: weight=0.131 CLEARED (dev=0.986)
Client 8: weight=1.303 CLEARED (dev=0.949)
Client 6: weight=0.100 CLEARED (dev=0.990)
Client 13: weight=0.100 PENALISED (dev=1.043)
Client 10: weight=1.162 CLEARED (dev=0.960)
Client 0: weight=1.247 CLEARED (dev=0.959)
Client 18: weight=0.100 PENALISED (dev=1.034)
Client 1: weight=1.070 CLEARED (dev=0.967)
Client 19: weight=0.100 PENALISED (dev=1.046)
→ Total weight: 11.697 | Avg weight: 0.585
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 90 accuracy recorded: 0.9469 | Δacc = 0.0027
INFO :
INFO : [ROUND 91]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0027 → REWARDING aligned clients

[ROUND 91] CFA Trust Weights:
Client 4: weight=0.100 NEUTRAL (sim=+0.009)
Client 0: weight=1.253 REWARDED (sim=+0.044)
Client 18: weight=0.100 MILD PENALTY (sim=-0.031)
Client 3: weight=1.393 REWARDED (sim=+0.057)
Client 2: weight=1.310 REWARDED (sim=+0.054)
Client 6: weight=0.100 REWARDED (sim=+0.011)
Client 11: weight=1.437 REWARDED (sim=+0.063)
Client 14: weight=0.100 MILD PENALTY (sim=-0.039)
Client 12: weight=0.489 NEUTRAL (sim=-0.004)
Client 7: weight=0.131 REWARDED (sim=+0.014)
Client 15: weight=0.100 MILD PENALTY (sim=-0.042)
Client 19: weight=0.100 MILD PENALTY (sim=-0.041)
Client 5: weight=1.184 REWARDED (sim=+0.045)
Client 17: weight=0.100 MILD PENALTY (sim=-0.028)
Client 1: weight=1.073 REWARDED (sim=+0.035)
Client 10: weight=1.166 REWARDED (sim=+0.040)
Client 9: weight=0.100 NEUTRAL (sim=+0.001)
Client 8: weight=1.310 REWARDED (sim=+0.050)
Client 16: weight=0.100 MILD PENALTY (sim=-0.045)
Client 13: weight=0.100 MILD PENALTY (sim=-0.040)
→ Total weight: 11.747 | Avg weight: 0.587
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 91 accuracy recorded: 0.9449 | Δacc = -0.0020
INFO :
INFO : [ROUND 92]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15354) [NON-IID] Client 14 | Total: 2530 samples | Distribution: {0: 6, 1: 328, 2: 560, 3: 371, 4: 155, 5: 589, 6: 141, 7: 93, 8: 262, 9: 25} [repeated
80x across cluster]
(ClientAppActor pid=15354) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 32x across cluster]
(ClientAppActor pid=15349) [ATTACK] Client 18 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0020 → PENALISING divergent clients

[ROUND 92] CFA Trust Weights:
Client 12: weight=0.415 PENALISED (dev=1.004)

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Client 7: weight=0.131 CLEARED (dev=0.986)
Client 3: weight=1.393 CLEARED (dev=0.941)
Client 8: weight=1.310 CLEARED (dev=0.945)
Client 4: weight=0.100 CLEARED (dev=0.989)
Client 9: weight=0.100 PENALISED (dev=0.992)
Client 19: weight=0.100 PENALISED (dev=1.043)
Client 10: weight=1.166 CLEARED (dev=0.960)
Client 5: weight=1.184 CLEARED (dev=0.955)
Client 14: weight=0.100 PENALISED (dev=1.037)
Client 18: weight=0.100 PENALISED (dev=1.034)
Client 13: weight=0.100 PENALISED (dev=1.041)
Client 11: weight=1.437 CLEARED (dev=0.936)
Client 17: weight=0.100 PENALISED (dev=1.034)
Client 0: weight=1.253 CLEARED (dev=0.955)
Client 15: weight=0.100 PENALISED (dev=1.041)
Client 2: weight=1.310 CLEARED (dev=0.947)
Client 6: weight=0.100 CLEARED (dev=0.990)
Client 16: weight=0.100 PENALISED (dev=1.045)
Client 1: weight=1.073 CLEARED (dev=0.968)
→ Total weight: 11.673 | Avg weight: 0.584
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:28:36,589 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3742 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 92 accuracy recorded: 0.9476 | Δacc = 0.0027
INFO :
INFO :      [ROUND 93]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0027 → REWARDING aligned clients

[ROUND 93] CFA Trust Weights:
Client 7: weight=0.131 REWARDED (sim=+0.014)
Client 4: weight=0.100 REWARDED (sim=+0.011)
Client 16: weight=0.100 MILD PENALTY (sim=-0.043)
Client 0: weight=1.258 REWARDED (sim=+0.045)
Client 8: weight=1.317 REWARDED (sim=+0.052)
Client 11: weight=1.446 REWARDED (sim=+0.063)
Client 2: weight=1.317 REWARDED (sim=+0.053)
Client 5: weight=1.189 REWARDED (sim=+0.047)
Client 1: weight=1.077 REWARDED (sim=+0.032)
Client 19: weight=0.100 MILD PENALTY (sim=-0.045)
Client 15: weight=0.100 MILD PENALTY (sim=-0.040)
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 17: weight=0.100 MILD PENALTY (sim=-0.033)
Client 6: weight=0.100 REWARDED (sim=+0.011)
Client 10: weight=1.171 REWARDED (sim=+0.040)
Client 18: weight=0.100 MILD PENALTY (sim=-0.033)
Client 14: weight=0.100 MILD PENALTY (sim=-0.040)
Client 3: weight=1.401 REWARDED (sim=+0.058)
Client 9: weight=0.100 NEUTRAL (sim=+0.004)
Client 12: weight=0.415 NEUTRAL (sim=+0.001)
→ Total weight: 11.723 | Avg weight: 0.586
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 93 accuracy recorded: 0.9469 | Δacc = -0.0007
INFO :
INFO :      [ROUND 94]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15348) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 8
0x across cluster]
(ClientAppActor pid=15347) [SETUP] Client 12 is MALICIOUS (label_flip) [repeated 31x across cluster]
(ClientAppActor pid=15350) [ATTACK] Client 12 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0007 → PENALISING divergent clients

[ROUND 94] CFA Trust Weights:
Client 17: weight=0.100 PENALISED (dev=1.029)
Client 19: weight=0.100 PENALISED (dev=1.043)
Client 1: weight=1.077 CLEARED (dev=0.967)
Client 2: weight=1.317 CLEARED (dev=0.950)
Client 15: weight=0.100 PENALISED (dev=1.037)
Client 12: weight=0.353 PENALISED (dev=0.994)
Client 3: weight=1.401 CLEARED (dev=0.943)
Client 13: weight=0.100 PENALISED (dev=1.046)
Client 6: weight=0.100 CLEARED (dev=0.989)
Client 16: weight=0.100 PENALISED (dev=1.047)
Client 11: weight=1.446 CLEARED (dev=0.939)
Client 0: weight=1.258 CLEARED (dev=0.955)
Client 5: weight=1.189 CLEARED (dev=0.956)
Client 18: weight=0.100 PENALISED (dev=1.035)
Client 8: weight=1.317 CLEARED (dev=0.948)
Client 9: weight=0.100 PENALISED (dev=0.996)
Client 7: weight=0.131 CLEARED (dev=0.987)
Client 14: weight=0.100 PENALISED (dev=1.041)
Client 10: weight=1.171 CLEARED (dev=0.961)
Client 4: weight=0.100 CLEARED (dev=0.989)
→ Total weight: 11.661 | Avg weight: 0.583
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 94 accuracy recorded: 0.9497 | Δacc = 0.0028
INFO :
INFO :      [ROUND 95]
INFO :      configure_fit: strategy sampled 20 clients (out of 20)
INFO :      aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0028 → REWARDING aligned clients

[ROUND 95] CFA Trust Weights:
Client 8: weight=1.323 REWARDED (sim=+0.052)
Client 16: weight=0.100 MILD PENALTY (sim=-0.046)
Client 14: weight=0.100 MILD PENALTY (sim=-0.039)
Client 1: weight=1.081 REWARDED (sim=+0.036)
Client 6: weight=0.100 REWARDED (sim=+0.013)
Client 5: weight=1.195 REWARDED (sim=+0.047)
Client 0: weight=1.264 REWARDED (sim=+0.044)
Client 7: weight=0.131 REWARDED (sim=+0.015)
Client 2: weight=1.324 REWARDED (sim=+0.053)
Client 18: weight=0.100 MILD PENALTY (sim=-0.037)
Client 19: weight=0.100 MILD PENALTY (sim=-0.048)
Client 3: weight=1.409 REWARDED (sim=+0.060)
Client 11: weight=1.456 REWARDED (sim=+0.063)
Client 13: weight=0.100 MILD PENALTY (sim=-0.044)
Client 4: weight=0.100 REWARDED (sim=+0.013)
Client 15: weight=0.100 MILD PENALTY (sim=-0.036)
Client 10: weight=1.176 REWARDED (sim=+0.041)
Client 9: weight=0.100 NEUTRAL (sim=+0.006)
Client 12: weight=0.353 NEUTRAL (sim=+0.003)
Client 17: weight=0.100 MILD PENALTY (sim=-0.032)
→ Total weight: 11.713 | Avg weight: 0.586
INFO :      configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 6
6x across cluster]
(ClientAppActor pid=15352) [SETUP] Client 14 is MALICIOUS (label_flip) [repeated 26x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 16 submitted poisoned update [repeated 16x across cluster]
INFO :      aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 95 accuracy recorded: 0.9481 | Δacc = -0.0016
INFO :

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INFO : [ROUND 96]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:28:46,609 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3814 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0016 → PENALISING divergent clients

[ROUND 96] CFA Trust Weights:
Client 4: weight=0.100 CLEARED (dev=0.992)
Client 11: weight=1.456 CLEARED (dev=0.937)
Client 6: weight=0.100 CLEARED (dev=0.986)
Client 0: weight=1.264 CLEARED (dev=0.953)
Client 5: weight=1.195 CLEARED (dev=0.954)
Client 14: weight=0.100 PENALISED (dev=1.038)
Client 1: weight=1.081 CLEARED (dev=0.966)
Client 15: weight=0.100 PENALISED (dev=1.036)
Client 18: weight=0.100 PENALISED (dev=1.040)
Client 12: weight=0.300 PENALISED (dev=1.005)
Client 3: weight=1.409 CLEARED (dev=0.941)
Client 2: weight=1.324 CLEARED (dev=0.948)
Client 19: weight=0.100 PENALISED (dev=1.043)
Client 8: weight=1.323 CLEARED (dev=0.946)
Client 7: weight=0.131 CLEARED (dev=0.986)
Client 9: weight=0.100 PENALISED (dev=0.997)
Client 10: weight=1.176 CLEARED (dev=0.959)
Client 16: weight=0.100 PENALISED (dev=1.045)
Client 17: weight=0.100 PENALISED (dev=1.030)
Client 13: weight=0.100 PENALISED (dev=1.038)
→ Total weight: 11.659 | Avg weight: 0.583
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 96 accuracy recorded: 0.9486 | Δacc = 0.0005
INFO :
INFO : [ROUND 97]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0005 → REWARDING aligned clients

[ROUND 97] CFA Trust Weights:
Client 6: weight=0.100 REWARDED (sim=+0.011)
Client 14: weight=0.100 MILD PENALTY (sim=-0.038)
Client 15: weight=0.100 MILD PENALTY (sim=-0.035)
Client 18: weight=0.100 MILD PENALTY (sim=-0.039)
Client 4: weight=0.100 REWARDED (sim=+0.014)
Client 11: weight=1.465 REWARDED (sim=+0.064)
Client 19: weight=0.100 MILD PENALTY (sim=-0.044)
Client 3: weight=1.417 REWARDED (sim=+0.057)
Client 0: weight=1.270 REWARDED (sim=+0.046)
Client 8: weight=1.330 REWARDED (sim=+0.052)
Client 5: weight=1.201 REWARDED (sim=+0.050)
Client 16: weight=0.100 MILD PENALTY (sim=-0.052)
Client 13: weight=0.100 MILD PENALTY (sim=-0.049)
Client 1: weight=1.085 REWARDED (sim=+0.036)
Client 12: weight=0.300 NEUTRAL (sim=+0.003)
Client 9: weight=0.100 NEUTRAL (sim=+0.006)
Client 7: weight=0.132 REWARDED (sim=+0.014)
Client 2: weight=1.331 REWARDED (sim=+0.055)
Client 17: weight=0.100 MILD PENALTY (sim=-0.030)
Client 10: weight=1.181 REWARDED (sim=+0.042)
→ Total weight: 11.712 | Avg weight: 0.586
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15349) [NON-IID] Client 10 | Total: 3053 samples | Distribution: {0: 997, 1: 736, 2: 102, 3: 112, 4: 144, 5: 815, 6: 29, 7: 4, 8: 33, 9: 81} [repeated 7
4x across cluster]
(ClientAppActor pid=15354) [SETUP] Client 16 is MALICIOUS (label_flip) [repeated 30x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 16 submitted poisoned update [repeated 16x across cluster]
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 97 accuracy recorded: 0.9499 | Δacc = 0.0013
INFO :
INFO : [ROUND 98]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0013 → REWARDING aligned clients

[ROUND 98] CFA Trust Weights:
Client 8: weight=1.338 REWARDED (sim=+0.055)
Client 17: weight=0.100 MILD PENALTY (sim=-0.032)
Client 2: weight=1.338 REWARDED (sim=+0.053)
Client 19: weight=0.100 MILD PENALTY (sim=-0.043)
Client 10: weight=1.186 REWARDED (sim=+0.042)
Client 1: weight=1.089 REWARDED (sim=+0.038)
Client 4: weight=0.100 REWARDED (sim=+0.011)
Client 3: weight=1.426 REWARDED (sim=+0.062)
Client 7: weight=0.132 REWARDED (sim=+0.019)
Client 16: weight=0.100 MILD PENALTY (sim=-0.049)
Client 6: weight=0.101 REWARDED (sim=+0.012)
Client 12: weight=0.300 NEUTRAL (sim=-0.006)
Client 13: weight=0.100 MILD PENALTY (sim=-0.039)
Client 9: weight=0.100 NEUTRAL (sim=+0.006)
Client 18: weight=0.100 MILD PENALTY (sim=-0.035)
Client 14: weight=0.100 MILD PENALTY (sim=-0.040)
Client 0: weight=1.276 REWARDED (sim=+0.047)
Client 11: weight=1.474 REWARDED (sim=+0.063)
Client 5: weight=1.206 REWARDED (sim=+0.047)
Client 15: weight=0.100 MILD PENALTY (sim=-0.042)
→ Total weight: 11.765 | Avg weight: 0.588
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 98 accuracy recorded: 0.9500 | Δacc = 0.0001
INFO :
INFO : [ROUND 99]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(ClientAppActor pid=15347) [NON-IID] Client 17 | Total: 3252 samples | Distribution: {0: 523, 1: 294, 2: 34, 3: 544, 4: 2, 5: 875, 6: 454, 7: 465, 8: 3, 9: 58} [repeated 75
x across cluster]
(ClientAppActor pid=15347) [SETUP] Client 17 is MALICIOUS (label_flip) [repeated 29x across cluster]
(ClientAppActor pid=15349) [ATTACK] Client 18 submitted poisoned update [repeated 12x across cluster]
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = +0.0001 → REWARDING aligned clients

[ROUND 99] CFA Trust Weights:
Client 13: weight=0.100 MILD PENALTY (sim=-0.043)
Client 16: weight=0.100 MILD PENALTY (sim=-0.044)
Client 12: weight=0.300 NEUTRAL (sim=-0.006)
Client 18: weight=0.100 MILD PENALTY (sim=-0.041)
Client 2: weight=1.345 REWARDED (sim=+0.053)
Client 14: weight=0.100 MILD PENALTY (sim=-0.036)
Client 1: weight=1.093 REWARDED (sim=+0.037)
Client 15: weight=0.100 MILD PENALTY (sim=-0.036)
Client 6: weight=0.101 REWARDED (sim=+0.014)
Client 8: weight=1.345 REWARDED (sim=+0.054)
Client 10: weight=1.191 REWARDED (sim=+0.044)
Client 5: weight=1.212 REWARDED (sim=+0.047)
Client 3: weight=1.435 REWARDED (sim=+0.061)
Client 17: weight=0.100 MILD PENALTY (sim=-0.038)
Client 19: weight=0.100 MILD PENALTY (sim=-0.045)
Client 0: weight=1.282 REWARDED (sim=+0.048)
Client 11: weight=1.484 REWARDED (sim=+0.065)
Client 7: weight=0.132 REWARDED (sim=+0.016)
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Client 4: weight=0.101 REWARDED (sim=+0.013)
Client 9: weight=0.100 NEUTRAL (sim=+0.009)
→ Total weight: 11.819 | Avg weight: 0.591
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 99 accuracy recorded: 0.9492 | Δacc = -0.0008
INFO :
INFO : [ROUND 100]
INFO : configure_fit: strategy sampled 20 clients (out of 20)
(raylet) [2026-04-21 13:28:56,625 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12
.3897 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
INFO : aggregate_fit: received 20 results and 0 failures
[CFA] Δacc = -0.0008 → PENALISING divergent clients

[ROUND 100] CFA Trust Weights:
Client 4: weight=0.101 CLEARED (dev=0.986)
Client 14: weight=0.100 PENALISED (dev=1.041)
Client 1: weight=1.093 CLEARED (dev=0.962)
Client 6: weight=0.101 CLEARED (dev=0.982)
Client 5: weight=1.212 CLEARED (dev=0.950)
Client 0: weight=1.282 CLEARED (dev=0.952)
Client 18: weight=0.100 PENALISED (dev=1.033)
Client 2: weight=1.345 CLEARED (dev=0.943)
Client 15: weight=0.100 PENALISED (dev=1.044)
Client 11: weight=1.484 CLEARED (dev=0.934)
Client 12: weight=0.254 PENALISED (dev=1.010)
Client 8: weight=1.345 CLEARED (dev=0.942)
Client 17: weight=0.100 PENALISED (dev=1.037)
Client 16: weight=0.100 PENALISED (dev=1.050)
Client 13: weight=0.100 PENALISED (dev=1.039)
Client 7: weight=0.132 CLEARED (dev=0.982)
Client 19: weight=0.100 PENALISED (dev=1.046)
Client 10: weight=1.191 CLEARED (dev=0.955)
Client 9: weight=0.100 PENALISED (dev=0.992)
Client 3: weight=1.435 CLEARED (dev=0.938)
→ Total weight: 11.774 | Avg weight: 0.589
INFO : configure_evaluate: strategy sampled 20 clients (out of 20)
INFO : aggregate_evaluate: received 20 results and 0 failures
[CFA] Round 100 accuracy recorded: 0.9518 | Δacc = 0.0026
INFO :
INFO : [SUMMARY]
INFO : Run finished 100 round(s) in 281.53s
INFO : History (loss, distributed):
INFO : round 1: 0.1661
INFO : round 2: 0.4089
INFO : round 3: 0.5095
INFO : round 4: 0.5509
INFO : round 5: 0.565
INFO : round 6: 0.5911
INFO : round 7: 0.6213
INFO : round 8: 0.6241
INFO : round 9: 0.6796
INFO : round 10: 0.6861
INFO : round 11: 0.7047
INFO : round 12: 0.7015
INFO : round 13: 0.7254
INFO : round 14: 0.7527
INFO : round 15: 0.7557
INFO : round 16: 0.7786
INFO : round 17: 0.7905
INFO : round 18: 0.7873
INFO : round 19: 0.8184
INFO : round 20: 0.8256
INFO : round 21: 0.8324
INFO : round 22: 0.8322
INFO : round 23: 0.8492
INFO : round 24: 0.8438
INFO : round 25: 0.861
INFO : round 26: 0.8669
INFO : round 27: 0.8805
INFO : round 28: 0.8793
INFO : round 29: 0.8801
INFO : round 30: 0.8803
INFO : round 31: 0.8864
INFO : round 32: 0.886
INFO : round 33: 0.8889
INFO : round 34: 0.8971
INFO : round 35: 0.8985
INFO : round 36: 0.9022
INFO : round 37: 0.9015
INFO : round 38: 0.8977
INFO : round 39: 0.9061
INFO : round 40: 0.9044
INFO : round 41: 0.9134
INFO : round 42: 0.912
INFO : round 43: 0.9058
INFO : round 44: 0.9157
INFO : round 45: 0.9169
INFO : round 46: 0.9189
INFO : round 47: 0.9159
INFO : round 48: 0.9178
INFO : round 49: 0.9219
INFO : round 50: 0.9235
INFO : round 51: 0.922
INFO : round 52: 0.9247
INFO : round 53: 0.9269
INFO : round 54: 0.9257
INFO : round 55: 0.9281
INFO : round 56: 0.9084
INFO : round 57: 0.9274
INFO : round 58: 0.9273
INFO : round 59: 0.9056
INFO : round 60: 0.9297
INFO : round 61: 0.9305
INFO : round 62: 0.9306
INFO : round 63: 0.9313
INFO : round 64: 0.9315
INFO : round 65: 0.9309
INFO : round 66: 0.9324
INFO : round 67: 0.9327
INFO : round 68: 0.9342
INFO : round 69: 0.9324
INFO : round 70: 0.9335
INFO : round 71: 0.9347
INFO : round 72: 0.9354
INFO : round 73: 0.9379
INFO : round 74: 0.9342
INFO : round 75: 0.9348
INFO : round 76: 0.9339
INFO : round 77: 0.9357
INFO : round 78: 0.9352
INFO : round 79: 0.9381
INFO : round 80: 0.9399
INFO : round 81: 0.9409
INFO : round 82: 0.9411
INFO : round 83: 0.9418
INFO : round 84: 0.9428
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INFO : round 85: 0.9425
INFO : round 86: 0.9419
INFO : round 87: 0.9448
INFO : round 88: 0.9447
INFO : round 89: 0.9442
INFO : round 90: 0.9469
INFO : round 91: 0.9449
INFO : round 92: 0.9476
INFO : round 93: 0.9469
INFO : round 94: 0.9497
INFO : round 95: 0.9481
INFO : round 96: 0.9486
INFO : round 97: 0.9499
INFO : round 98: 0.95
INFO : round 99: 0.9492
INFO : round 100: 0.9518
INFO : History (metrics, distributed, evaluate):
INFO : {'accuracy': [(1, 0.1661),
INFO : (2, 0.4089),
INFO : (3, 0.5095),
INFO : (4, 0.5509),
INFO : (5, 0.565),
INFO : (6, 0.5911),
INFO : (7, 0.6213),
INFO : (8, 0.6241),
INFO : (9, 0.6796),
INFO : (10, 0.6861),
INFO : (11, 0.7047),
INFO : (12, 0.7015),
INFO : (13, 0.7254),
INFO : (14, 0.7527),
INFO : (15, 0.7557),
INFO : (16, 0.7786),
INFO : (17, 0.7905),
INFO : (18, 0.7873),
INFO : (19, 0.8184),
INFO : (20, 0.8256),
INFO : (21, 0.8324),
INFO : (22, 0.8322),
INFO : (23, 0.8492),
INFO : (24, 0.8438),
INFO : (25, 0.861),
INFO : (26, 0.8669),
INFO : (27, 0.8805),
INFO : (28, 0.8793),
INFO : (29, 0.8801),
INFO : (30, 0.8803),
INFO : (31, 0.8864),
INFO : (32, 0.886),
INFO : (33, 0.8889),
INFO : (34, 0.8971),
INFO : (35, 0.8985),
INFO : (36, 0.9022),
INFO : (37, 0.9015),
INFO : (38, 0.8977),
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INFO : (41, 0.9134),
INFO : (42, 0.912),
INFO : (43, 0.9058),
INFO : (44, 0.9157),
INFO : (45, 0.9169),
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INFO : (53, 0.9269),
INFO : (54, 0.9257),
INFO : (55, 0.9281),
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INFO : (58, 0.9273),
INFO : (59, 0.9056),
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INFO : (61, 0.9305),
INFO : (62, 0.9306),
INFO : (63, 0.9313),
INFO : (64, 0.9315),
INFO : (65, 0.9309),
INFO : (66, 0.9324),
INFO : (67, 0.9327),
INFO : (68, 0.9342),
INFO : (69, 0.9324),
INFO : (70, 0.9335),
INFO : (71, 0.9347),
INFO : (72, 0.9354),
INFO : (73, 0.9379),
INFO : (74, 0.9342),
INFO : (75, 0.9348),
INFO : (76, 0.9339),
INFO : (77, 0.9357),
INFO : (78, 0.9352),
INFO : (79, 0.9381),
INFO : (80, 0.9399),
INFO : (81, 0.9409),
INFO : (82, 0.9411),
INFO : (83, 0.9418),
INFO : (84, 0.9428),
INFO : (85, 0.9425),
INFO : (86, 0.9419),
INFO : (87, 0.9448),
INFO : (88, 0.9447),
INFO : (89, 0.9442),
INFO : (90, 0.9469),
INFO : (91, 0.9449),
INFO : (92, 0.9476),
INFO : (93, 0.9469),
INFO : (94, 0.9497),
INFO : (95, 0.9481),
INFO : (96, 0.9486),
INFO : (97, 0.9499),
INFO : (98, 0.95),
INFO : (99, 0.9492),
INFO : (100, 0.9518)]}]

```

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--- cfa_trust_noniid ---
Round 1: 0.1661
Round 2: 0.4089
Round 3: 0.5095
Round 4: 0.5509
Round 5: 0.5650
Round 6: 0.5911
Round 7: 0.6213

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Round 8: 0.6241
Round 9: 0.6796
Round 10: 0.6861
Round 11: 0.7047
Round 12: 0.7015
Round 13: 0.7254
Round 14: 0.7527
Round 15: 0.7557
Round 16: 0.7786
Round 17: 0.7905
Round 18: 0.7873
Round 19: 0.8184
Round 20: 0.8256
Round 21: 0.8324
Round 22: 0.8322
Round 23: 0.8492
Round 24: 0.8438
Round 25: 0.8610
Round 26: 0.8669
Round 27: 0.8805
Round 28: 0.8793
Round 29: 0.8801
Round 30: 0.8803
Round 31: 0.8864
Round 32: 0.8860
Round 33: 0.8889
Round 34: 0.8971
Round 35: 0.8985
Round 36: 0.9022
Round 37: 0.9015
Round 38: 0.8977
Round 39: 0.9061
Round 40: 0.9044
Round 41: 0.9134
Round 42: 0.9120
Round 43: 0.9058
Round 44: 0.9157
Round 45: 0.9169
Round 46: 0.9189
Round 47: 0.9159
Round 48: 0.9178
Round 49: 0.9219
Round 50: 0.9235
Round 51: 0.9220
Round 52: 0.9247
Round 53: 0.9269
Round 54: 0.9257
Round 55: 0.9281
Round 56: 0.9084
Round 57: 0.9274
Round 58: 0.9273
Round 59: 0.9056
Round 60: 0.9297
Round 61: 0.9305
Round 62: 0.9306
Round 63: 0.9313
Round 64: 0.9315
Round 65: 0.9309
Round 66: 0.9324
Round 67: 0.9327
Round 68: 0.9342
Round 69: 0.9324
Round 70: 0.9335
Round 71: 0.9347
Round 72: 0.9354
Round 73: 0.9379
Round 74: 0.9342
Round 75: 0.9348
Round 76: 0.9339
Round 77: 0.9357
Round 78: 0.9352
Round 79: 0.9381
Round 80: 0.9399
Round 81: 0.9409
Round 82: 0.9411
Round 83: 0.9418
Round 84: 0.9428
Round 85: 0.9425
Round 86: 0.9419
Round 87: 0.9448
Round 88: 0.9447
Round 89: 0.9442
Round 90: 0.9469
Round 91: 0.9449
Round 92: 0.9476
Round 93: 0.9469
Round 94: 0.9497
Round 95: 0.9481
Round 96: 0.9486
Round 97: 0.9499
Round 98: 0.9500
Round 99: 0.9492
Round 100: 0.9518
[PLOT] Saved → /Users/apple/Desktop/Msc Project/Artefact/project-main/results/attack1

Round-by-round accuracy:

Round 1: 0.1661
Round 2: 0.4089
Round 3: 0.5095
Round 4: 0.5509
Round 5: 0.5650
Round 6: 0.5911
Round 7: 0.6213
Round 8: 0.6241
Round 9: 0.6796
Round 10: 0.6861
Round 11: 0.7047
Round 12: 0.7015
Round 13: 0.7254
Round 14: 0.7527
Round 15: 0.7557
Round 16: 0.7786
Round 17: 0.7905
Round 18: 0.7873
Round 19: 0.8184
Round 20: 0.8256
Round 21: 0.8324
Round 22: 0.8322
Round 23: 0.8492
Round 24: 0.8438
Round 25: 0.8610
Round 26: 0.8669
Round 27: 0.8805
Round 28: 0.8793
Round 29: 0.8801
Round 30: 0.8803
Round 31: 0.8864

Round 32: 0.8860
Round 33: 0.8889
Round 34: 0.8971
Round 35: 0.8985
Round 36: 0.9022
Round 37: 0.9015
Round 38: 0.8977
Round 39: 0.9061
Round 40: 0.9044
Round 41: 0.9134
Round 42: 0.9120
Round 43: 0.9058
Round 44: 0.9157
Round 45: 0.9169
Round 46: 0.9189
Round 47: 0.9159
Round 48: 0.9178
Round 49: 0.9219
Round 50: 0.9235
Round 51: 0.9220
Round 52: 0.9247
Round 53: 0.9269
Round 54: 0.9257
Round 55: 0.9281
Round 56: 0.9084
Round 57: 0.9274
Round 58: 0.9273
Round 59: 0.9056
Round 60: 0.9297
Round 61: 0.9305
Round 62: 0.9306
Round 63: 0.9313
Round 64: 0.9315
Round 65: 0.9309
Round 66: 0.9324
Round 67: 0.9327
Round 68: 0.9342
Round 69: 0.9324
Round 70: 0.9335
Round 71: 0.9347
Round 72: 0.9354
Round 73: 0.9379
Round 74: 0.9342
Round 75: 0.9348
Round 76: 0.9339
Round 77: 0.9357
Round 78: 0.9352
Round 79: 0.9381
Round 80: 0.9399
Round 81: 0.9409
Round 82: 0.9411
Round 83: 0.9418
Round 84: 0.9428
Round 85: 0.9425
Round 86: 0.9419
Round 87: 0.9448
Round 88: 0.9447
Round 89: 0.9442
Round 90: 0.9469
Round 91: 0.9449
Round 92: 0.9476
Round 93: 0.9469
Round 94: 0.9497
Round 95: 0.9481
Round 96: 0.9486
Round 97: 0.9499
Round 98: 0.9500
Round 99: 0.9492
Round 100: 0.9518

(raylet) [2026-04-21 13:29:06,632 E 15344 723413] (raylet) file_system_monitor.cc:116: /tmp/ray/session_2026-04-21_13-24-14_497870_15335 is over 95% full, available space: 12.3885 GB; capacity: 460.432 GB. Object creation will fail if spilling is required.
(ClientAppActor pid=15353) [NON-IID] Client 18 | Total: 3829 samples | Distribution: {0: 28, 1: 625, 2: 13, 3: 1019, 4: 255, 5: 465, 6: 1172, 7: 225, 8: 18, 9: 9} [repeated 64x across cluster]
(ClientAppActor pid=15353) [SETUP] Client 18 is MALICIOUS (label_flip) [repeated 27x across cluster]
(ClientAppActor pid=15354) [ATTACK] Client 15 submitted poisoned update [repeated 12x across cluster]
(fl-env) apple@MacBook-Pro-10 project-main %